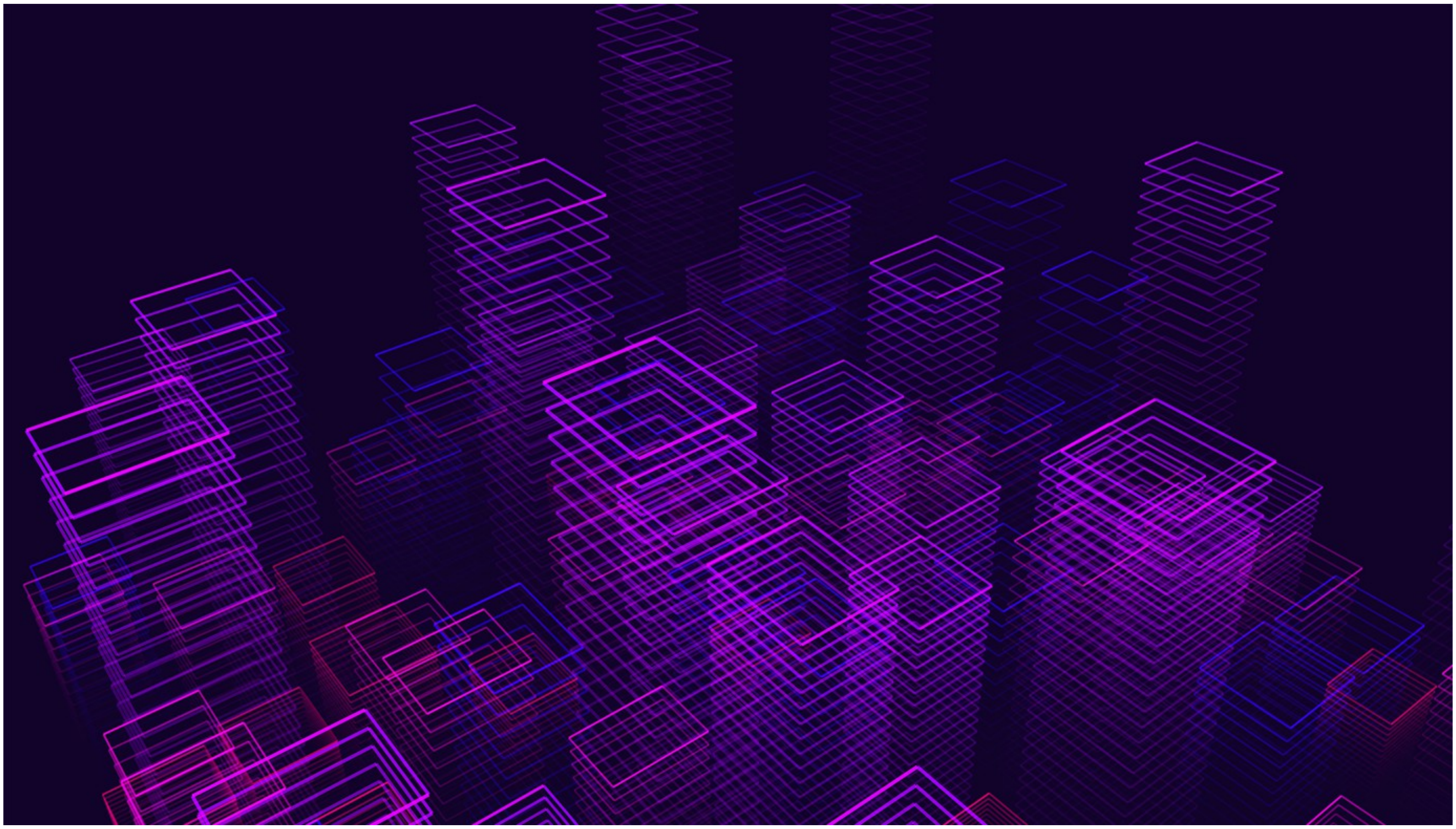
October 18, 2023 10:12 PM GMT

Nu Holdings

The Path to $100 Billion

Nubank could reach US$100 billion valuation by 2026. Increasing product penetration in Brazil and success in Mexico and Colombia are highly underappreciated. Our new product-by-product and country-by-country model suggests US$65 billion in market cap could be unlocked. Our new 2024 PT is US$16, offering 107% upside. NU is our Top Pick.



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Summary & Conclusions

We believe Nubank could reach US$100 billion valuation by 2026. The company is well underway in building a sizable and profitable operation in Brazil, reaching more than 80 million clients and a profit run rate close to US$1 billion. Yet, we think it can become an even larger business, growing revenues threefold in the next five years by cross-selling products to its very large and loyal client base. In addition, we believe management can successfully replicate its business model in Mexico and Colombia, reaching 45 million clients in the next five years and growing revenues to one-fourth the size of Brazil over the same period.

The market is significantly underestimating Nubank's ability to deliver growth and profitability — increasing product penetration in Brazil and success in Mexico and Colombia are not priced in and remain key stock catalysts, in our view. Our conversations with investors and current sell-side consensus estimates suggest that the market is underestimating the opportunity in Brazil and not pricing in much related to Mexico and Colombia. Based on our unique, proprietary, and granular work in this report, we are increasing our net income estimates as much as 46% by 2028 and putting our 2025 through 2028 numbers 30 to 50% above consensus. We are also rolling over our price target to year-end 2024 and increasing it to US$16 per share, versus US$11 before — at the top of the Street and offering an attractive 107% upside. NU is our Latam financials Top Pick.

The cross-selling opportunity in Brazil could unlock US$43 billion in market cap. In our view, Nubank has a unique opportunity to win greater share of wallet among its clients in Brazil by cross-selling products that still show low penetration (e.g., personal, payroll, and BNPL loans). For instance, we expect Nubank can cross-sell payroll loans to 8% of its active client base in Brazil and capture 14% market share five years from now. Our deep dive work in this report and over the past two years underscores the attractive cross-selling opportunity in Brazil. Must-read reports include The Rise of Digital Banks, The Power of Cards, The Rise of Digital Payroll Loans, and The Rise of Latam BNPL.

Mexico and Colombia could add US$22 billion to Nubank's market cap, unlocking significant value no one seems to be pricing in. Nubank is well positioned for success in Mexico and Colombia, in our view, as it can leverage its powerful brand, best-in-class customer experience, tech superiority, advanced underwriting processes, and low-cost structure in these inefficient banking systems. Mexico and Colombia have a combined population of +180 million compared to +200 million in Brazil, similar levels of GDP per capita adjusted for purchasing power parity, and a large underserved and unbanked portion of the population. We think Nubank's operations in those two countries are poised to grow at an accelerated pace. We see a huge opportunity to capture new clients (45 million by 2028) and increase profitability by cross-selling new products that are expected to roll out at a faster pace relative to what occurred in Brazil. For instance, the company already launched personal loans in Mexico last month and is planning to roll out payroll loans before year-end. And we think it can cross-sell personal and payroll loans to 10% and 5%, respectively, of its active client base in Mexico, capturing 12% and 9% market share, respectively, five years from now.

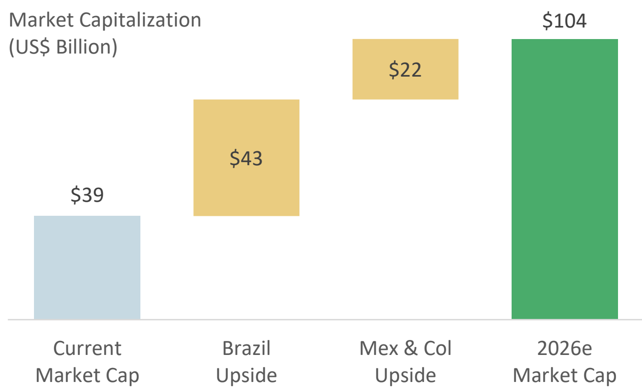
Why is this report unique? Unfortunately, the company provides limited disclosure on a product-by-product and country-by-country basis, making it very difficult for the market to understand the opportunities to increase cross-selling of multiple products in Brazil and successfully grow clients and cross-sell products in Mexico and Colombia. However, we believe our work sheds light on many unanswered questions, and it provides a distinctive framework and proprietary analyses that will allow investors to size up these opportunities. For example: i) Our proprietary unit economics builds in Mexico and Brazil provide greater visibility on six products that Nubank offers or will roll out soon — credit cards, personal loans, payroll loans, BNPL, insurance, and investments. ii) Our proprietary TAM models underscore the massive revenue pool available for disruption of ~US$100 billion, leveraging industry data and our unit economics builds. iii) The insights from our work on rightsizing the client base and product penetration in Mexico over the next five years is something investors are probably going to see for the first time. iv) Our new and improved financial model for Nubank incorporates product-by-product and country-by-country build-ups, making it the first of its kind in the Street, we think.

THE TOP 10 FINDINGS & CONCLUSIONS FROM OUR WORK ARE SUMMARIZED BELOW:

1) Nubank could reach over US$100 billion valuation by 2026, as higher product penetration in Brazil and success in Mexico and Colombia could unlock US$65 billion in market cap. Our new product-by-product and country-by-country model suggests higher product penetration in Brazil — mainly personal, payroll, and BNPL loans — and successful customer growth and product penetration in Mexico and Colombia could add an additional US$65 billion in market cap.

Methodology: We use our discount rate to push forward our residual income-based 2024 PT of US$16 to year-end 2026 — implying Nubank's fair value will reach US$104 billion, or US$20.9 per share, three years from now, implying a P/E multiple of 22.5x. We calculate Brazil will be worth US$82 billion, or US$16.4 per share, using our discount rate to push forward the NPV from our previous standalone Brazil model to year-end 2026. Then we calculate the fair value of Mexico and Colombia as the difference between the value of our new consolidated model (Brazil + Mexico + Colombia) and the value of our previous standalone Brazil model. We get to a 2026 value of US$22 billion, or US$4.5 per share.

Exhibit 1: Increasing penetration in Brazil and success in Mexico and Colombia...



Source: Morgan Stanley Research Estimates (e).

Exhibit 2: ... can make Nubank a US$100 billion market cap stock by 2026.

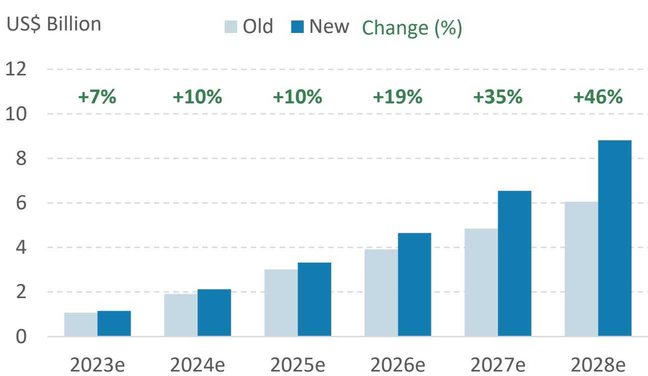
Discount Rate	14.2%

US$	2024e	2025e	2026e	2027e	2028e
Fair Value / Share	16.0	18.3	20.9	23.9	27.3
EPS	0.42	0.66	0.93	1.31	1.76
P/E	37.8x	27.5x	22.5x	18.3x	15.5x
Market Cap ($Bn)	80	91	**104**	119	136

Source: Morgan Stanley Research Estimates (e).

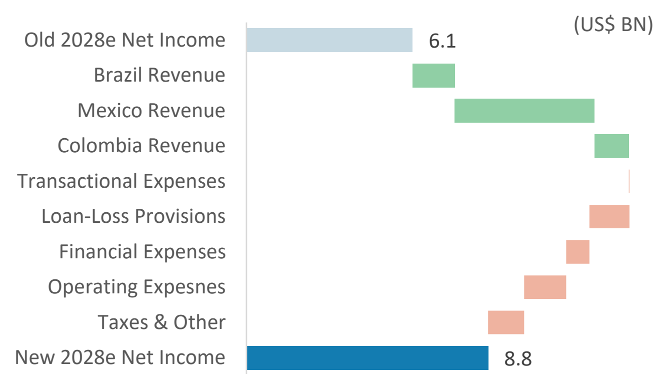
2) We are increasing our Nubank earnings and ROE estimates on the back of our new bottom-up buildup models by country. The outputs of our detailed models suggest that our consolidated numbers for Nubank looked low. Hence, we are gradually increasing our net income estimates, starting with 7% in 2023 and as much as 46% by 2028. As a result, we now see a sustainable ROE of ~36% between 2025 and 2028, versus 30% before. Note, while our prior model tried to capture some upside from Mexico and Colombia, it was not based on the detailed bottom-up product-by-product and country-by-country revamp we undertook this time around.

Exhibit 3: We are increasing our 2028 NI estimate by 46%...



Source: Morgan Stanley Research Estimates (e).

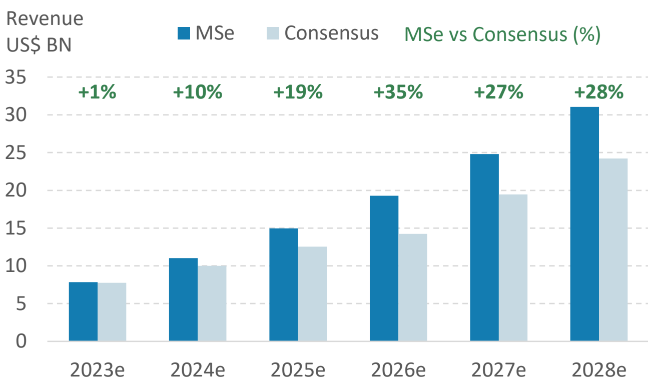
Exhibit 4: ... as we include Mexico and Colombia to our model.



Source: Morgan Stanley Research Estimates (e).

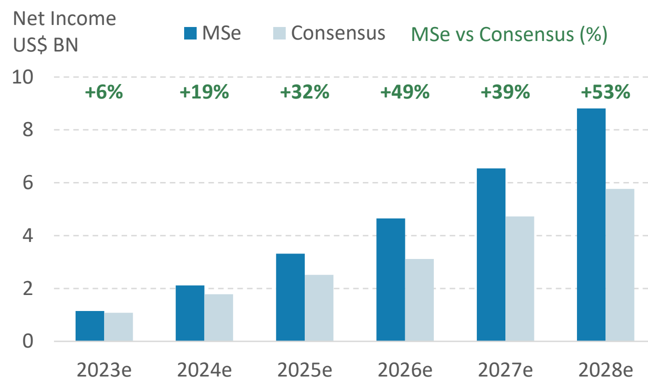
3) Not only is the market underestimating the opportunity to increase product penetration in Brazil, but we also think consensus is not pricing in much related to Mexico and Colombia. Based on our granular product-by-product and country-by-country model, our 2025 through 2028 consolidated net income numbers are 30 to 50% above consensus. In addition, our detailed work on unit economics gave us greater conviction to forecast attractive gross profit margin expansion going forward. And this, combined with operating leverage on the back of Nubank's highly scalable business model, explains why our new net income estimates are well above consensus.

Exhibit 5: Our new 2025 to 2028 revenue estimates for Nubank are 20-30% above consensus...



Source: Visible Alpha, Morgan Stanley Research Estimates (e).

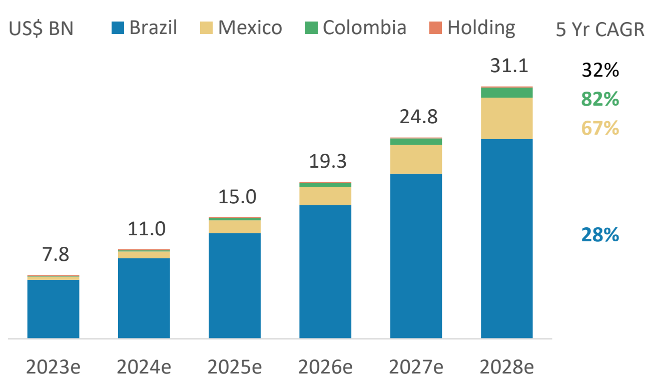
Exhibit 6: ... and our net income numbers are 30-50% higher, as we see attractive margin expansion.



Source: Visible Alpha, Morgan Stanley Research Estimates (e).

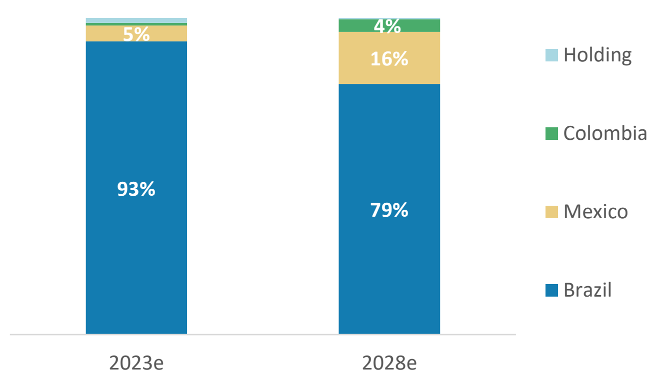
4) We see Mexico and Colombia making up 20% of total revenue in 2028, up from 6% currently. This means Mexico and Colombia revenue should grow at five-year CAGRs of 67% and 82%, respectively, reaching US$5.1 and US$1.2 billion in 2028. So, our total revenue forecast is now 34% higher than before, reaching US$31 billion in 2028. Our revenue forecasts are based on our unit economics models and reasonable customer growth and product penetration assumptions in each country. More on this below.

Exhibit 7: We increased our 2028 revenue forecast by 30%, reaching US$31 billion in 2028.



Source: Morgan Stanley Research Estimates (e).

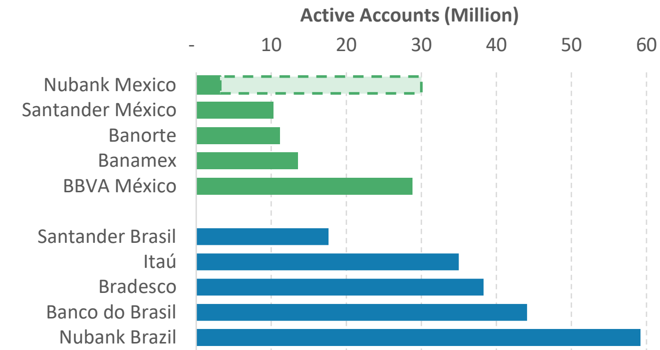
Exhibit 8: As Mexico and Colombia ramp-up, we see them reaching 20% of revenue in 2028.



Source: Morgan Stanley Research Estimates (e).

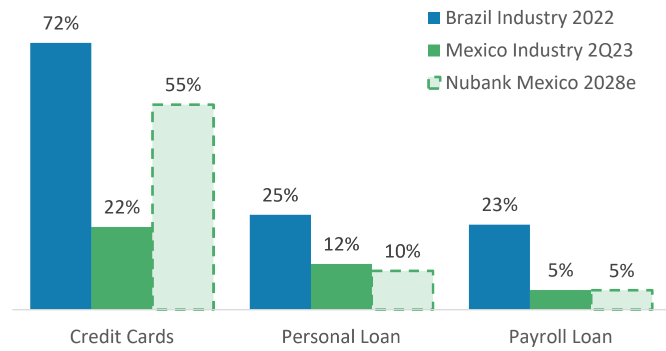
5) Rightsizing the opportunity in Mexico, we calculate Nubank can capture ~30 million active clients and increase penetration of products significantly over the next five years. We came to these conclusions by considering the following: the number of clients owned by looking at Nubank's performance relative to the incumbent banks in Mexico and Brazil, including the number of active clients and product penetration. We also took into account demographics and smartphone penetration in each country. All told, we are optimistic on the growth prospects for Nubank in Mexico given the powerful drivers of financial inclusion and electronification of financial services. We estimate ~11.5 million individuals in Mexico already applied to Nubank, supporting our view that its 3.6 million client base could grow much faster.

Exhibit 9: Nubank can capture ~30 million active customers in Mexico by 2028...



Source: Company Data, Morgan Stanley Research Estimates (e).

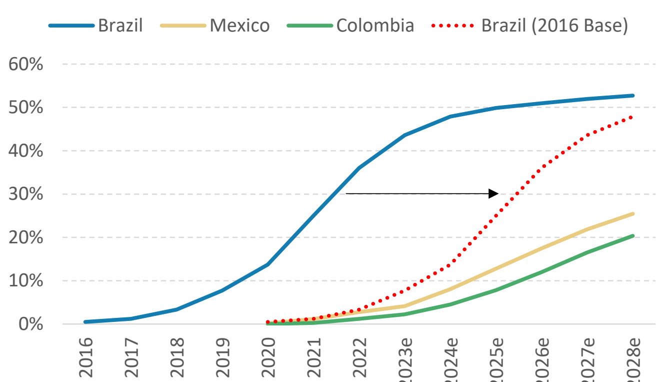
Exhibit 10: ... increasing penetration of personal and payroll loans from 0% to 5-10%.



Source: BCB, CNBV, Morgan Stanley Research Estimates (e).

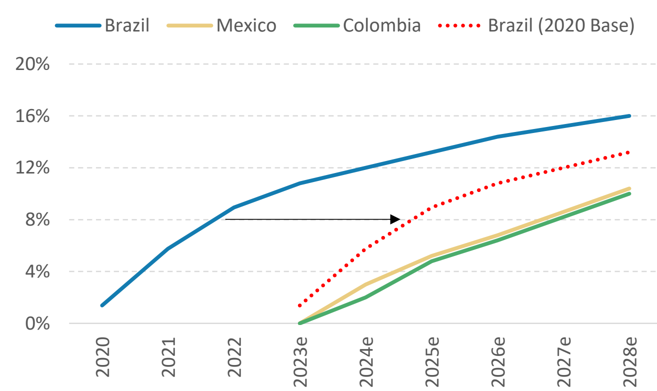
And we think our estimates for client growth and product penetration outside of Brazil are quite reasonable, if not conservative. Indeed, our current projections for client growth in Mexico and Colombia imply a much more gradual ramp-up versus what occurred in Brazil over the past handful of years. We are also modeling to a more gradual increase in penetration of key lending products. Looking at personal loans, which were launched back in 2019 in Brazil, we estimate that reaching 8% penetration will take almost 2 years longer outside of Brazil, a conservative view.

Exhibit 11: Active Clients By Country (% of Adult Population).



Source: Company data, Morgan Stanley Research Estimates (e).

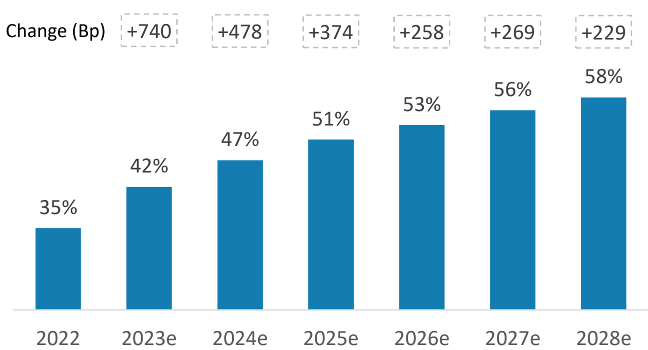
Exhibit 12: Penetration of Personal Loans (% of Active Clients).



Source: Company data, Morgan Stanley Research Estimates (e).

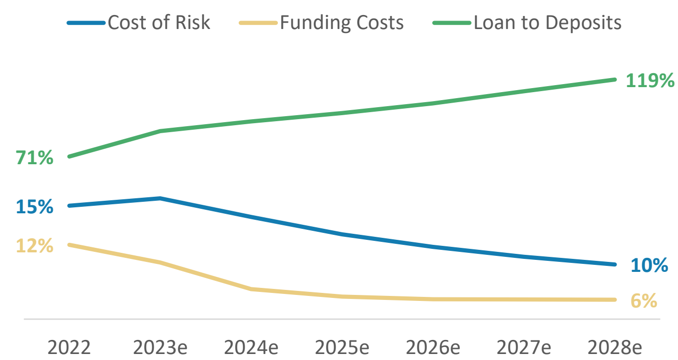
6) Our new estimates build in attractive gross profit margin expansion. Our model yields a gross profit margin of 58% in 2028, up from 42% this year. The expansion is mostly due to revenue growth outpacing that of loan-loss provisions and financial expenses, as a result of the following assumptions: lower cost of risk, lower cost of funding, and higher leverage. First, our long-term cost of risk forecasts in all countries trend lower toward, but slightly above, the historical industry average of each product. Mix shift is also an important driver of lower cost of risk. We forecast lower funding costs on the back of falling interest rates and stable remuneration on a growing deposit base. And we assume leverage will increase as loan growth continues to outpace deposits.

Exhibit 13: We forecast attractive gross profit margin expansion over the next 5 years...



Source: Morgan Stanley Research Estimates (e).

Exhibit 14: ... on the back of lower cost of risk, lower cost of funding, and higher leverage.

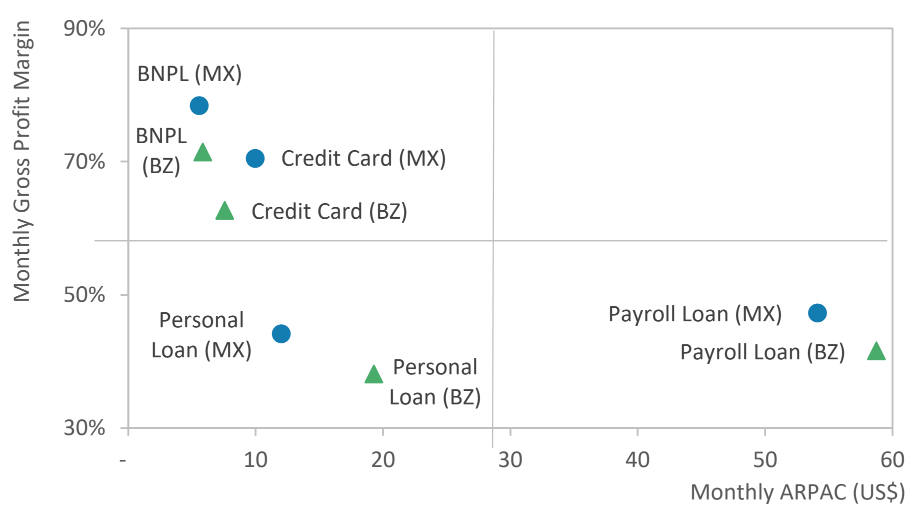


Source: Company Data, Morgan Stanley Research Estimates (e).

7) Our unit economics work shows that lending is more profitable in Mexico relative to Brazil, with higher gross profit margins on a product-by-product basis. Our proprietary unit economics models suggest that Nubank's credit offering should provide higher gross profit margin returns in Mexico — across credit cards, personal loans, payroll loans, and BNPL. Higher share of interest earning-balances in cards, along with lower funding cost and lower cost of risk in personal and payroll loans, are a few examples of the drivers of higher gross profit margins in Mexico.

The assumptions in our models are tailored to Nubank's offering and clients base, aiming to replicate its unit economics in Mexico and Brazil. Our estimates are supported by data from the central bank, channel checks, and scraping of Nubank's disclosures. Importantly, the company does not provide unit economics by product or country. Hence, investors will be seeing many of these numbers for the first time.

Exhibit 15: Nubank's credit offering offers greater profitability in Mexico vis-à-vis Brazil.



Source: Morgan Stanley Research Estimates (e).

8) Credit cards can generate exceptional returns in Mexico, on the back of higher share of interest-earning balances. Our unit economics work show that credit cards in Mexico have 50% greater gross profit contribution compared to Brazil, with gross margins of 70% versus 63%. The main reason behind this gap is that Mexican issuers have a higher share of interest-earning balances and collect interest income for more days, which more than offsets higher financial expense due to greater working capital needs relative to Brazil.

To illustrate this, we ran a sensitivity analysis of Nubank's credit card unit economics in Mexico. Evidence from our analysis shows Nubank's gross profit would be 5% higher for every 5% increase in share of revolving loans and 10% higher for every 5% increase in share of interest with installment. For instance, the gross profit be ~27% higher if we assume interest-earning balances represent ~75% of card expenditures, compared to 55% in our base case.

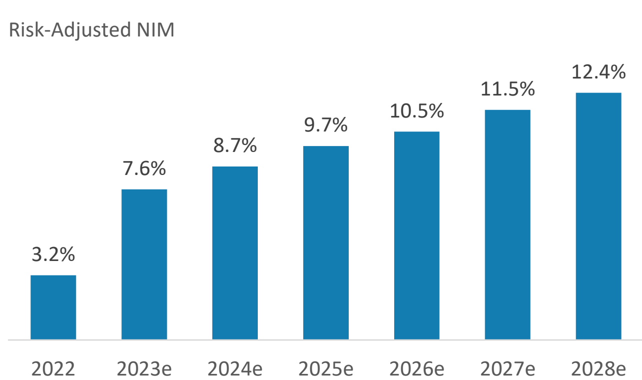
Exhibit 16: Mexico — Nubank's monthly Gross Profit sensitivity to interest-earning balances.

		Revolving (% of Total)				Change in Gross Profit vs Base Case			
Gross Profit		40%	45%	50%	55%				
Installments with interest (% of Total)	5%	6.1	6.5	6.9	7.2	-13%	-8%	-2%	3%
	10%	6.6	**7.0**	7.4	7.8	-5%	**0%**	5%	11%
	15%	7.2	7.6	8.0	8.3	2%	8%	13%	19%
	20%	7.8	8.1	8.5	**8.9**	10%	16%	21%	**27%**

Source: Morgan Stanley Research Estimates (e).

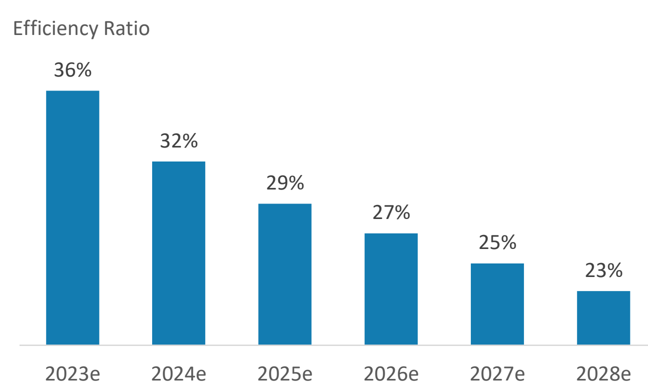
9) Nubank's scalable business model with a high percentage of fixed costs offers further operating leverage as it expands beyond Brazil. Indeed, we see the efficiency ratio improving from 36% in 2023 to 23% in 2028. And based on our unit economics framework, we forecast that the monthly ARPAC should jump from US$9.5 to 21.3 over the next five years, while the monthly cost to serve per client should increase only from US$0.8 to 1.5 over the same period.

Exhibit 17: We expects gradual NIM expansion going forward, after strong performance this year.



Source: Company Data, Morgan Stanley Research Estimates (e).

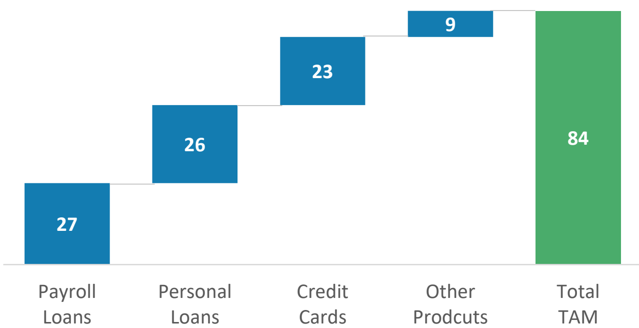
Exhibit 18: We also forecast operating leverage will boost earnings growth and profitability.



Source: Morgan Stanley Research Estimates (e).

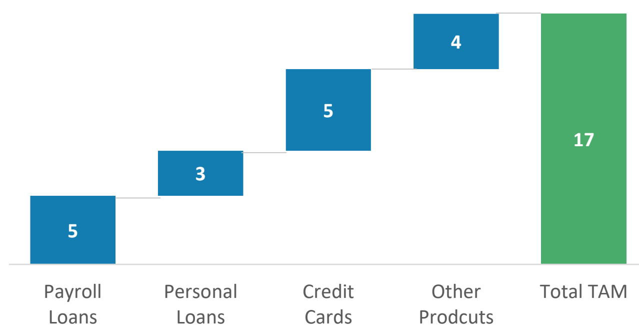
10) We calculate the current revenue TAM is US$84 billion in Brazil and US$17 billion in Mexico and we see Nubank's market share doubling to 16% by 2028. Our top-down TAM models focus on credit cards, personal loans, payroll loans, Buy-Now Pay-Later (BNPL), retail investments, and insurance. Given limited data on these addressable markets, our proprietary analysis leverages multiple sources along with our country-level expertise of banking and lending products in Brazil. Overall, we get to a combined revenue TAM in Brazil and Mexico of over US$100 billion, which represents a meaningful white space opportunity for Nubank looking at our combined revenue estimates in these countries — US$7.7 billion in 2023, or 8% market share, growing to US$29.7 billion in 2028, or 16% market share.

Exhibit 19: We estimate Brazil revenue TAM is currently US$84 Billion...



Source: Morgan Stanley Research Estimates (e). Note: Others include insurance, investments and BNPL.

Exhibit 20: ... four times greater than Mexico's US$17 Billion revenue TAM.



Source: Morgan Stanley Research Estimates (e). Note: Others include insurance, investments and BNPL.

What's Included in This Report?

THE PATH TO US$100 BILLION

1. Snapshot of key figures by country. We present a summary of Nubank's key figures in each country and for the consolidated business, looking at our estimates for this year and five years out, including number of active customers, penetration of the adult population, product penetration, revenue by product, revenue per active client, market shares, and cost of risk.

2. Product unit economics in Mexico and Brazil. We provide our proprietary unit economics models for six products that Nubank offers or will roll out soon in Mexico and we benchmark these to the unit economics in Brazil. Our unit economics work providse greater visibility on revenue per active client, gross profit contribution, and gross profit margins, among other things. Our analysis covers credit cards, personal loans, payroll loans, BNPL, and other non-credit products like insurance and investment brokerage.

3. Sensitivity analysis of Nubank's credit card unit economics. Leveraging our unit economics models, we ran different scenarios changing the mix of credit card receivables. Assuming higher and lower share of interest-earning receivables — revolving and installments with interest — we can quantify the impact in revenue per active client, gross profit contribution, and gross profit margin. Given the early stage development of Nubank Mexico, we think this sensitivity analysis provides insight into the different potential outcomes.

4. Total addressable market estimates by product and by country. We detail the key assumptions used to calculate the revenue TAM for each of the six products mentioned before in both Mexico and Brazil. Leveraging industry data and our unit economics models, our TAM estimates underscore the massive revenue pool available for disruption of ~US$100 billion. We also calculate the gross profit TAM and the attractive growth opportunity going forward for selected products.

5. Rightsizing the opportunity in Mexico. We take an in-depth look at the financial system in Mexico to measure how many clients Nubank can reasonably capture and what level of product penetration is achievable over the next five years. We analyze and benchmark the performance of incumbent banks and Nubank, as well as other country level metrics. For example, we look at the evolution of clients owned by incumbents and Nubank. We compare financial inclusion, demographics, and smartphone penetration levels between Mexico and Brazil. And we put side by side product penetration levels of incumbent banks and Nubank in Mexico and Brazil.

7. Nubank's financial model overhaul. We believe our new product-by-product and country-by-country model is the first of its kind in the Street. And the outputs of our model underscore our view that the market is massively underestimating Nubank's opportunity to increase product penetration in Brazil and not pricing in much related to Mexico and Colombia. In addition, we provide bull and bear scenarios that provide more visibility on the potential growth paths given the early state of product penetration in Brazil and the business in Mexico and Colombia.

8. The US$100 billion market cap methodology. We value the company based on residual income analysis, measuring the spread between return on equity and cost of equity and the time period in which management can generate returns in excess of the cost of equity. Then we use our discount rate to push forward the fair value of the company. We also calculate the fair value of the Brazil operation and the contribution from Mexico and Brazil. Lastly, we provide implied P/E multiples and benchmark them to global profitable & mature tech-platforms.

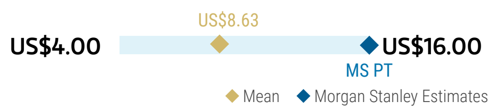
Risk Reward – Nu Holdings Ltd. (NU.N) Top Pick

Overweight on attractive growth avenues and secular tailwinds

PRICE TARGET US$16.00

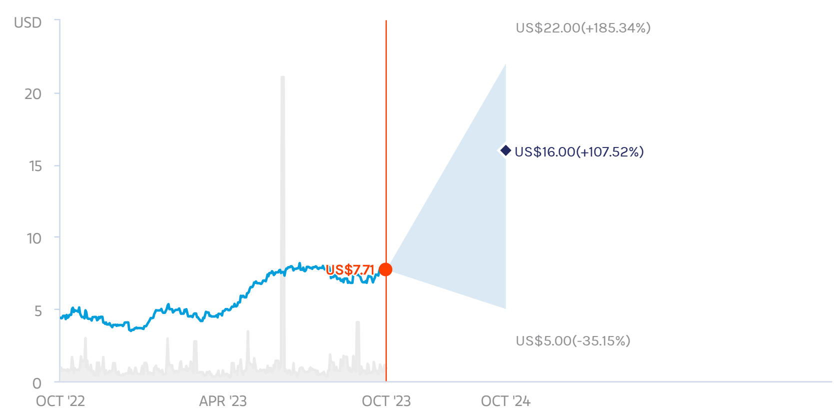
Our valuation work and price targets are based on residual income analysis. In our residual income model, the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). Our valuation model uses a discount rate of 14.2% and long-term ROE of 35%.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



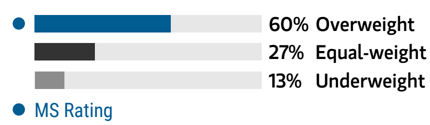
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

Nubank is well-positioned to build one of the most valuable banking franchises in Latam with superior technology, best-in-class customer satisfaction, and strong cohort performance and unit economics. The company has many attractive avenues to increase top line growth: roll out & cross-sell of new products to a huge and loyal client base, new geographies, and M&A. We think the market is massively underestimating its ability to deliver growth and profitability — particularly cross-selling personal and payroll loans in Brazil and building a successful franchise in Mexico and Colombia. We see adjusted ROE reaching ~35% in 2028 on the back of strong revenue growth and operating leverage.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian:	*Positive*
Disruption:	*Positive*
Market Share:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE US$22.00

Faster industry growth & successful cross selling

Secular industry trends surprise to the upside, digitalization and penetration of financial products grow faster than expected. Nubank can capitalize on its huge and loyal customer base by cross selling an increasing number of products, significantly improving client monetization. As a result, top line growth accelerates, while operating leverage allows additional revenues to flow down to the bottom line, supporting higher sustainable ROEs.

BASE CASE US$16.00

Secular tailwinds & client base expansion

Financial inclusion and credit penetration continue to improve on the back of currently low bancarization levels in Latam and ongoing digitalization trends. Nubank can successfully expand into new geographies, continuing to add new clients at a healthy pace, and launching new products in parallel. The company navigates competition effectively and gains market share. Its highly scalable business model drives attractive operating leverage and sustainable ROEs.

BEAR CASE US$5.00

Regulatory headwinds & tougher competition

Macro outlook deteriorates and industry growth slows down more than expected. Competition gets tougher, as new players and incumbents successfully expand digital banking initiatives. Expansion to new geographies and product launches take longer than expected to gain traction, resulting in relatively slower revenue growth. Regulatory hurdles increase pressuring profitability in the short term.

Risk Reward – Nu Holdings Ltd. (NU.N)

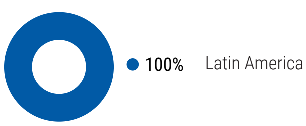
KEY EARNINGS INPUTS

Drivers	2022	2023e	2024e	2025e
Active Clients (MM)	61.2	76.0	88.7	99.0
Net Revenue (YoY, %) (%)	182.2	63.6	40.7	35.7
Gross Profit (YoY, %) (%)	126.9	98.5	56.7	46.5
Card Receivables (YoY, %) (%)	72.2	61.0	40.2	23.3
Loan to Customers (YoY, %) (%)	40.1	142.7	107.6	77.2

INVESTMENT DRIVERS

- Financial inclusion and credit penetration
- New client additions
- Roll out & cross sell of products
- Geographical expansion
- M&A
- New business verticals and products

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	5/5 MOST	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Increasing penetration of financial products in LatAm
- Successful geographical expansion
- New products added to the platform
- Improving cross-sell initiatives
- M&A

RISKS TO DOWNSIDE

- Asset quality deterioration
- Regulatory headwinds
- Tougher competition could impact margins
- Execution risk of international expansion
- Potential conflicts of interest between controlling and minority holders

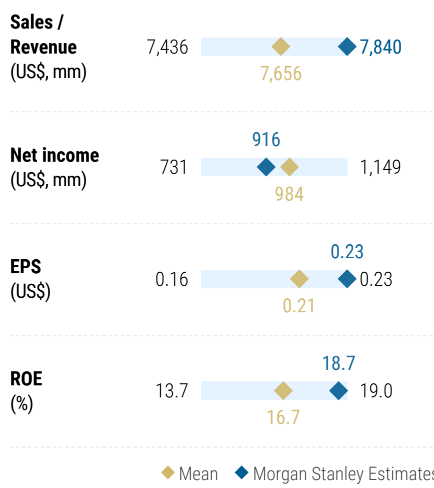
OWNERSHIP POSITIONING

Inst. Owners, % Active	94.4%
HF Sector Long/Short Ratio	1.9x
HF Sector Net Exposure	10.9%

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2023e



Source: Refinitiv, Morgan Stanley Research

NU's Key Figures by Country

Exhibit 21: Nubank — Customers and Revenue by Country.

	2023e	2028e	2023e	2028e	2023e	2028e	2023e	2028e
Active Customers	71	90	4	27	1	8	76	125
As % of Adult Population	44%	53%	4%	25%	2%	20%	25%	40%
Penetration of Active Customers								
Savings Accounts	90%	90%	40%	90%	10%	90%	86%	90%
Credit Cards	53%	60%	88%	55%	90%	55%	55%	58%
Personal Loans	11%	16%	0%	10%	0%	10%	10%	14%
Payroll Loans	1%	8%	0%	5%	0%	5%	1%	8%
BNPL	1%	11%	0%	19%	-	-	1%	12%
Investments	18%	25%	1%	19%	-	-	17%	22%
Insurance	2%	12%	0%	11%	-	-	2%	11%
Revenue (US$ MM)	7,260	24,586	393	5,099	63	1,250	7,840	31,059
Credit Cards	3,518	7,434	382	2,682	57	573	3,957	10,688
Personal Loans	1,476	5,989	-	725	0	313	1,476	7,027
Payroll Loans	60	4,426	-	576	0	183	60	5,185
BNPL	8	457	-	320	-	-	8	777
Investments	120	517	-	101	-	-	120	618
Insurance	23	293	-	76	-	-	23	369
Securities and Others	2,055	5,470	11	620	7	181	2,196	6,395
Revenue Mix								
Credit Cards	48%	30%	97%	53%	90%	46%	50%	34%
Personal Loans	20%	24%	-	14%	-	25%	19%	23%
Payroll Loans	1%	18%	-	11%	-	15%	1%	17%
BNPL	0%	2%	-	6%	-	-	0%	3%
Investments	2%	2%	-	2%	-	-	2%	2%
Insurance	0%	1%	-	1%	-	-	0%	1%
Securities and Others	28%	22%	3%	12%	10%	14%	28%	21%

Source: Morgan Stanley Research Estimates.

Exhibit 22: Nubank — Unit Economics by Country.

								
	2023e	2028e	2023e	2028e	2023e	2028e	2023e	2028e
Monthly ARPAC (US$)	9.3	23.0	10.4	17.2	8.4	13.9	9.5	21.3
Credit Cards	8.6	15.8	11.1	16.5	8.1	11.6	8.8	15.7
Personal Loans	18.7	35.9	-	25.6	-	38.2	18.7	34.5
Payroll Loans	27.5	54.4	-	41.2	-	43.2	27.5	52.0
BNPL	3.1	4.3	-	6.3	-	-	3.1	4.9
Investments	1.0	1.9	-	2.0	-	-	1.0	1.9
Insurance	1.7	2.5	-	2.5	-	-	1.7	2.5
Implicit Market Share								
Credit Cards	13%	20%	4%	14%	3%	14%		
Personal Loans	6%	18%	-	12%	-	6%		
Payroll Loans	1%	14%	-	9%	-	5%		
BNPL	1%	17%	-	11%	-	-		
Investments	2%	3%	0%	2%	-	-		
Insurance	1%	7%	-	2%	-	-		
Deposits	3%	6%	0%	3%	0%	2%		
Cost of Risk	16.4%	9.7%	12.6%	8.5%	13.4%	9.0%	16.2%	9.5%
Credit Cards	14.0%	8.7%	12.6%	7.5%	13.4%	8.5%	13.9%	8.5%
Personal Loans	28.0%	23.0%	-	13.8%	-	13.9%	28.0%	21.1%
Payroll Loans	4.4%	2.8%	-	7.4%	-	3.6%	4.4%	3.3%
BNPL	6.2%	5.0%	-	5.0%	-	-	6.2%	5.0%

Source: Morgan Stanley Research Estimates

Nubank's Financial Model

Exhibit 23: NU – Consolidated Income Statement.

US$ Millions	2018	2019	2020	2021	2022	2023e	2024e	2025e	2026e	2027e	2028e
Interest Income	162	338	383	1,047	3,555	6,225	8,710	11,827	15,312	19,819	24,999
Fee and commission income	157	274	354	651	1,237	1,615	2,322	3,143	3,986	4,968	6,060
Total Revenue	**319**	**612**	**737**	**1,698**	**4,792**	**7,840**	**11,032**	**14,969**	**19,298**	**24,787**	**31,059**
Transaction Expenses	43	79	127	117	176	207	283	356	422	493	557
Loan-Loss Provisions	119	175	169	481	1,405	2,457	3,561	4,559	5,549	6,684	7,924
Interest and Other Financial Expenses	45	110	114	367	1,548	1,875	2,017	2,477	3,061	3,758	4,509
Total Costs of Services	**207**	**364**	**410**	**965**	**3,129**	**4,539**	**5,860**	**7,392**	**9,032**	**10,935**	**12,990**
Gross Profit	**112**	**248**	**327**	**733**	**1,663**	**3,301**	**5,171**	**7,577**	**10,265**	**13,852**	**18,069**
Customer Support and Operations	47	116	124	191	335	486	653	861	1,074	1,332	1,609
Marketing Expenses	6	42	19	80	153	149	245	334	432	555	696
General and Administrative Expenses	75	181	210	403	695	735	896	1,076	1,269	1,472	1,678
Share-based Compensation	9	19	56	225	282	345	457	579	690	814	928
Other Operating Income/(Expense)	(7)	(20)	(10)	(4)	(150)	(222)	(305)	(401)	(499)	(617)	(742)
Operating Expenses	**145**	**377**	**419**	**903**	**1,616**	**1,937**	**2,557**	**3,250**	**3,964**	**4,791**	**5,653**
Other Income/(Expense)	-	-	(101)	-	-	-	-	-	-	-	-
Income Before Income Taxes	**(33)**	**(129)**	**(193)**	**(170)**	**47**	**1,364**	**2,614**	**4,327**	**6,302**	**9,062**	**12,415**
Income Taxes	(5)	(37)	(22)	(5)	56	448	863	1,471	2,206	3,172	4,345
Non-controlling interests	-	-	-	(0)	(0)	-	-	-	-	-	-
Net Income	**(29)**	**(93)**	**(171)**	**(165)**	**(9)**	**916**	**1,752**	**2,856**	**4,096**	**5,890**	**8,070**
Full Diluted Shares Outstanding (mm)	3,670	3,800	4,393	4,871	4,851	4,996	4,996	4,996	4,996	4,996	4,996
EPS (US$)	**(0.01)**	**(0.02)**	**(0.04)**	**(0.04)**	**(0.00)**	**0.18**	**0.35**	**0.57**	**0.82**	**1.18**	**1.62**
(+) Share-based Compensation	9	19	56	225	282	345	457	579	690	814	928
(-) Tax effect on share-based comp	-	0	13	60	73	86	91	116	138	163	186
(-) One-Time Gain / (Loss)	-	-	(101)	(6)	(4)	26	-	-	-	-	-
Adjusted Gain / Loss for the year	**(19)**	**(74)**	**(27)**	**6**	**204**	**1,148**	**2,117**	**3,319**	**4,649**	**6,541**	**8,812**
Adjusted EPS (US$)	**(0.01)**	**(0.02)**	**(0.01)**	**0.00**	**0.04**	**0.23**	**0.42**	**0.66**	**0.93**	**1.31**	**1.76**
Profitability											
Gross Margin	35%	41%	44%	43%	35%	42%	47%	51%	53%	56%	58%
Efficiency Ratio	69%	91%	88%	77%	55%	36%	32%	29%	27%	25%	23%
Operating Margin	-10%	-21%	-26%	-10%	1%	17%	24%	29%	33%	37%	40%
Net Margin	-9%	-15%	-23%	-10%	0%	12%	16%	19%	21%	24%	26%
Adj Net Margin	-6%	-12%	-4%	0%	4%	15%	19%	22%	24%	26%	28%
Adj ROA	-1%	-2%	0%	0%	1%	3%	4%	5%	6%	6%	7%
Adj ROE	-6%	-16%	-5%	0%	4%	21%	30%	36%	36%	37%	36%

Source: Company data, Morgan Stanley Research Estimates (e)

Exhibit 24: NU – Consolidated Balance Sheet.

US$ Millions	2018	2019	2020	2021	2022	2023e	2024e	2025e	2026e	2027e	2028e
Assets											
Cash and cash equivalents	379	1,247	2,344	2,706	4,172	6,544	6,061	5,825	5,824	6,053	7,188
Securities	656	2,401	4,422	10,020	12,859	12,770	17,844	22,671	27,371	32,053	36,541
Credit card receivables, net	1,624	2,787	2,909	4,781	8,233	13,257	18,580	22,907	27,697	33,692	40,051
Loans to customers, net	-	58	175	1,195	1,673	4,061	8,431	14,939	22,863	32,424	43,598
Deferred tax assets	55	94	125	361	811	1,598	2,474	3,445	4,583	5,975	7,545
PP&E + Intangible Assets	7	10	22	86	210	319	388	472	573	697	847
Miscelaneous assets & collateral	36	85	135	641	958	1,143	1,189	1,238	1,288	1,340	1,395
Other financial assets at amortized cost	29	79	23	69	1,000	1,918	2,682	3,400	4,055	4,856	5,697
Other assets	-	-	-	-	-	-	-	-	-	-	-
Total Assets	**2,786**	**6,760**	**10,154**	**19,859**	**29,917**	**41,610**	**57,650**	**74,897**	**94,255**	**117,090**	**142,862**
Liabilities											
Deposits	629	2,693	5,585	9,667	15,809	22,998	33,270	43,875	54,996	66,894	79,268
Payables to credit card network	1,676	2,977	3,331	4,882	7,055	9,955	13,370	16,443	19,826	24,038	28,498
Borrowings and financing	115	303	177	157	586	1,011	1,535	2,167	2,840	3,585	4,358
Other liabilities	69	175	623	709	1,577	1,453	1,530	1,611	1,696	1,786	1,881
Total Liabilities	**2,489**	**6,148**	**9,716**	**15,416**	**25,026**	**35,417**	**49,705**	**64,096**	**79,358**	**96,303**	**114,005**
Shareholders' Equity	**297**	**612**	**438**	**4,441**	**4,891**	**6,194**	**7,945**	**10,801**	**14,897**	**20,787**	**28,857**
Equity attributable to non-controlling interests	-	-	-	2	-	-	-	-	-	-	-
Total Liabilities and Equity	**2,786**	**6,760**	**10,154**	**19,859**	**29,917**	**41,610**	**57,650**	**74,897**	**94,255**	**117,090**	**142,862**

Source: Company data, Morgan Stanley Research Estimates (e)

Exhibit 25: NU – Key Ratios.

US$ Millions	2018	2019	2020	2021	2022	2023e	2024e	2025e	2026e	2027e	2028e
Summary Financials											
Total Clients (MM)	6.0	20.1	33.3	53.9	74.7	93.0	109.6	123.5	136.6	149.5	161.2
Active Clients (MM)	5.1	12.1	21.8	41.1	61.2	76.0	88.7	99.0	108.4	117.5	125.4
Net Revenue	319	612	737	1,698	4,792	7,840	11,032	14,969	19,298	24,787	31,059
Interest Income	162	338	383	1,047	3,555	6,225	8,710	11,827	15,312	19,819	24,999
Fee and commission	157	274	354	651	1,237	1,615	2,322	3,143	3,986	4,968	6,060
Gross Profit	112	248	327	733	1,663	3,301	5,171	7,577	10,265	13,852	18,069
Operating Expenses	145	377	419	903	1,616	1,937	2,557	3,250	3,964	4,791	5,653
Net Income	(29)	(93)	(171)	(165)	(9)	916	1,752	2,856	4,096	5,890	8,070
Adj. Net Income	(19)	(74)	(27)	6	204	1,148	2,117	3,319	4,649	6,541	8,812
EPS (US$)	(0.01)	(0.02)	(0.04)	(0.04)	(0.00)	0.18	0.35	0.57	0.82	1.18	1.62
Adj. EPS (US$)	(0.01)	(0.02)	(0.01)	0.00	0.04	0.23	0.42	0.66	0.93	1.31	1.76
Card Receivables	1,624	2,787	2,909	4,781	8,233	13,257	18,580	22,907	27,697	33,692	40,051
Loans to Customers	-	58	175	1,195	1,673	4,061	8,431	14,939	22,863	32,424	43,598
Deposits	629	2,693	5,585	9,667	15,809	22,998	33,270	43,875	54,996	66,894	79,268
Shareholders Equity	297	612	438	4,441	4,891	6,194	7,945	10,801	14,897	20,787	28,857
Growth Rates											
Total Clients (MM)	NA	235%	66%	62%	38%	25%	18%	13%	11%	9%	8%
Active Clients (MM)	NA	137%	80%	89%	49%	24%	17%	12%	9%	8%	7%
Net Revenue	NA	92%	20%	130%	182%	64%	41%	36%	29%	28%	25%
Interest Income	NA	109%	13%	173%	240%	75%	40%	36%	29%	29%	26%
Fee and commission	NA	74%	29%	84%	90%	31%	44%	35%	27%	25%	22%
Gross Profit	NA	122%	32%	124%	127%	98%	57%	47%	35%	35%	30%
Operating Expenses	NA	160%	11%	116%	79%	20%	32%	27%	22%	21%	18%
Net Income	NA	224%	85%	-4%	-95%	-10270%	91%	63%	43%	44%	37%
Adj. Net Income	NA	285%	-64%	-124%	3129%	463%	84%	57%	40%	41%	35%
EPS (US$)	NA	213%	60%	-9%	-95%	-9958%	91%	63%	43%	44%	37%
Adj. EPS (US$)	NA	272%	-69%	-120%	3252%	452%	84%	57%	40%	41%	35%
Card Receivables	NA	72%	4%	64%	72%	61%	40%	23%	21%	22%	19%
Loans to Customers	NA	NA	201%	584%	40%	143%	108%	77%	53%	42%	34%
Deposits	NA	328%	107%	73%	64%	45%	45%	32%	25%	22%	18%
Shareholders Equity	NA	106%	-28%	914%	10%	27%	28%	36%	38%	40%	39%
Profit Margins & Key Ratios											
Gross Margin	35%	41%	44%	43%	35%	42%	47%	51%	53%	56%	58%
Efficiency Ratio	69%	91%	88%	77%	55%	36%	32%	29%	27%	25%	23%
Operating Margin	-10%	-21%	-12%	-10%	1%	17%	24%	29%	33%	37%	40%
LLP/Avg Loans	NA	7.8%	5.7%	11.3%	17.3%	18.6%	16.5%	14.3%	12.8%	11.6%	10.7%
Net Margin	-9%	-15%	-23%	-10%	0%	12%	16%	19%	21%	24%	26%
Adj Net Margin	-6%	-12%	-4%	0%	4%	15%	19%	22%	24%	26%	28%
Adj ROA	-1%	-2%	0%	0%	1%	3%	4%	5%	6%	6%	7%
Adj ROE	-6%	-16%	-5%	0%	4%	21%	30%	36%	36%	37%	36%
ARPAC - Yearly (US$)	63	71	43	55	93	114	134	159	186	220	256
ARPAC - Monthly (US$)	5.2	5.9	3.6	4.6	7.7	9.5	11.1	13.3	15.5	18.3	21.3
Cost to Serve - Yearly (US$)	18	23	15	10	10	10	11	13	14	16	18
Cost to Serve - Monthly (US$)	1.5	1.9	1.2	0.8	0.8	0.8	0.9	1.1	1.2	1.3	1.5
Revenue By Country	**319**	**612**	**737**	**1,698**	**4,792**	**7,840**	**11,032**	**14,969**	**19,298**	**24,787**	**31,059**
Brazil	319	612	736	1,645	4,506	7,260	9,918	13,023	16,437	20,324	24,586
Mexico	-	0	1	30	201	393	837	1,545	2,279	3,540	5,099
Colombia	-	-	0	1	20	63	136	275	462	809	1,250
Other / Holding	-	-	-	23	65	123	141	126	120	114	123
Revenue Contribution											
Brazil	100%	100%	100%	97%	94%	93%	90%	87%	85%	82%	79%
Mexico	-	0%	0%	2%	4%	5%	8%	10%	12%	14%	16%
Colombia	-	-	0%	0%	0%	1%	1%	2%	2%	3%	4%
Other / Holding	-	-	-	1%	1%	2%	1%	1%	1%	0%	0%

Source: Company data, Morgan Stanley Research Estimates (e)

Mexico & Brazil Unit Economics

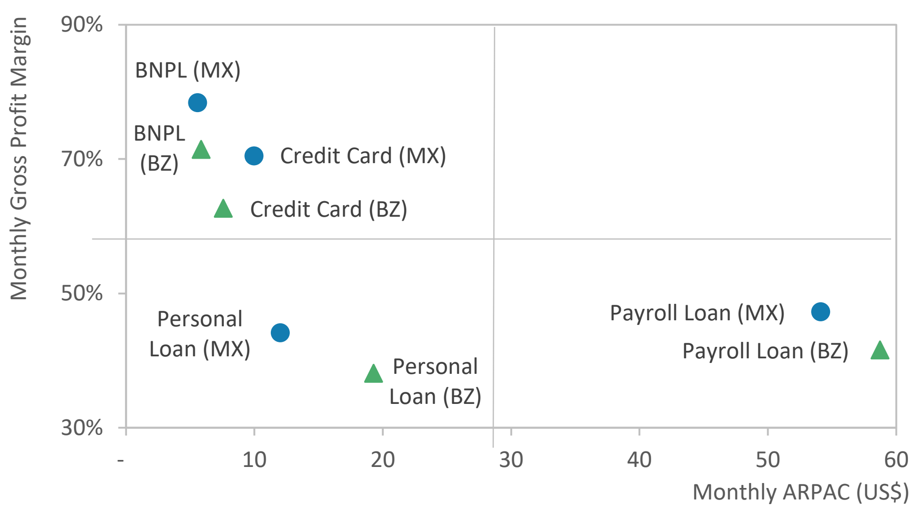
We build bottom-up unit economics models for Nubank in Mexico and Brazil. In this section, we introduce our proprietary models for six products Nubank now offers or will roll-out soon in both Mexico and Brazil, and we build to an estimated gross profit or credit contribution — equal to interest income and fees less funding expense and loan-loss provisions. Our analysis covers credit cards, personal loans, payroll loans, BNPL, and two non-credit products: insurance and investment brokerage.

We think our models provide an insightful and differentiated approach to understand the profitability gap between products and countries and how mix shift will impact Nubank's consolidated KPIs The assumptions in our models are tailored to Nubank's offering and clients base, aiming to replicate its unit economics in Mexico and Brazil. Our estimates are supported by data from the central bank, channel checks, and scraping of Nubank's disclosures. Importantly, the company does not disclose any unit economics by product or country; hence, we think many market participants will be seeing these numbers for the first time.

KEY TAKEAWAYS

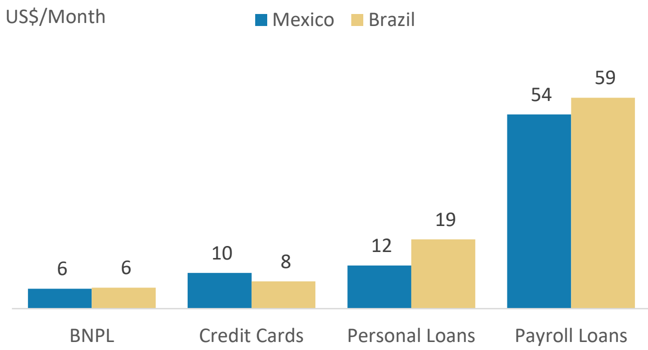
1) Lending is more profitable in Mexico on a product-by-product basis. Indeed, our models suggest that Nubank's credit offering should provide higher gross margin returns in Mexico — across credit cards, personal loans, payroll loans, and BNPL. Higher share of interest earning-balances in cards, along with lower funding cost and cost of risk on personal loans, are a few examples of the drivers of higher margins in Mexico. Also, our models demonstrate that BNPL and credit cards are the most profitable products, while payroll loans offer the lowest margin but largest ARPAC. More details by product are included below.

Exhibit 26: Nubank's credit offering offers greater gross margin in Mexico vis-à-vis Brazil.



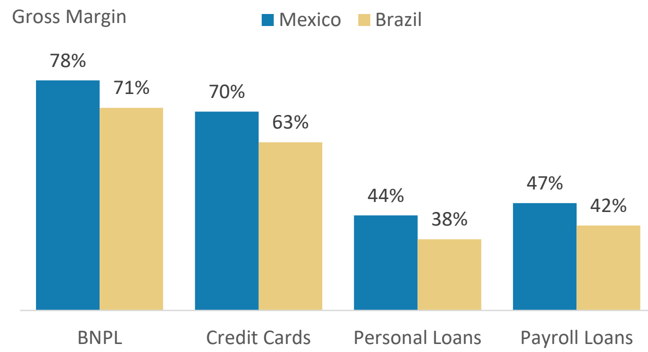
Source: Morgan Stanley Research Estimates.

Exhibit 27: Nubank — Monthly ARPAC by Product.



Source: Morgan Stanley Research Estimates.

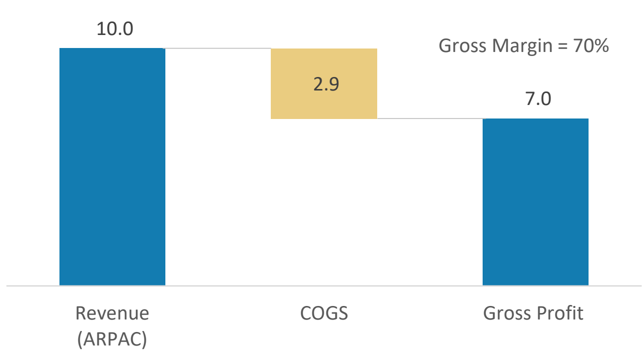
Exhibit 28: Nubank — Monthly Gross Profit Margin by Product.



Source: Morgan Stanley Research Estimates.

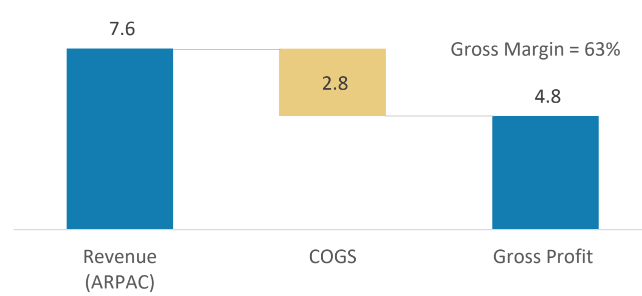
2) Nubank's credit cards are more profitable in Mexico compared to Brazil, because cards in Mexico have a higher share of interest-earning balances and collect interest income for more days. We estimate the gross profit margin for cards in Mexico is 70%, compared to 63% in Brazil. In addition, the monthly gross profit contribution per credit card holder is almost 50% higher, at US$7.0 in Mexico versus US$4.8 in Brazil. The main reason behind the gap is that Mexican issuers have a higher share of interest-earning balances and collect interest income for more days, which more than offsets higher financial expense due to greater working capital needs relative to Brazil. The Brazilian card market, where 75-80% of card balances are in interest-free installments, limits the profitability of the business. While interest-free installments do exist in Mexico (Meses sin Intereses), they are limited to specific short-term promotions, they are not a permanent feature of the card business, as in Brazil.

Exhibit 29: Nubank Mexico — Credit Card Unit Economics (US$/Month)



Source: Morgan Stanley Research Estimates.

Exhibit 30: Nubank Brazil — Credit Card Unit Economics (US$/Month)

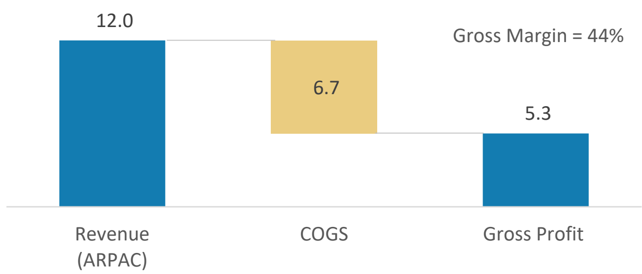


Source: Morgan Stanley Research Estimates.

3) We ran a sensitivity analysis of Nubank's credit card unit economics in Mexico to better understand how the share of interest-earning balances impacts the profitability of the product. Evidence from our analysis shows Nubank's ARPAC and gross profit would be 5% higher for every 5% increase in the share of revolving loans and 10% higher for every 5% increase in the share of interest with installment. For instance, ARPAC and gross profit would increase ~30% if we assume interest-earning balances represent ~75% of card expenditures, compared to 55% in our base case.

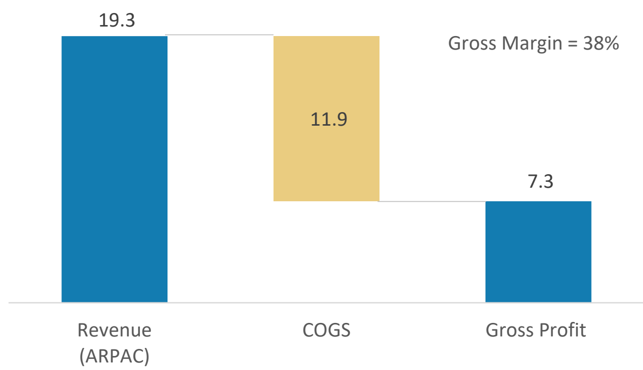
4) Our unit economics models show personal loans also have higher gross profit margins in Mexico versus Brazil. We think personal loans can earn Nubank a higher gross profit margin in Mexico, at 44% versus 38% in Brazil. The gap is due to lower cost of funding and better cost of risk in Mexico, which allow lenders like Nubank to offer lower prices and still earn healthier returns. That said, the absolute gross profit contribution per customer is almost 40% greater in Brazil at US$7.3 per month, versus US$5.3 per month in Mexico because rates are much higher in Brazil.

Exhibit 31: Nubank Mexico — Personal Loan Unit Economics (US$/Month)



Source: Morgan Stanley Research Estimates.

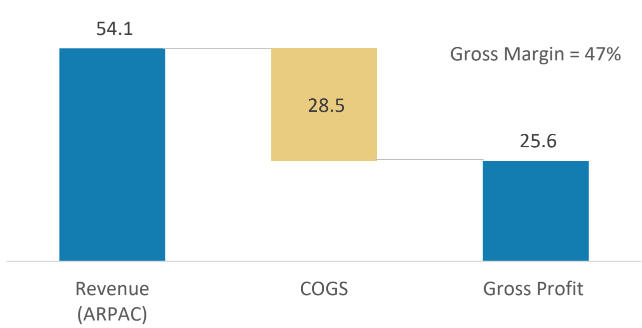
Exhibit 32: Nubank Brazil — Personal Loan Unit Economics (US$/Month)



Source: Morgan Stanley Research Estimates.

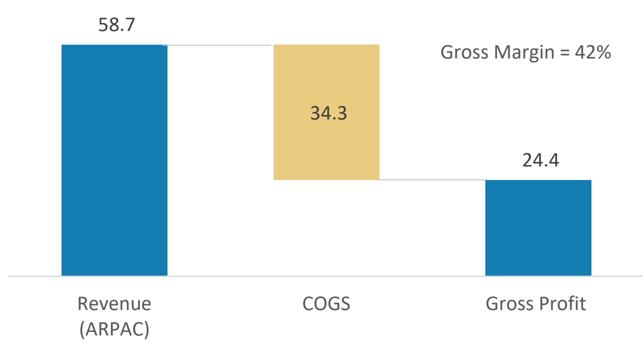
5) Payroll loans also have higher gross margin in Mexico, albeit the gap is small. Our unit economics build suggests the gross profit margin is not that different between Mexico and Brazil — 47% in Mexico versus 50% for INSS retirees in Brazil. However, the gross margin for public sector employees in Brazil is a tad lower at 42%, mostly due to higher prices and lower cost of funding in Mexico. We also estimate that Nubank can earn, on average, a gross profit contribution of US$26 per month in Mexico, 5% higher compared to US$24 per month for public sector employees in Brazil and 3x higher compared to US$8.5 per month for INSS retirees in Brazil.

Exhibit 33: Nubank Mexico — Payroll Loan Unit Economics (US$/Month)



Source: Morgan Stanley Research Estimates.

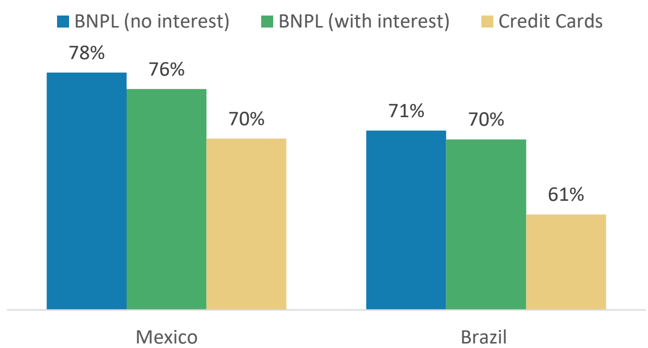
Exhibit 34: Nubank Brazil — Payroll Loan Unit Economics (US$/Month)



Source: Morgan Stanley Research Estimates. Note: Unit economics for public sector employees.

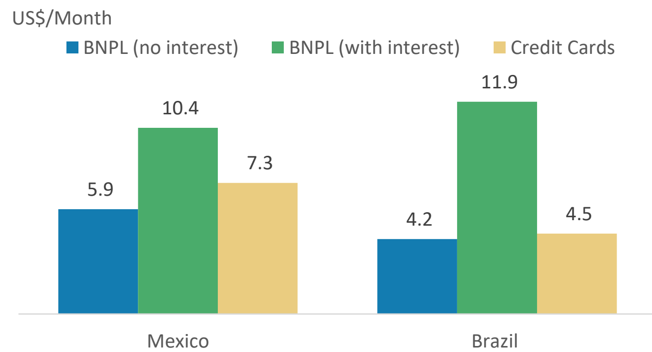
6) We think BNPL could offer superior unit economics compared to credit cards. Based on reasonable assumptions, corroborated with market participants, our BNPL models show higher gross profit margins compared to credit cards in both Mexico and Brazil. The gap in margins is mostly due to better asset quality in both countries. On the other hand, the monthly ARPAC is significantly greater for BNPL transactions with interest rates, but smaller for BNPL transaction with no interest. Lastly, our model assumes customers make four BNPL transactions per year, reaching an annual spend ~80% below that of credit cards. For a deep dive on BNPL in Latin America, please see our dispatch: The Rise of Latam BNPL.

Exhibit 35: BNPL with and without interest has higher gross margins compared to credit cards.



Source: Morgan Stanley Research Estimates.

Exhibit 36: But only BNPL with interest generates greater monthly ARPAC relative to credit cards.



Source: Morgan Stanley Research Estimates.

1) CREDIT CARD UNIT ECONOMICS

Our unit economics models show Nubank's credit cards are more profitable in Mexico compared to Brazil. We estimate a gross profit margin of 70% in Mexico, compared to 63% in Brazil. In addition, the monthly gross profit contribution per credit card holder is almost 50% higher, at US$7.0 in Mexico versus US$4.8 in Brazil. The main reason behind this gap is that Mexican issuers have a higher share of interest-earning balances and collect interest income for more days, which allows them to earn more revenue and more than offset higher financial expense due to greater working capital needs relative to Brazil.

Below, we provide a summary of our credit card model in Mexico and Brazil, and we discuss in detail the key assumptions in our models:

Exhibit 37: Nubank Unit Economics — Credit Cards.

USD / Month	Mexico	Brazil	MX vs BZ
Interest Income	6.0	4.7	29%
Loan	5.8	4.3	36%
Floating	0.2	0.4	-42%
Interest Expense	1.5	0.6	153%
Net Interest Income	**4.6**	**4.1**	**12%**
Loan-Losses	1.5	2.3	-34%
NII Risk Adjusted	**3.1**	**1.8**	**68%**
Interchange Fee	2.1	2.5	-17%
Late Fee	1.9	0.4	354%
Gross Profit Contribution	**7.0**	**4.8**	**48%**
Key Ratios (Annualized)			
Monthly ARAPC	10.0	7.6	32%
Excluding Float	9.7	7.2	36%
NIM	21%	15%	594 bp
Cost of Risk	6.9%	8.4%	-151 bp
Gross Margin	70%	63%	780 bp
Loan Details			
Card Expenditures / Year	2,000	2,000	
Tenure (Months)	12	12	
Average Loan Balance	259	322	
Interest Earning Portfolio*	126	56	
IEP as % of Avg Loan	49%	17%	
Annual Interest Income Yield	54%	84%	

Source: Morgan Stanley Research Estimates. Note: *Includes revolving loans and installments with interests.

Card Expenditures

We estimate Nubank's average cardholder expenditure in Mexico and Brazil is ~US$2,000 per year. Using company data we calculate the average TPV per active card holder over the last 12 months was close to US$2,000. In addition, we looked at industry data in both Mexico and Brazil and found that using the same level of card expenditure in both countries is a reasonable assumption.

We split card expenditures in five categories: revolving, installments with interest, installments without interest, paid in full, and nonperforming. Below, we discuss the main assumptions for each type and we explain the differences between Mexico and Brazil:

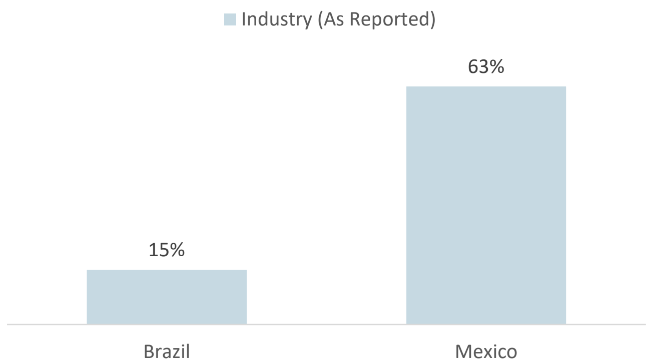
1) Revolving

The balance of credit card receivables that are revolving and generate interest income is significantly larger in Mexico than Brazil. At the industry level, data from the central banks shows that revolving loans represent 63% of outstanding credit card receivables in Mexico vs. 15% in Brazil. Now, these figures are not entirely correct. According to our channel checks, the central bank data includes revolving loans that do not generate interest income (i.e., clients that pay the minimum balance and installments without interest). At Banorte, for instance, it's true revolving balance represents 60% of outstanding loans, 15 percentage points below the central bank figure, suggesting that the industry number could be closer to 45% instead of 63%.

We assume Nubank's revolving balance is equal to 45% of card expenditures. A conservative assumption relative to Banorte's 60% and considering the potential composition of its client base in Mexico. Indeed, Nubank's client base could hypothetically revolve more than the industry average if it has a higher portion of unbancarized younger and lower-income individuals that tend to revolve more than bancarized older and higher income individuals. We provide a sensitivity analysis later in this section to better understand the impact that higher revolving balances could have on revenues and gross margins.

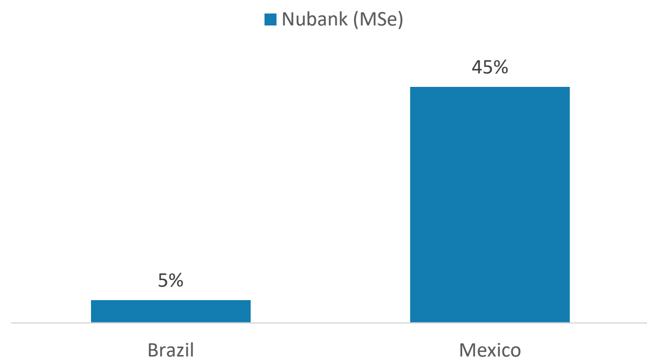
A similar issue takes place in Brazil, were the regulator includes non-performing revolving loans that do not generate interest income. According to Nubank, the portion of revolving loans that generates interest income in its Brazilian operations is less than 5% of card expenditures, and the difference between this and the reported figures by the company is mostly non-performing revolving loans.

Exhibit 38: Revolving Loans — Percentage of Outstanding Credit Card Receivables.



Source: Banxico, BCB.

Exhibit 39: Revolving Loans — Percentage of Annual Credit Card Expenditure.



Source: Morgan Stanley Research Estimates.

2) Installments with interest

In Brazil, the company discloses that installments with interest represent ~16% of the outstanding credit card receivables. However, it includes products such as Pix and Boleto financing that have been driving the growth of this category, which are not really credit card transactions, and are not available in Mexico. Hence, to make a fair comparison of the credit card unit economics we exclude these products. We assume Nubank's installments with interest represent 9% of the annual card expenditure, instead of the 16% figure reported by the company.

Below, we provide more color on products included in this category:

- In July 2021, Nubank enabled customers to pay boletos using interest-bearing credit card installments, a common feature offered by incumbent banks (more details here).
- In July 2021, Nubank allowed its customers to pay debit transactions in interest-bearing credit card installments (more details here).
- In March 2022, the company launched a feature allowing clients to pay in interest-bearing credit card installments for products or services that are not included in the merchant's interest-free installment offering (more details here).
- In July 2022, the company allowed customers to make PIX transfers using their credit card, paying in up to 12 installments (more details here).

In Mexico, we assume installments with interest represent a similar share of card expenditures to Brazil. Unfortunately, the company does not disclose this figure, and we did not find any disclosure at the central bank and other regulators. However, our channel checks with market participants suggest that it is not more than 10% at the industry level, so we run our Nubank unit economics model with this input.

3) Installments without interest (*"Parcelados" & "Meses sin Intereses"*)

In Brazil, credit card interest-free installment plans, also known as *parcelados*, are a staple of consumption. Interest-free installment plans are widely offered by merchants to consumers. Nearly 62 million Brazilians have at least one outstanding interest-free installment plan, according to a 2021 survey by CNDL/SPC. Consumers can access these interest-free plans through credit cards or boletos (payment slips), with credit cards being the top choice among 71% of consumers, according to the CNDL/SPC survey. Indeed, interest-free installment plans are by far the most popular modality of credit card payment in Brazil.

In Mexico, credit card interest-free installment plans, also known as *Meses sin Intereses*, are seasonal promotions banks offer to credit cards customers. Unlike *parcelados* in Brazil, *meses sin intereses* are not a permanent universal feature of credit cards. Consumer demand and merchant adoption unlocked the success of *meses sin intereses* in Mexico. In 1999, American Express introduced interest-free installment plans in Mexico in partnership with upscale retailer El Palacio de Hierro. Since then, hundreds of merchants enabled *meses sin intereses* in a similar fashion, working with one or multiple card issuers.

Our unit economic models assume that *parcelados* represent 40% of card expenditures in Brazil and *meses sin intereses* represent 20% of card expenditures in Mexico.

- **In Brazil,** Nubank's disclosure shows that installments without interest represent 37% of card balances. However, the contribution increases slightly over 40% when we adjust the balance of card receivables to exclude PIX and Boleto receivables, as mentioned before, and to include nonperforming loans. Hence, we think it's reasonable to assume *parcelados* represent 40% of card balances. As a reference, data from the Brazilian central bank is not useful, since the regulator only disclosed non-interest earning receivables (~80% of credit card loans), that include both paid in full and *parcelados*.
- **In Mexico,** meses sin intereses represent ~25% of credit card loans, interest-bearing installment promotions represent ~20%, and single installment transactions represent the remaining ~50%, according to the central bank. The regulator also shows that ~40% of card balances are paid in full and ~60% are not-paid in full. But the latter includes clients that do not earn interest income such as those that pay the minimum balance or have *meses sin intereses*. As mentioned above, our channel checks suggest that the mix is slightly different, so in our unit economics model we assume *meses sin intereses* at Nubank will represent 20% of card expenditures.

4) Non-Performing

We used industry and company data to proxy non-performing balances in our unit economics models:

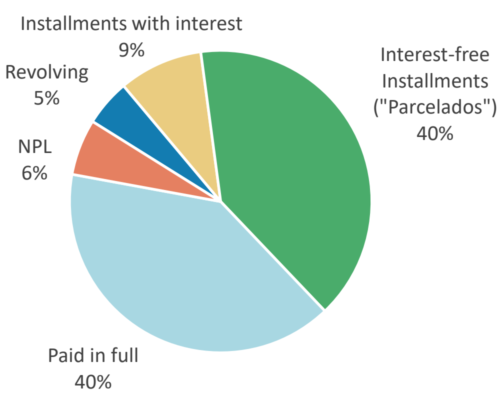
- **In Brazil,** the company provides disclosure on NPLs excluding the mix impact from personal loans — currently 90 day NPLs are close to 6% — and at the historical industry level credit card 90 day NPLs averaged 7% since 2011.
- **In Mexico,** the 90 day NPL ratio of credit cards at the industry level averaged 5.4% since 2001, but is currently close to 3%. However, we also note that best-in-class banks historically deliver NPLs 50 bp below the average.

To be conservative and taking into account the above, our unit economic models assume that nonperforming loans as a percentage of card expenditures represent 6% in Brazil and 5% in Mexico.

5) Paid in full

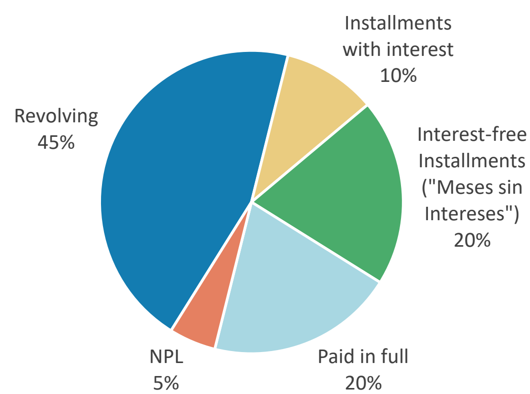
We assume all remaining balances are paid in full in both countries — 20% of card expenditures in Mexico and 40% in Brazil.

Exhibit 40: Nubank — Brazil Credit Card Expenditure Mix



Source: Morgan Stanley Research Estimates.

Exhibit 41: Nubank — Mexico Credit Card Expenditure Mix



Source: Morgan Stanley Research Estimates.

Interest Income

Our unit economics models for Nubank show that credit cards in Mexico generate more interest income relative to Brazil. We calculate an average cardholder earns a monthly interest income of US$5.8 in Mexico, 45% greater compared to US$4.0 per month in Brazil.

We calculate interest income for each type of interest-earning balance (revolving and installments with interest). Our assumptions include prices, term at origination, effective duration, and number of days revolving, most of which are disclosed by regulators and the company or we fined tuned with our channel checks. Non-interest-earning balances include: nonperforming loans, *parcelados, meses sin intereses,* or paid in full.

i) The assumptions for revolving balances in Mexico and Brazil are very different. Revolving balances represent only 5% of card expenditures in Brazil, compared to ~45% in Mexico. And revolving interest rates are close to 13% per month in Brazil, 3x higher compared to Mexico. In addition, Brazilian card issuers charge interest for only 15 days, on average, compared to 45 days in Mexico — 15 past due days plus the entire 30-day billing period — because Mexican issuers need to cover greater financial costs, given a shorter two-day settlement period. In contrast, Brazilian card issuers only charge interest for 15 past due days, on average, because the settlement takes place at the end of the 30-day billing period.

Exhibit 42: Revolving Balance — Key Assumptions.

	Brazil	Mexico
Share of Annual Card Expenditure	5%	45%
Monthly Interest Rate	13%	4%
Revolving (Days)	15	45
Collection (Days)	45	45
(-) Settlement (Days)	30	2
(=) Working Capital (Days)	15	43

Source: Morgan Stanley Research Estimates.

ii) Installments with interest are slightly more similar. Balances represent only 9-10% of card expenditures in both countries, while our channel checks suggest that the term at origination and effective duration should be the same. Instead, the main difference is on price. Nubank's average monthly interest rates in Brazil is 8%, or 3x higher compared to Mexico.

Exhibit 43: Installments with Interest — Key Assumptions.

	Brazil	Mexico
Share of Annual Card Expenditure	9%	10%
Monthly Interest Rate	9%	3%
Term at Origination (Months)	5	5
Effective Term (Months)	4	4
Collection (Days)	37	37
(-) Settlement (Days)	30	2
(=) Working Capital (Days)	7	35

Source: Morgan Stanley Research Estimates.

Interest Expense

Our credit card models show that interest expense is also higher in Mexico. We calculate an average cardholder costs a monthly interest expense of US$1.5 in Mexico, or 2.5x greater compared to US$0.6 per month in Brazil.

However, net interest margin is greater too. Indeed, our credit card model yields an average net interest margin of 21% in Mexico versus 15% in Brazil, as the interest income portion more than offsets the interest expense.

We calculate interest expense based on the following assumptions:

i) Funding costs. In Brazil, we assume Nubank's funding cost is equal to 80% of Selic, similar to the consolidated funding cost disclosed by the company. In Mexico, we assume funding costs equal to 50% of the benchmark interest rate (TIIE). A conservative assumption, in our view, as it is well above the incumbent's funding cost of 30-40%.

ii) Term at origination and effective duration. For installment balances, both with and without interest, we assume the average term at origination is five months and the effective duration is four months in both Mexico and Brazil. Our estimate is based on our conversations with other market participants in the issuing and acquiring space.

iii) Number of working capital days. As mentioned before, we use the number of working capital days (the period between the settlement to the merchant and the collection from the cardholder) to calculate the amount of interest expense. In Brazil, we assume revolving balances have 15 days of working capital and other balances seven days. In Mexico, revolving balances have 43 days of working capital and other balances 35 days. Other balances include: installments with interest, *parcelados, meses sin intereses,* and paid in full.

Exhibit 44: Interest Expense — Key Assumptions.

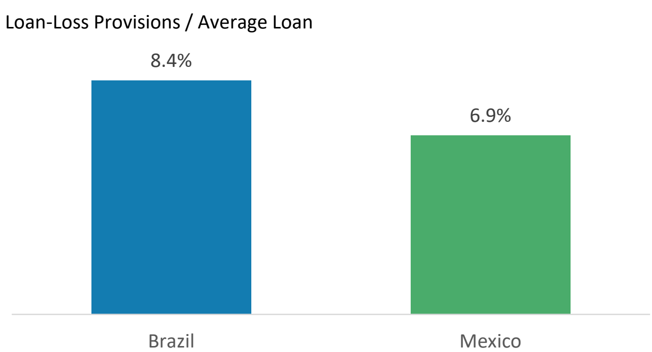
	Brazil	Mexico
Funding Cost (% Benchmark Rate)	80%	50%
Annual Funding Rate	9%	6%
Term at Origination (Months)	5	5
Effective Term (Months)	4	4
Revolving Working Capital (Days)	15	43
Other Working Capital (Days)	7	35
Net Interest Margin (NIM)	15%	21%

Source: Morgan Stanley Research Estimates.

Loan-Loss Provisions

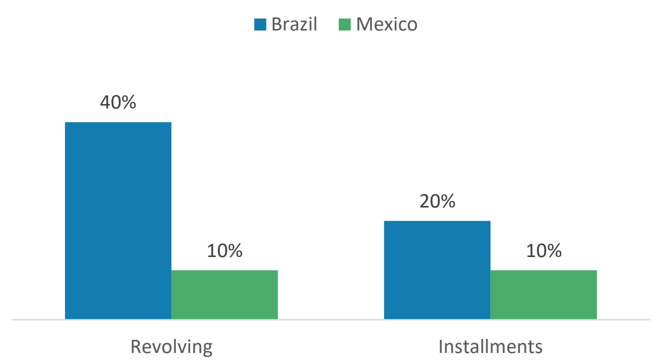
We estimate Nubank's credit card cost of risk is 7-8%. Given the lack of disclosure and data around loan-loss provisions and cost of risk by type of balance, we corroborated our estimates with market participants in both Mexico and Brazil and we assume Nubank is at the same level of best-in-class banks. Notably, the cost of risk is significantly higher for revolving loans in Brazil, at 40% vs. 10% in Mexico, but the impact in the overall amount of provisions is not as relevant because revolving loans represent only 5% of card expenditure in the country.

Exhibit 45: Total Cost of Risk (Annualized).



Source: Morgan Stanley Research Estimates.

Exhibit 46: Cost of Risk by Type of Balance.



Source: Morgan Stanley Research Estimates. Note: installments includes balances with and without interest.

Fee Income

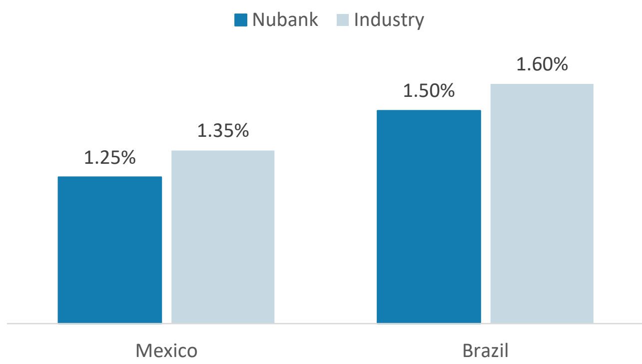
We calculate interchange and late fees contribute 35-40% of the monthly ARPAC, equal to US$3 in Brazil and US$4 in Mexico. The interchange fee contribution is slightly greater in Brazil, mostly because of prices. In contrast, the late fee contribution is greater in Mexico because it has a higher amount of late transactions (revolving and past due balances). Below, we provide more details for each type of fee:

- **We looked at the industry average interchange fees based on data from the central banks and we adjust it based on the mix of the company.** For instance, the average interchange fee in Brazil is ~1.60%. But this includes a higher share of premium cards relative to the mix at Nubank. And premium cards charge a higher interchange fee, on average, ~1.70%. Hence, we use a lower fee for Nubank in Brazil of 1.50%, in-line with the average of intermediate and basic cards. Using the same logic in Mexico, we use an interchange fee of 1.25%, compared to the average interchange for the industry of 1.35%.

- **For late fees, we considered regulatory limits in Brazil, our channel checks in Mexico, and company disclosure.** In Brazil, card issuers are allowed to charge two types of late fees: i) a late fee prorated by the number of days outstanding in the month and capped at 1% of the monthly outstanding balance; and ii) a delinquency fee capped at 2% of the monthly outstanding balance. We assume Nubank will charge the maximum fee. And we corroborate our assumptions looking at company disclosure of late fees over the last 12 months as a percentage of total TPV, equal to 0.21% and compared to 0.24% in our unit economics model. In Mexico, banks typically charge a one-time fixed amount between M$400 and 500, regardless of the outstanding balance. We assume Nubank will charge M$400, at the low end of the industry. That is equal to 1.1% of TPV, well above the 0.2% in Brazil because Mexico has more late balances.

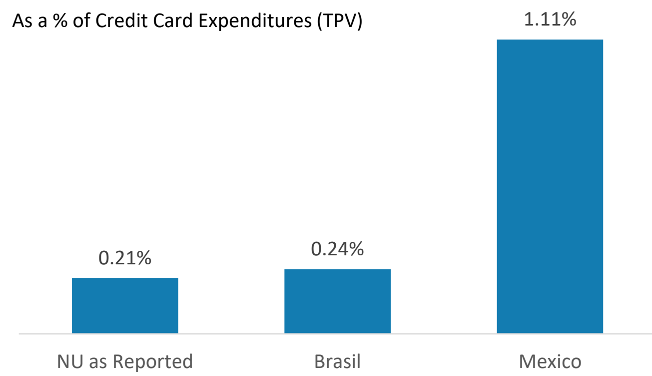
- **Our unit economics model does not consider rewards fees, recharge fees, among others.** In our view, most of these fees are too small and have little to no impact on the profitability of the product, as they are offset by the cost of the service. For instance, rewards fees are prices to cover the cost of points and benefits. In addition, we done expect Nubank will charge its long standing fee strategy of not charging other fees like annual memberships.

Exhibit 47: Credit Card Interchange Fees



Source: Banxico, Morgan Stanley Research Estimates.

Exhibit 48: Credit Card Late Fees



Source: BCB, Morgan Stanley Research Estimates.

Mexico Credit Card — Sensitivity Analysis to Card Expenditure Mix

We ran a sensitivity analysis of Nubank's credit card unit economics in Mexico to better understand how card expenditure mix impacts the ARPAC and gross profit. Specifically, we show different scenarios with higher and lower share of interest-earning balances — revolving and installments with interest. Our base case scenario assumes interest-earning balances represent ~55% of card expenditures — revolving 45% and interest with installments 10%. However, we think our base case could be conservative because Nubank's client base could have a higher than average interest-earning balance, particularly revolving loans.

ARPAC and gross profit increase 26-27% if we assume interest-earning balances represent ~75% of card expenditures, up from 55% before. Evidence from our analysis shows Nubank's ARPAC and gross profit would be 5% higher for every 5% increase in share of revolving loans and 10% higher for every 5% increase in share of interest with installment.

Exhibit 49: Mexico — Nubank's monthly ARPAC and Gross Profit sensitivity to interest-earning balances.

Monthly ARPAC	Installments with interest (% of Total)	Revolving (% of Total) 40%	45%	50%	55%
	5%	8.7	9.1	9.6	10.1
	10%	9.5	**10.0**	10.4	10.9
	15%	10.3	10.8	11.3	11.7
	20%	11.2	11.6	12.1	**12.6**

Change in ARPAC vs Base Case				
5%	-13%	-8%	-4%	1%
10%	-5%	**0%**	5%	9%
15%	4%	8%	13%	18%
20%	12%	17%	21%	**26%**

Gross Profit	Installments with interest (% of Total)	Revolving (% of Total) 40%	45%	50%	55%
	5%	6.1	6.5	6.9	7.2
	10%	6.6	**7.0**	7.4	7.8
	15%	7.2	7.6	8.0	8.3
	20%	7.8	8.1	8.5	**8.9**

Change in Gross Profit vs Base Case				
5%	-13%	-8%	-2%	3%
10%	-5%	**0%**	5%	11%
15%	2%	8%	13%	19%
20%	10%	16%	21%	**27%**

Source: Morgan Stanley Research Estimates.

2) PERSONAL LOAN UNIT ECONOMICS

Our unit economic models yield higher gross profit margins from personal loans in Mexico versus Brazil. Our unit economics build for personal loan suggests that Nubank can earn a gross profit margin of 44% in Mexico, versus 38% in Brazil. The gap is due to lower funding costs and better cost of risk in Mexico, which allow lenders like Nubank to offer lower prices and still earn healthier returns. That said, the gross profit per customer is almost 40% greater in Brazil at US$7.3 per month versus US$5.3 per month in Mexico, given the higher interest rates in Brazil.

We estimate an average net interest income per borrower of US$10 per month in Mexico and US$15 per month in Brazil. Note, personal loans earn greater NII in Brazil because prices are higher, despite lower funding costs in Mexico that partially offset the gap. To calculate net interest income we make the following assumptions in our unit economics models:

i) Loan Size. We assume the average loan size per borrower is ~US$600 in both Mexico and Brazil, based on Nubank's disclosure — outstanding personal loans and active personal loan borrowers — and conversations with multiple market participants in Mexico.

ii) Interest Rates. We use monthly interest rates of 3% in Mexico and 5% in Brazil. Our estimates are based on company data available at the Brazilian central bank and industry data from the Mexican regulator.

iii) Term & Duration. We assume the term at origination is 6 months and the effective duration is four months, and we corroborate these with our channel checks. Notably, both assumptions are significantly below than the industry average in Mexico and Brazil because Nubank's personal loans is tailored to a specific client/purchase and risk profile.

iv) Past-due Loans. We assume a past-due ratio of 12% in Mexico and 20% in Brazil, based on our channel checks in each country and looking at NPL ratios from the central banks. Importantly, our models assume that past due loans do not earn interest income.

v) Funding Costs. Similar to our credit card model, we assume Nubank's funding cost in Brazil is equal to 80% of Selic and in Mexico equal to 50% of TIIE. We also assume loans are fully funded — i.e., loan to deposit ratio of 100%. Our model implies that the average interest expense per month in Mexico is 49% lower compared to Brazil.

vi) Floating Income. We calculate floating income based on excess capital and the average Selic rate, considering the risk weighted assets (RWA) and capital requirements (Basel ratio). The RWA requirement for personal loans is 75% in Mexico and 100% in Brazil, and we use a 13% Basel ratio in both countries.

We also calculate an average loan-loss provision of US$5 per month in Mexico and US$8 per month in Brazil. To calculate loan-losses, we multiply the nonperforming loans over 90 days by the average loan balance in each month. Our model implies an annual cost of risk of 13% in Mexico and 22% in Brazil. Importantly, the higher cost of risk explains why prices are much higher in Brazil.

Exhibit 50: Nubank Unit Economics — Personal Loans.

USD / Month	Mexico	Brazil	MX vs BZ
Interest Income	12.0	19.3	-38%
Loan	11.5	18.5	-38%
Floating	0.5	0.8	-28%
Interest Expense	2.1	4.2	-49%
Net Interest Income	**9.9**	**15.1**	**-34%**
Loan-Losses	4.6	7.7	-41%
NII Risk Adjusted	**5.3**	**7.3**	**-28%**
Key Ratios (Annualized)			
Monthly ARAPC	12.0	19.3	-38%
Excluding Float	11.5	18.5	-38%
NIM	29%	44%	-1459 bp
Cost of Risk	13%	22%	-890 bp
Gross Margin	44%	38%	601 bp
Loan Details			
Loan Size	600	600	
Tenure (Months)	4	4	
Monthly Interest Rate	2.9%	5.0%	

Source: Morgan Stanley Research Estimates.

3) PAYROLL LOAN UNIT ECONOMICS

To better understand how profitable payroll loans can be, we recently took a deep dive into the unit economics of the product: "The Rise of Digital Payroll Loans". In that report, we calculate the unit economics for two types of payroll loans (public sector employees and INSS retirees) and for two distribution models (direct-to-consumer and correspondents in Brazil). However, in this section we tailor the unit economics of payroll loans in Brazil to Nubank, focusing only on direct-to-consumer channels for public sector and INSS retirees.

Unfortunately, there is no disclosure by type of payroll loan in Mexico. According to our channel checks, incumbent banks in Mexico do not use correspondents, but non-bank lenders do because they focus on lower income borrowers that are harder to reach. We assume Nubank will continue to operate exclusively through digital direct-to-consumer channels. In addition, our channel checks mentioned that the average loan ticket is slightly higher for private sector employees, compared to public sector employees and retirees. And like Brazil, public sector payrolls are auctioned by the government typically to incumbent banks that offer lower interest rates to the employees. Lastly, only banks that own a pension distribution business can disburse and collect payroll loans to employees.

An important difference between payroll loans in Mexico and Brazil is how banks collect the loans and the implied default risk. In Brazil, banks collect principal and interest before the salary or pension is paid to the borrower, hence reducing the default risk. And even in the case of death or termination of employment there are additional guarantees that materially reduce the cost of risk. In contrast, Mexican lenders are not allowed to collect before the salary or pension is paid. However, the standard it to collect principal and interest payments almost instantly after paying the salary or pension, reducing the cost of risk because the borrower does not have time to withdraw. The problem is that portability and a lack of interoperability increase the cost of risk, as customers can transfer their payroll and the lender no longer has the ability to collect instantly, making the risk profile of the loan look more like an unsecured personal loan in Mexico.

Our payroll loan unit economic models also yield similar gross profit margins in Mexico versus Brazil. Our unit economics build for payroll loan suggests that Nubank can earn a gross profit margin of 47% in Mexico versus 50% for INSS retirees and 42% for public sector employees in Brazil. The gap versus public sector employees in Brazil is mostly due to higher prices and lower funding costs in Mexico. We also estimate that Nubank can earn, on average, a gross profit of US$26 per month in Mexico, 34% higher compared public sector employees and 3x higher compared to INSS retirees in Brazil.

We estimate an average net interest income per borrower of US$41 per month in Mexico, compared to US$31 per month for public sector employees in Brazil and only US$10 per month for INSS retirees. Note, payroll loans earn greater NII in Mexico because prices are higher and funding costs are lower. Given the early stage of the product and lack of company disclosure, we mostly use industry data and color from our channel checks. To calculate net interest income, we make the following assumptions in our unit economics models:

i) Loan Size. We assume the average loan size per borrower is ~US$4,000 in both Mexico and Brazil public sector employees. For INSS retirees, the average ticket is much lower at ~US$1,000. As a reference, our assumptions are similar to the average ticket at origination in both countries, based on industry data in Mexico and according to conversations with industry participants in Brazil, as such data is not publicly available.

ii) Interest Rates. We use monthly interest rates of 2.0% in Mexico, 1.6% in Brazil public sector employees, and 1.9% for INSS retirees. For Mexico, we used the industry average. For Brazil public sector employees, we use Nubank's average rate over the last six months, available at the central bank. And for INSS retirees we also use the industry average, as the company has yet to launch this product.

iii) Term & Duration. We use the industry average term at origination and effective duration for all models. For the term at origination: 53 months in Mexico, 91 months for public payroll in Brazil, and 77 months for INSS retirees. For the effective duration: 45 months in Mexico, 38 months for public payroll in Brazil, and 30 months for INSS retirees.

iv) Past-due Loans. We assume a past-due ratio of 7% in Mexico, well above the 2% in Brazil. We used data from the local regulators and we confirmed with our channel checks in each country. As a reference, NPL levels are higher in Mexico because payroll portability and lack of interoperability make the collection process more complex, making the product operate like personal loans. Importantly, our models assume that past due loans do not earn interest income.

v) Funding Costs. Similar to our previous models, we assume Nubank's funding cost in Brazil is equal to 80% of Selic and in Mexico equal to 50% of TIIE. We also assume loans are fully funded — i.e., loan to deposit ratio of 100%.

vi) Floating Income. We calculate floating income based on excess capital and the average Selic rate, considering the risk-weighted assets (RWA) and capital requirements (Basel ratio). The RWA requirement for payroll loans is 75% in Mexico, 75% for public sector employees in Brazil, and 50% for INSS retirees. Lastly, we use a 13% Basel ratio in our three models.

We also calculate an average loan-loss provision of US$16 per month in Mexico, compared to US$6.5 per month for public sector employees in Brazil and only US$1.6 per month for INSS retirees. To calculate loan-losses, we multiply the nonperforming loans over 90 days by the average loan balance in each month. Our model implies an annual cost of risk of 7% in Mexico and only 2% in Brazil. Importantly, the higher cost of risk explains why prices are much higher in Mexico.

Exhibit 51: Nubank Unit Economics — Payroll Loans.

		Brazil		MX vs BZ	
USD / Month	**Mexico**	**Public**	**INSS**	**Public**	**Public**
Interest Income	54.1	58.7	17.1	-8%	216%
Loan	50.4	54.9	16.5	-8%	206%
Floating	3.7	3.8	0.6	-4%	474%
Interest Expense	12.7	27.8	7.0	-54%	81%
Net Interest Income	**41.4**	**30.9**	**10.1**	**34%**	**310%**
Loan-Losses	15.8	6.5	1.6	143%	907%
NII Risk Adjusted	**25.6**	**24.4**	**8.5**	**5%**	**200%**
Key Ratios (Annualized)					
Monthly ARAPC	54.1	58.7	17.1	-8%	216%
Excluding Float	50.4	54.9	16.5	-8%	206%
NIM	19%	10%	14%	809 bp	500 bp
Cost of Risk	7%	2%	2%	489 bp	499 bp
Gross Margin	47%	42%	50%	569 bp	-255 bp
Loan Details					
Loan Size	4,000	4,000	1,000		
Tenure (Months)	45	45	30		
Monthly Interest Rate	2.0%	1.6%	1.9%		

Source: Morgan Stanley Research Estimates.

4) BNPL UNIT ECONOMICS

We recently took a deep dive into how BNPL works and the opportunities and risks it offers in Brazil and Mexico. We also provide an overview of Nubank's BNPL offering, a best-in-class — simple and secure — checkout experience that solves the common pain points of online shopping in Latin America. The solution provides consumers with higher credit limits, longer installments vs. credit cards, and attractive discounts and cashback. For more details, please see The Rise of Latam BNPL and The Upsurge of Latam BNPL.

Nubank's BNPL transactions take place through a closed-loop payment arrangement. Each transaction simply matches a customer's Nubank checking account ID with the merchant's Nubank business account ID, with Nubank serving as the "issuer" and NuPay as the "acquirer." In addition, NuPay integrates with merchant's checkout using gateways or its own API. And this disintermediates the card networks (like Visa and Mastercard) and the merchant acquirers (like Cielo and Stone), allowing Nubank to capture 100% of the economics. The solution is only available for e-commerce transactions, but eventually it will be available in-store purchases also.

Nubank's BNPL offering is expected to provide superior economics compared to its credit card business. With BNPL, Nubank receives a single fee from the merchant equivalent to the sum of interchange, acquiring, and prepayment fees — all transactions settle in T+1, giving no room to third party players to prepay the installments. In addition, BNPL transactions can earn interest on 100% of the loan disbursement and can leverage real-time data to make an enhanced underwriting decision with 100% marginal utilization of cost of risk. Indeed, BNPL transactions could have lower credit risk vis-à-vis credit cards, given the nature of underwriting a one-time credit line, knowing the exact type of purchase, and the risk associated to the specific product and industry.

There are three purchase options through NuPay: i) immediate debit (using the customer's NuAccount), ii) interest-free BNPL, and iii) interest-bearing BNPL. In this section, we focus only on the unit economics of BNPL transactions, leveraging our conversations with market participants due to limited company disclosure for the product. Note, the company has yet to announce the launch of BNPL in Mexico, but it thinks the opportunity for BNPL may be even bigger than in Brazil.

Our unit economics models support our view that BNPL offers higher gross profit margins compared to credit cards in both countries. We estimate the gross margin of interest-free BNPL is ~78% in Mexico and ~71% in Brazil, compared to 70% and 63% for credit cards, respectively. The gap in margins is mostly due to better asset quality assumption in both countries. To calculate gross profits we make the following assumptions in our unit economics models:

i) BNPL Expenditures. In both Mexico and Brazil, we assume the average loan size per borrower is ~US$50 for BNPL with no interest and ~US$100 for BNPL with interest. We also assume customers make four transactions per year, reaching an annual spend US$200 and US$400, or ~80% below that of credit cards. Notably, over 95% of clients who have used NuPay continue to use the product in the following months when it is available.

ii) Interest Rates. For BNPL installments with interest, we use monthly interest rates of 6.0% in Mexico and 8.5% in Brazil. Our estimates similar to those charged for credit card installments with interest.

iii) Fees. As mentioned before, Nubank will collect a single fee for interchange, acquiring, and prepayment. We used market prices in both Mexico and Brazil: interchange fee of 1.25% in Mexico and 1.50% in Brazil; acquiring fee of 2.10% in Mexico and 2.30% in Brazil; and prepayment of 2% in both Mexico and Brazil. In addition, we assume the company collects a fixed late fee of US$10 per transaction with no interest and that only 10% of expenditures are late.

iv) Term & Duration. In both countries, we assume the term at origination is 6 months for BNPL with no interest and 12 months for BNPL with interest. And we use an effective duration of four months and eight months, respectively.

v) Past-due Loans. Currently, BNPL delinquency is in line with that of credit cards; however, Nubank sees NPLs improving as it gains more experience with product and eventually incorporates SKU data into the underwriting process. As such, we assume the 90-day non-performing ratio of all BNPL transactions is 4%. Our model implies an annual cost of risk of 4 to 5% in Mexico and Brazil.

vi) Funding Costs. Similar to previous models, we assume Nubank's funding cost in Brazil is equal to 80% of Selic and in Mexico equal to 50% of TIIE. We also assume loans are fully funded — i.e., loan to deposit ratio of 100%. Our model implies that the average interest expense per month in Mexico is ~50% lower compared to Brazil.

vii) Floating Income. We calculate floating income based on excess capital and the average Selic rate, considering the risk weighted assets (RWA) and capital requirements (Basel ratio). We assume the RWA requirement for BNPL is similar to personal loans — 75% in Mexico and 100% in Brazil — and we use a 13% Basel ratio in both countries.

Exhibit 52: Nubank Unit Economics — BNPL Interest Free.

USD / Month	Mexico	Brazil	MX vs BZ
Interest Income	0.2	0.3	-28%
Loan	-	-	NM
Floating	0.2	0.3	-28%
Interest Expense	0.7	1.2	-40%
Net Interest Income	**(0.5)**	**(0.9)**	**-43%**
Loan-Losses	0.5	0.5	0%
NII Risk Adjusted	**(1.0)**	**(1.4)**	**-28%**
Net MDR	1.7	1.9	-12%
Prepayment Fee	3.5	3.5	0%
Late Fee	0.2	0.2	0%
Gross Profit Contribution	**4.4**	**4.2**	**4%**
Key Ratios (Annualized)			
Monthly ARAPC	5.6	5.9	-5%
Excluding Float	5.4	5.6	-4%
Revenue Yield	11%	12%	-59 bp
Cost of Risk	5%	5%	0 bp
Gross Margin	78%	71%	695 bp
Loan Details			
Loan Size	200	200	
Tenure (Months)	4	4	
Monthly Interest Rate	-	-	

Source: Morgan Stanley Research Estimates.

Exhibit 53: Nubank Unit Economics — BNPL With Interest.

USD / Month	Mexico	Brazil	MX vs BZ
Interest Income	2.4	3.5	-31%
Loan	2.2	3.2	-31%
Floating	0.2	0.3	-28%
Interest Expense	1.4	2.5	-42%
Net Interest Income	**1.0**	**1.0**	**-5%**
Loan-Losses	1.0	1.0	-3%
NII Risk Adjusted	**(0.1)**	**(0.0)**	**38%**
Net MDR	0.8	1.0	-12%
Prepayment Fee	7.2	7.5	-4%
Late Fee	-	-	NM
Gross Profit Contribution	**8.0**	**8.4**	**-5%**
Key Ratios (Annualized)			
Monthly ARAPC	10.4	11.9	-12%
Excluding Float	10.2	11.6	-12%
Revenue Yield	21%	24%	-292 bp
Cost of Risk	4%	4%	1 bp
Gross Margin	76%	70%	602 bp
Loan Details			
Loan Size	400	400	
Tenure (Months)	8	8	
Monthly Interest Rate	6.0%	8.5%	

Source: Morgan Stanley Research Estimates.

5) INSURANCE & INVESTMENTS UNIT ECONOMICS

Our insurance models yield a monthly ARPAC of US$0.6 in both Mexico and Brazil, well below Nubank's credit products. To calculate the ARPAC we make the following assumptions:

i) First, we looked at prices and found that the basic life insurance plan that Nubank offers is ~US$1.4/month. We also found that industry average mobile phone insurance is close to ~US$2/month, home insurance ~US$6-8/month, and basic auto insurance plans can be as low as US$8-10/month.

ii) To simplify, we assume the average monthly premium is US$2 in both Mexico and Brazil, given limited company disclosure and considering that life insurance represents over 90% of Nubank's 1.1 million policies and that higher ticket auto insurance plans will only launch in coming months.

iii) Nubank collects a brokerage fee from the insurance underwriter of the policy, around 30% according to our channel checks.

Our investment brokerage models suggest the monthly ARPAC is ~US$1.4 in both Mexico and Brazil, also at the low end of Nubank's product offering. To calculate the ARPAC we make the following assumptions:

i) We calculate the average AUC per customer was ~US$1,550 in 2022, using the following numbers disclosed by the company: total AUC of ~US$8.0 billion and, on average, 5.2 million NuInvest customers in 2022.

ii) We assume an average brokerage fee of 1.1%. We corroborated assumptions with data from local regulators and market participants.

Exhibit 54: Nubank Unit Economics — Insurance & Investments.

USD	Mexico	Brazil	MX vs BZ
Insurance			
Monthly Premiums	2.0	2.0	0%
Brokerage Fee	30%	30%	
Monthly ARPAC	0.6	0.6	0%
Investments			
AUC per Client	1,600	1,600	0%
Commission	1.1%	1.1%	
Monthly ARPAC	1.4	1.4	0%

Source: Morgan Stanley Research Estimates.

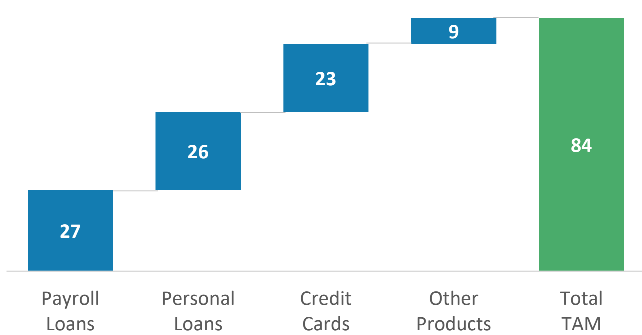
Defining Nubank's Brazil & Mexico TAM

Leveraging our proprietary unit economics models, we estimate the TAM for each of Nubank's products in Brazil and Mexico. To do so, we apply publicly available industry data along with information compiled via channel checks to our unit economics models. We think this bottom-up approach provides unique insight into the vast market opportunity available to Nubank.

Our analysis suggests that the current, total revenue TAM is US$84 billion in Brazil and US$17 billion in Mexico. As such, we get to a combined revenue TAM in Brazil and Mexico of over US$100 billion, distributed across several banking products including: cards, payroll and personal loans, Buy-Now Pay-Later (BNPL), retail investments, and insurance. In Brazil, the payroll loans market represents the biggest revenue opportunity, while credit cards are the top contributor to the revenue pool in Mexico.

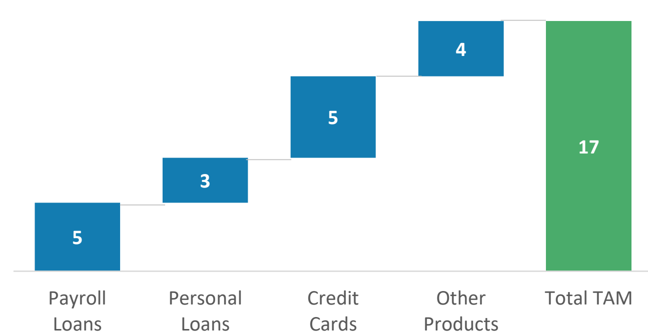
NU has 8% share, growing to 16% in 2028. Based on the our TAM calculations and our new P&L estimates, we calculate that Nubank currently has 8% market share of the total TAM, and can expand to 16% by 2028.

Exhibit 55: We estimate Brazil revenue TAM is currently US$84 Billion...



Source: Morgan Stanley Research Estimates (e). Note: Others include insurance, investments and BNPL.

Exhibit 56: ... four times greater than Mexico's US$17 Billion revenue TAM.



Source: Morgan Stanley Research Estimates (e). Note: Others include insurance, investments and BNPL.

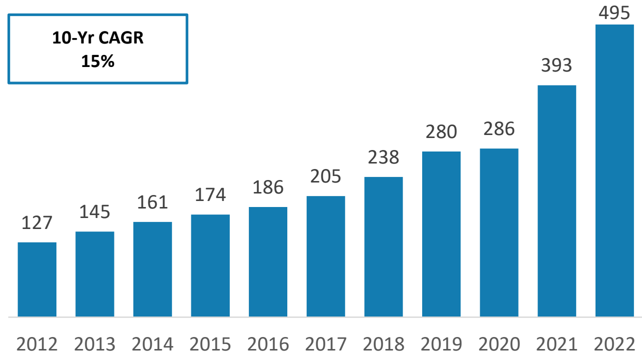
Credit Cards

We calculate that the current, total cards TAM between Brazil and Mexico is US$28 billion in revenues and US$22 billion in gross profit. This includes US$23 billion in revenues from credit, debit, and pre-paid cards in Brazil and US$5 billion in Mexico; US$18 billion in gross profit in Brazil and US$4 billion in Mexico.

Cards in Brazil. Credit cards are one of the most prevalent lending instrument for Brazilian consumers. This, of course, is no surprise given the attractive interest-free installments embedded in the product (i.e., *parcelados*). Our 2022 AlphaWise Consumer Survey revealed that about 80% of adults had bank-issued credit cards, which is likely higher today as the number of issued credit cards grew 9% last year to 431 million. Today, the total credit card balance outstanding in Brazil amounts to R$515 billion, having grown at a 15% CAGR over the past 10 years. Of course, debit and pre-paid cards are also significant products given the breadth of banking penetration in Brazil.

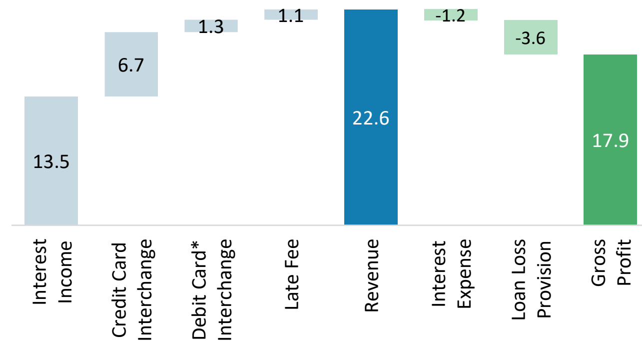
Brazil TAM. Using the latest industry data, and our unit economics models from the prior section, we were able to back into the various revenue streams from cards in Brazil. The main inputs include total card volumes, industry interest rates, NPLs, and interchange fees as published by the BCB. Note that we used the newly implemented interchange caps for debit and pre-paid cards for these full-year gross profit TAM calculations. Publicly available information was also supplemented by channel checks with industry participants. All told, we got to a total revenues pool of R$113 billion per year, or ~US$23 billion, and a gross profit pool of R$89 billion per year, or ~US$18 billion.

Exhibit 57: Brazil — Outstanding Credit Card Receivables (R$ Bn).



Source: BCB, Morgan Stanley Research.

Exhibit 58: Brazil — Credit and Debit Card TAM (US$ Bn).

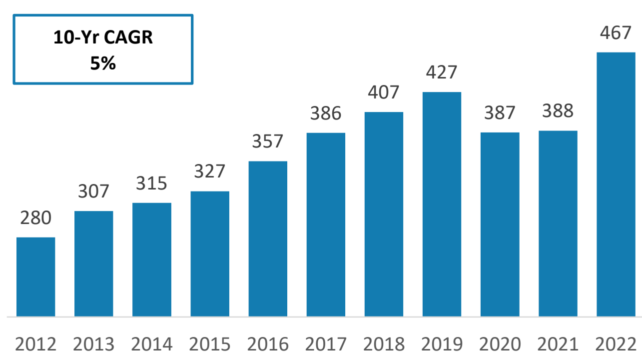


Source: Morgan Stanley Research Estimates. Note: *Includes pre-paid cards.

Cards in Mexico. The credit card industry in Mexico is very small. According to INEGI, only 11% of the adult population had access to a bank issued credit card — ~21 million active cards or 0.2 per inhabitant, compared to 2 or more cards per inhabitant in the US. This is in large part due to high informality rates and a lack of incentives for banks to offer products to higher risk individuals. While debit cards are more widely available, Mexico still falls short of Brazil, as bank account penetration is quite low — less than half of the adult population in Mexico had a bank account in 2021. For more details, please see: The Mexican Card Industry In Charts.

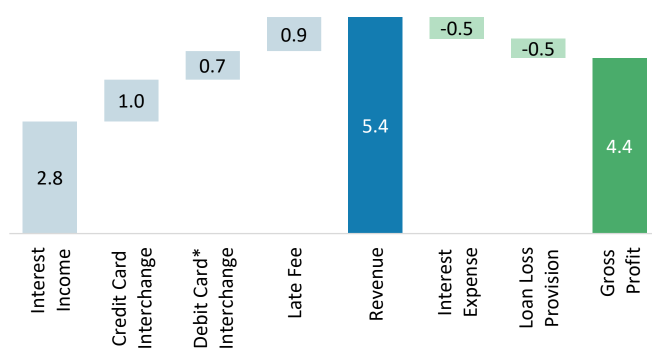
Mexico TAM. Using the latest industry data, and our unit economics models from the prior section, we were able to back into the various revenue streams from cards in Mexico. The main inputs include total card volumes, industry interest rates, NPLs, and interchange fees as published by Banxico. Publicly available information was also supplemented by channel checks with industry participants. All told, we got to a total revenues pool of M$97 billion per year, or ~US$5 billion, and a gross profit pool of M$79 billion per year, or ~US$4 billion.

Exhibit 59: Mexico — Outstanding Card Receivables (M$ Bn).



Source: Banxico, Morgan Stanley Research.

Exhibit 60: Mexico — Credit and Debit Card TAM (US$ Bn).



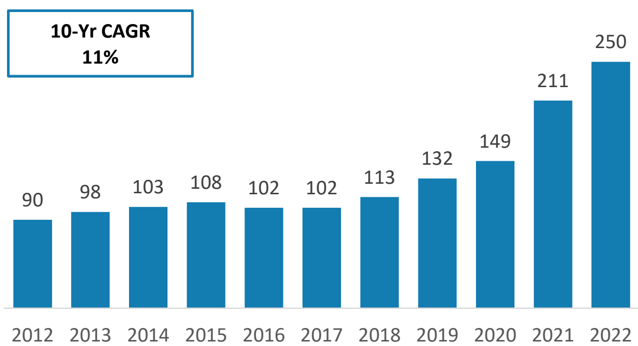
Source: Morgan Stanley Research Estimates. Note: *Includes pre-paid cards.

Personal Loans

We calculate that the personal loans TAM between Brazil and Mexico is US$29 billion in revenues and US$21 billion in gross profit. This includes US$26 billion in revenues from personal loans in Brazil and US$3 billion in Mexico; US$19 billion in gross profit in Brazil and US$2 billion in Mexico.

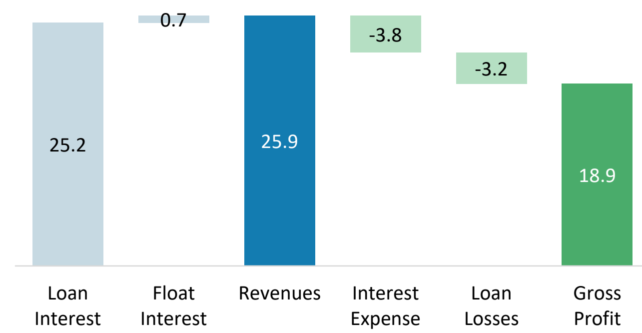
Brazil TAM. Today, there are approximately R$250 billion of outstanding personal loan balances in Brazil, having grown at a CAGR of 11% over the last 10 years and now accounting for close to 15% of all consumer loans. To get to the product's TAM, we leveraged our proprietary unit economics model and some of the latest industry data: total outstanding loans, NPLs, and interest rates. Importantly, since the average payroll loan in Brazil has an effective term of over two years (much longer than that of Nubank), we used average interest rates from the past 24 months and the average of our economist's projection for Selic over the coming 24 months. All told, we got to a total revenue pool of R$130 billion per year, or ~US$26 billion, and a gross profit pool of R$94 billion per year, or ~US$19 billion. Note that while personal loans are far less widespread than credit cards, the total revenue pool is about 20% larger given the big ticket sizes and high interest rates charged in Brazil.

Exhibit 61: Brazil — Outstanding Personal Loans (R$ Bn).



Source: BCB, Morgan Stanley Research.

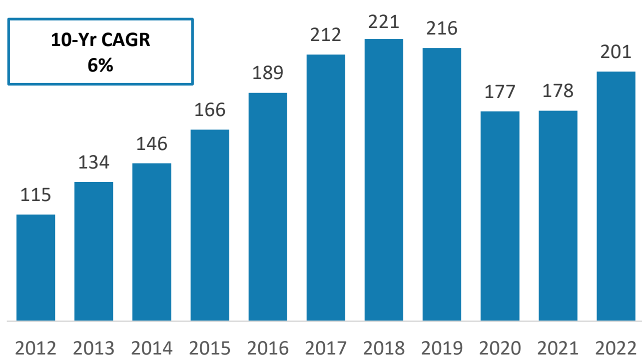
Exhibit 62: Brazil — Personal Loans TAM (US$ Bn).



Source: Morgan Stanley Research Estimates.

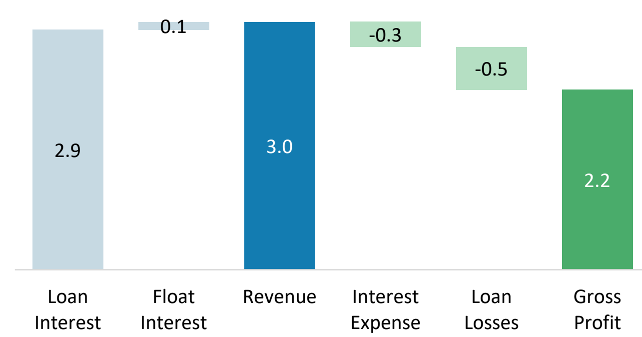
Mexico TAM. Today, there are approximately M$215 billion of outstanding personal loan balances in Mexico, having grown at a CAGR of 6% over the last 10 years and now accounting for over 15% of all consumer loans. To get to the product's TAM, we leveraged our proprietary unit economics model and some of the latest industry data, like total outstanding loans, NPLs, and interest rates. Importantly, since the average personal loan in Mexico has an effective term of over two years, we used average interest rates from the past 24 months and the average of our economist's projection for TIIE over the coming 24 months. All told, we got to a total revenues pool of M$54 billion per year, or ~US$3 billion, and a gross profit pool of M$39 billion per year, or ~US$2 billion.

Exhibit 63: Mexico — Outstanding Personal Loans (M$ Bn).



Source: Banxico, Morgan Stanley Research.

Exhibit 64: Mexico — Personal Loans TAM (US$ Bn).



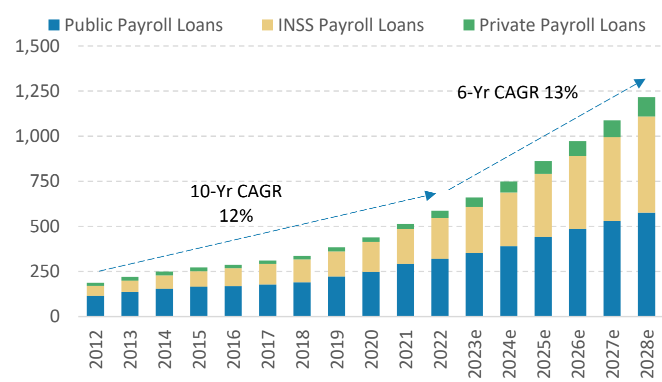
Source: Morgan Stanley Research Estimates.

Payroll Loans

We calculate that the payroll loans TAM between Brazil and Mexico is US$32 billion in revenues and US$14 billion in gross profit. This includes US$27 billion in revenues from payroll loans in Brazil and US$5 billion in Mexico; US$11 billion in gross profit in Brazil and US$3 billion in Mexico.

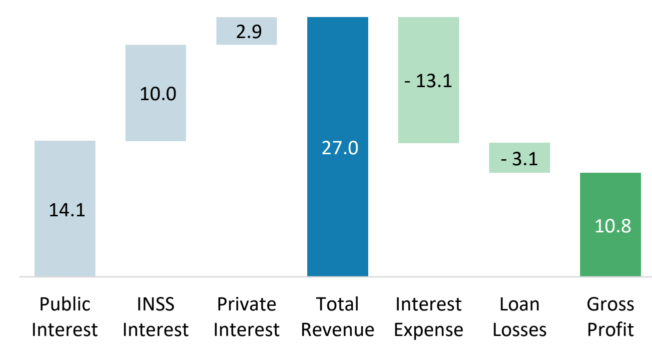
Brazil TAM. With over R$600 billion in payroll loans outstanding, the product accounts for 33% of total consumer loans, making it the largest source of consumer credit in Brazil. Payroll loans can be grouped in to three main categories — public, INSS retirees, and private — each of which has its own unique features and unit economics. Hence, we have built our own proprietary, bottom-up industry model that estimates the market opportunity by category: Morgan Stanley's Payroll Loan Industry Model. Adding interest on float to our model and an 80% Selic cost of funding (to make it comparable with our other unit economics builds), we got to a total revenues pool of R$135 billion per year, or ~US$27 billion, and a gross profit pool of R$54 billion per year, or ~US$11 billion. And we expect the gross profit pool to double over the next five years as funding costs normalize and total loans continue to grow. Note: We do not consider the cost of banking correspondents due to lack of industry data.

Exhibit 65: Brazil — Outstanding Payroll Loans (R$ Bn).



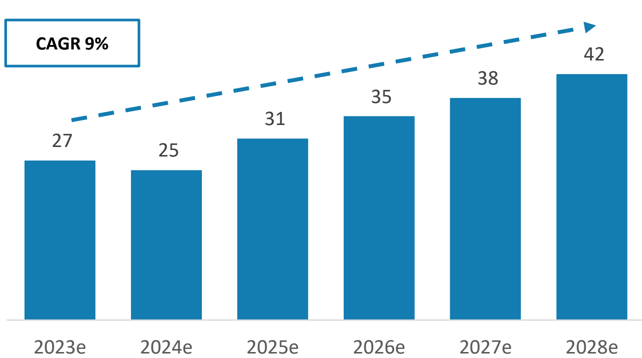
Source: BCB, Morgan Stanley Research Estimates (e).

Exhibit 66: Brazil — Payroll Loans TAM (US$ Bn).



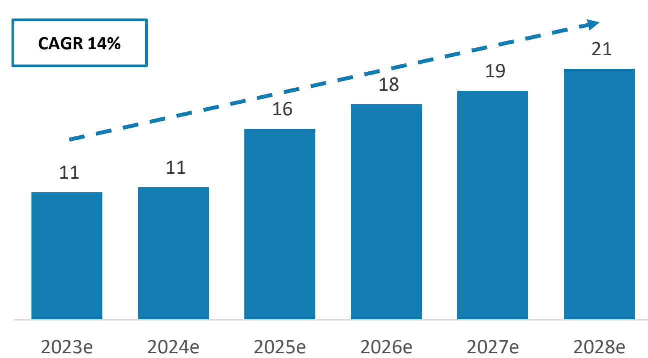
Source: Morgan Stanley Research Estimates.

Exhibit 67: Brazil — Payroll Loans Revenue TAM (US$ Bn).



Source: Morgan Stanley Estimates (e).

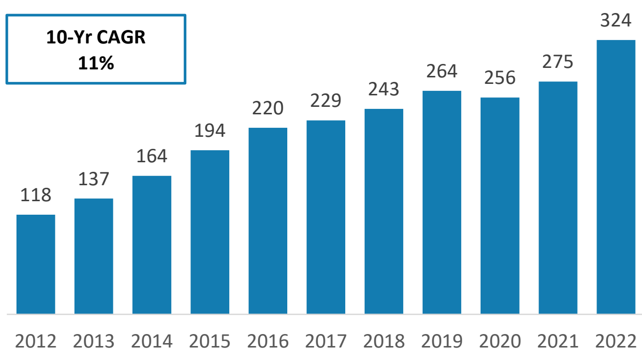
Exhibit 68: Brazil — Payroll Loans Gross Profit TAM (US$ Bn).



Source: Morgan Stanley Estimates (e).

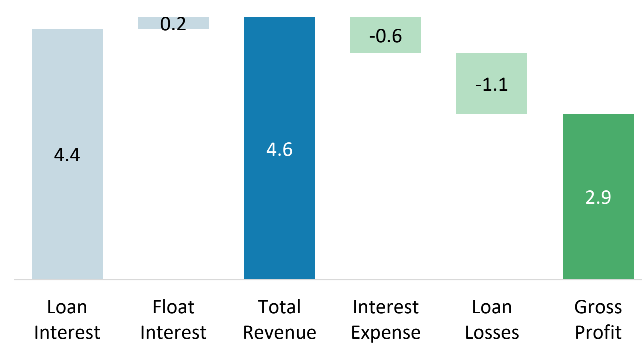
Payroll Loans in Mexico. Today, there are approximately M$350 billion of outstanding payroll loan balances in Mexico, having grown at a CAGR of 11% over the last 10 years and now accounting for over 25% of all consumer loans. To get to the product's TAM, we leveraged our proprietary unit economics model and some of the latest industry data, like total outstanding loans, NPLs, and interest rates. Importantly, since the average payroll loan in Mexico has an effective term of close to four years, we used average interest rates from the past 24 months and the average of our economist's projection for TIIE over the coming 4 years. All told, we got to a total revenues pool of M$82 billion per year, or ~US$5 billion, and a gross profit pool of M$52 billion per year, or ~US$3 billion.

Exhibit 69: Mexico — Total Payroll Loans (M$ Bn).



Source: Banxico, Morgan Stanley Research.

Exhibit 70: Mexico — Payroll Loans TAM (US$ Bn).



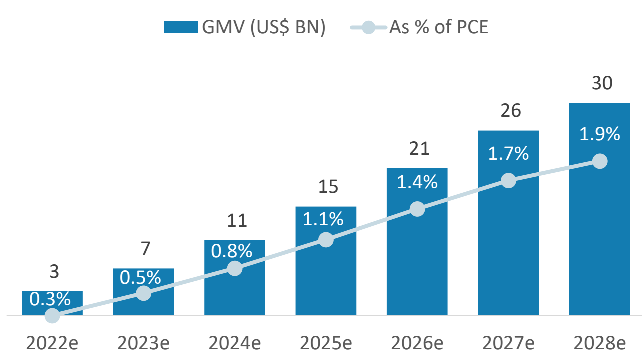
Source: Morgan Stanley Research Estimates.

Buy-Now Pay-Later (BNPL)

We calculate a combined BNPL revenue TAM between Brazil and Mexico of ~US$2 billion that could reach US$5-6 billion by 2028. This includes US$700 million in revenues from BNPL in Brazil and US$900 million in Mexico, this year. By 2028, we see US$2.4 billion in revenues from BNPL in Brazil and US$2.8 billion in Mexico.

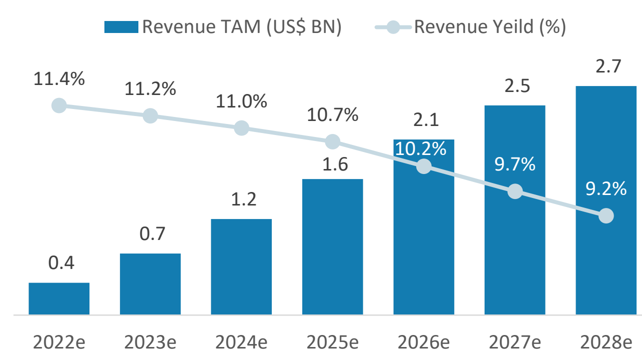
Brazil. While still small today, we think the BNPL opportunity is quite robust (as outlined in The Rise of Latam BNPL and The Upsurge of Latam BNPL). By our estimates, total BNPL volumes could reach US$30 billion in 2028. To get to our TAM, we leverage our unit economics builds an apply a revenue yield (i.e., gross take rate) to the GMV estimates. All told, we get to a total revenues TAM of US$700 million in 2023 and US$2.7 billion in 2028.

Exhibit 71: Brazil — BNPL GMV.



Source: Morgan Stanley Research Estimates (e).

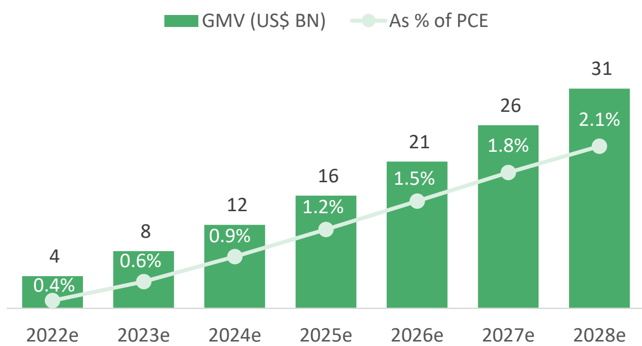
Exhibit 72: Brazil — BNPL Revenue TAM.



Source: Morgan Stanley Research Estimates (e).

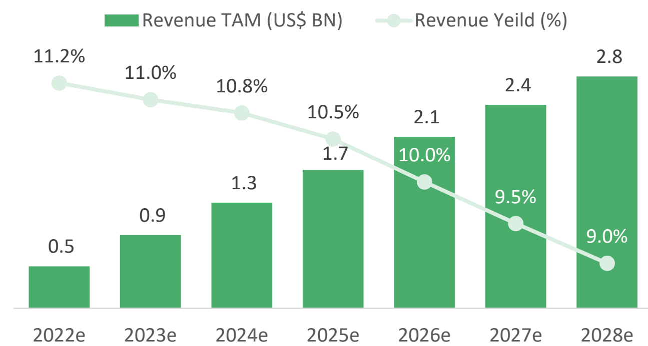
Mexico. We think the BNPL opportunity could be even larger in Mexico than Brazil give the under-banked nature of the market. By our estimates, total BNPL volumes could reach US$31 billion in 2028. To get to our TAM, we leverage our unit economics builds an apply a revenue yield (i.e., gross take rate) to the GMV estimates. All told, we get to a total revenues TAM of US$900 million in 2023 and US$2.8 billion in 2028.

Exhibit 73: Mexico — BNPL GMV.



Source: Morgan Stanley Research Estimates (e).

Exhibit 74: Mexico — BNPL Revenue TAM.



Source: Morgan Stanley Research Estimates (e).

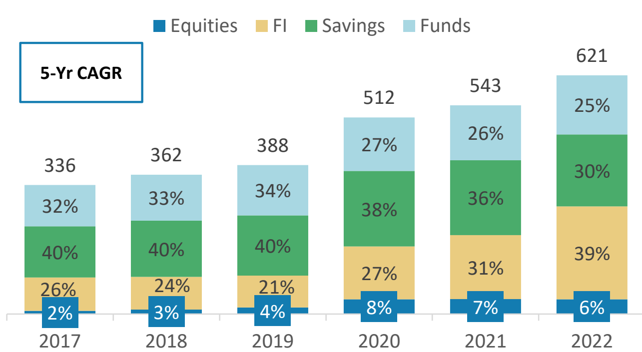
Retail Investments

We calculate that the retail investments TAM between Brazil and Mexico is US$8 billion in revenues. This includes US$4 billion in revenues in Brazil and US$4 billion in Mexico.

Brazil AUM. We calculate retail investment AUM reached ~US$600 billion in 2022, excluding pension and private assets, growing at a 5-year CAGR of 13%. Leveraging our proprietary asset management model, we estimate that retail investments accounted for ~30% of total industry AUM in 2022.

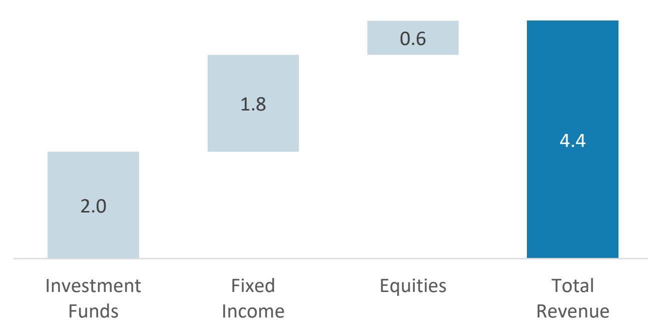
Brazil TAM. We calculate a total revenue pool of US$4.4 billion for retail investments in Brazil. We only consider assets invested in funds, equities, fixed income securities, and savings products. And we calculate the following management fees by asset class: 1.29% for investment funds, 1.66% for equities, 0.74% for fixed income, and 0.00% for savings.

Exhibit 75: Brazil — Retail Investments (US$ Bn).



Source: ANBIMA, Morgan Stanley Research Estimates. Note: excludes pension and private retail assets.

Exhibit 76: Brazil — Total Retail Investments TAM (US$ Bn).

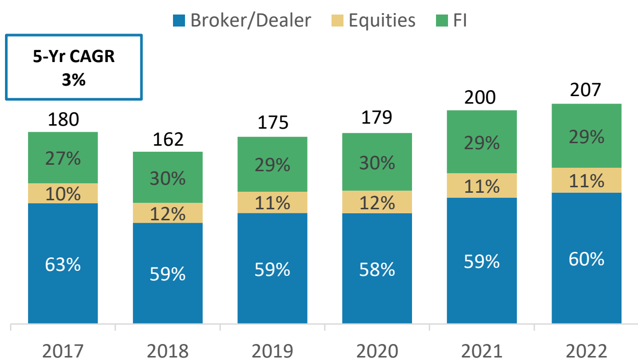


Source: ANBIMA, Morgan Stanley Research Estimates.

Mexico AUM. The retail investments industry in Mexico is smaller and has grown at a slower pace compared to Brazil. We estimate the retail AUM at brokers and mutual funds reached ~US$207 billion in 2022, of which US$123 billion is related to brokers and US$83 billion to mutual funds. For this analysis, we do no include AUM related to pension funds (*Afores*).

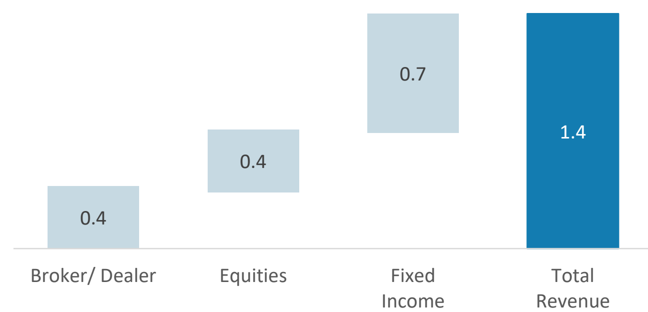
Mexico TAM. We calculate a total revenue pool of US$1.4 billion for retail investments in Mexico. We calculate retail AUM excluding pension assets, as we did for Brazil. We adjust the total AUM considering that individual investors account for 30% of AUM at broker/dealers and 60% at mutual funds AUM, according to our channel checks. And, we calculate the average broker commission is 0.3% and the following management fees by asset class: 1.43% for equities and 1.14% for fixed income.

Exhibit 77: Mexico — Retail Investments (US$ Bn).



Source: CNBV, Morgan Stanley Research.

Exhibit 78: Mexico — Total Retail Investments TAM (US$ Bn).



Source: CNBV, Morgan Stanley Research.

Insurance

We calculate that the life insurance brokerage TAM between Brazil and Mexico is US$8 billion in revenues. This includes US$3 billion in revenues in Brazil and US$1 billion in Mexico.

Insurance Brokerage in Brazil. In 2022, there were R$58 billion in total written life insurance premiums in Brazil (both group and individual), having grown at a 10-year CAGR of 10%. Assuming an up-front brokerage fee of around 30% on all policies we get to a total revenue TAM of R$17 billion, or US$3 billion.

Insurance Brokerage in Mexico. In 2022, there were M$84 billion in total written life insurance premiums in Mexico (both group and individual), having grown at a 10-year CAGR of 8%. Applying the same brokerage fee of around 30%, we get to a total revenue TAM of M$17 billion, or US$4 billion.

Rightsizing NU's Mexico Opportunity

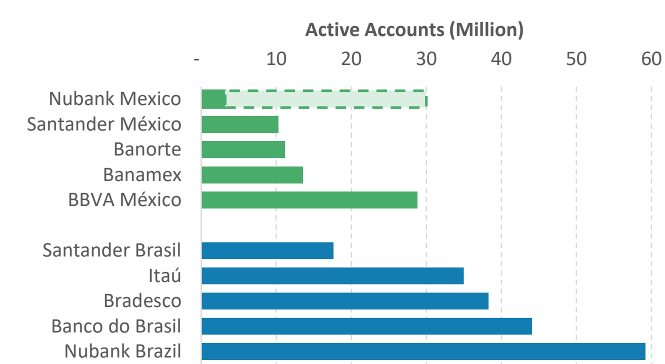
We are very optimistic on Nubank's growth prospects in Mexico given the powerful drivers of financial inclusion and electronification of financial services, amid a banking system with relatively weak incumbents (and we are not referring to Banorte and BBVA, which we see as the reference banks in the country). In addition, we think Nearshoring will likely provide a powerful tailwind for loan growth and overall banking activity over the next decade or so . For more details, please see our recent dispatch: Dream the Nearshoring Dream. In this section, we answer two important questions for NU: 1) How many clients can Nubank have in Mexico? and 2) What product penetration is achievable over time?

1. How many clients can Nubank have in Mexico?

We think Nubank can capture ~30 million active clients in Mexico over the next five years. We came to this conclusion by considering the following: i) the number of clients served by the incumbent banks in Mexico; ii) the unbancarized population in the country; iii) demographics and smartphone penetration; and iv) benchmarking versus Nu's development in Brazil.

We estimate ~11.5 million individuals already applied to Nubank, suggesting that its current 3.6 million client base could grow much faster. Indeed, 20% of the smartphone-connected adult population in Mexico already applied to receive a Nubank credit card or account, according to management here. And, we estimate ~57 million of adults have a smartphone, equal to 65% of adults 18 years or older.

Exhibit 79: We think Nubank can capture ~30 million active customers in Mexico by 2028.



Source: Company Data, Morgan Stanley Research Estimates (e).

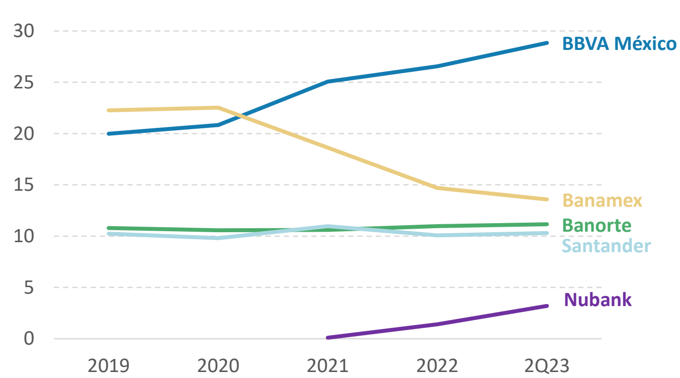
i) Nubank is disrupting Mexican banks' underserved clients.

The top four banks in Mexico have ~65 million clients... BBVA Mexico leads the group with 29 million, followed by Banamex with 14 million, Banorte with 12 million, and Santander with 10 million. Importantly, Banamex recently lost a significant amount of clients to BBVA Mexico, following reports of a potential sale of the bank and its persistent underperformance over the last 20 years. For more details, please see our dispatch on Banamex here.

... many of which are underserved and likely to open an account with Nubank. We think there is plenty of room for fintech companies like Nubank to disrupt and take share away from the established bank incumbents, particularly through risk-based pricing, leveraging data better, and offering superior digital customer experience. Also, without the overhead of a bricks-and-mortar retail network, fintech companies' operating costs are greatly reduced, allowing them to offer better pricing on banking products.

Nubank has captured 3.6 million clients, mostly individuals underserved by the incumbent banks. Nubank's client base in Mexico already reached one third of the size of leading banking franchises such as Banorte and Santander, an impressive feat considering that most of its clients already had access to banking services, according to management. Indeed, the bank is initially focusing on potential clients with historical credit bureau data, but it will expand its universe of potential clients as it leverages data from the recently launched savings account to improve its underwriting models in Mexico — allowing it to accelerate client growth going forward.

Exhibit 80: Mexico — Active accounts by bank (Million)



Source: CNBV, Morgan Stanley Research. Note: includes savings, checking, time deposits, and payroll accounts. Nubank figures are the number of customers reported by the company.

Measuring Bank Clients in Mexico

There is limited disclosure around the number of clients banks serve in Mexico. Local regulators only disclose the number of active accounts by bank, and only a couple of banks provide the number of unique clients. For instance, BBVA Mexico has 29 million clients; Banorte, 11.5 million; and Santander, almost 10 million. However, we note that the number of active accounts seems to be a good proxy to the number of clients reported by these banks, plus or minus 1 million looking at the past three years.

Important caveats:

i) We exclude pension fund clients. For instance, Banorte's Afore has 8.6 million clients in addition to the 11.5 million clients mentioned above, but many, if not most, do not have an account at the bank, according to management. So, we do not consider them as clients of the bank. As a reference, Mexican banks are not allowed to access the Afore's customer data, even if they are owned by the same financial group, which limits the cross-selling opportunity.

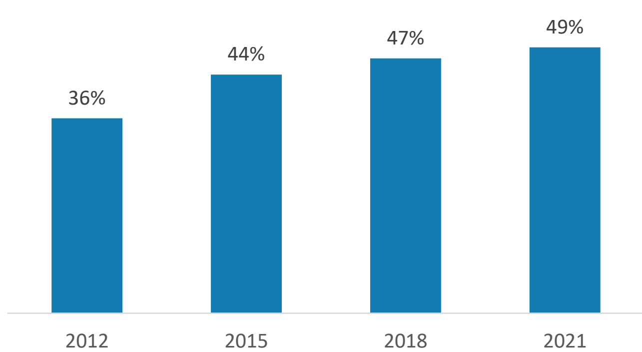
ii) The number of active bank accounts is not perfect. The numbers reported by the CNBV are not adjusted for customers with more than one account at the same bank, it includes both individuals and companies, and it considers active accounts those with at least one transfer, withdrawal, or deposits in the last three years. Evidently, it is not an accurate representation of unique customers for the entire system, but a good enough proxy for each bank. There are 109 million active bank accounts in Mexico, but only ~50% of the adult population, or 45 million individuals, have at least one formal bank account (see here).

iii) There is little to no disclosure on fintech clients. To our knowledge, only Nubank and Banorte's JV with Rappi disclose the number of active clients. As of 2Q23, Nubank had 3.6 million and Rappi had 1 million. News reports suggest that other players also reached over one million clients, including: Albo (4 million), Klar (2 million), Kueski (2 million), Flink (2 million), among others.

ii) Financial inclusion offers a major opportunity in Mexico.

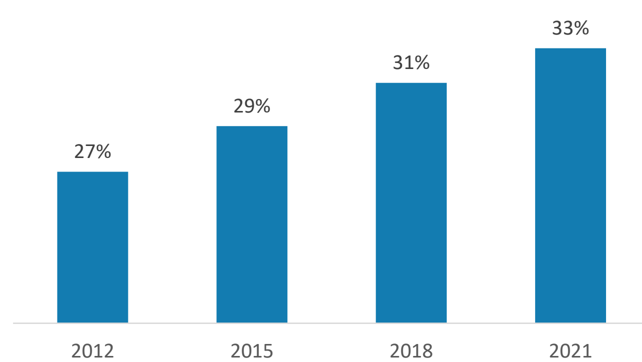
The financial system in Mexico has ignored or underserved a large portion of consumers. Only ~50% of the adult population, or 45 million individuals, have an active bank account, compared to +95% in Brazil. And only 33% of adults have at least one formal credit product.

Exhibit 81: Mexico — Bank Account Penetration (% Adults).



Source: INEGI, Morgan Stanley Research.

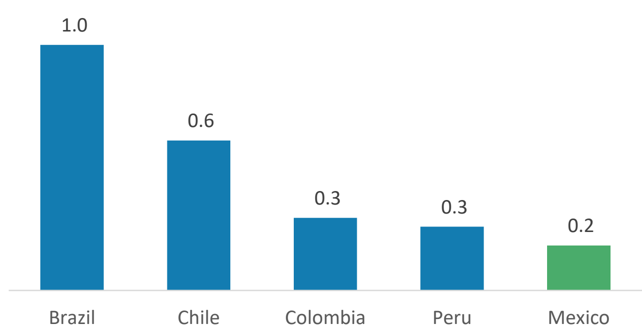
Exhibit 82: Mexico — Formal Credit Penetration (% Adults).



Source: INEGI, Morgan Stanley Research.

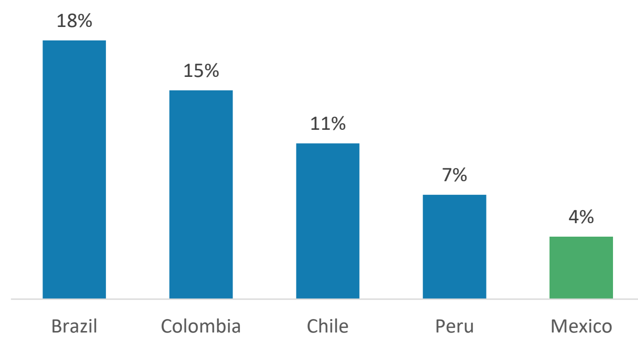
Low penetration of consumer loans offers attractive growth opportunities in key markets. For instance, Mexico has one of the lowest number of active credit cards per capita in the region, almost 80% lower than Brazil, despite similar levels of internet and smart-phone penetration and GDP per capita in both countries.

Exhibit 83: Latam — Active Credit Cards per Inhabitant (2022).



Source: Morgan Stanley Research.

Exhibit 84: Latam — Consumer Loans to GDP (2022).



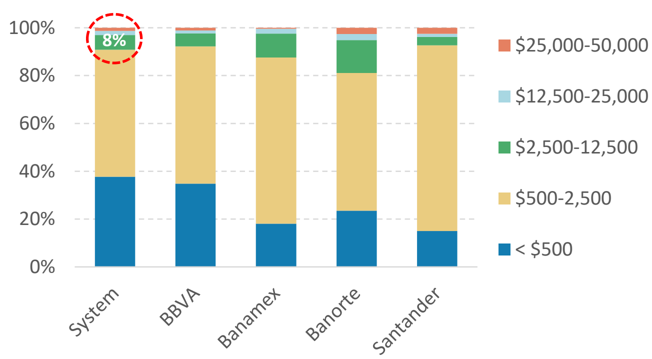
Source: Morgan Stanley Research. Note: Excludes mortgages.

iii) A large young digitalized population offers another path for fintech companies like Nubank.

Savings and deposits in the financial system are highly concentrated in a few accounts... Some 60% of all savings and deposits are concentrated in just 8% of active bank accounts (excluding large outliers with balances above US$50,000).

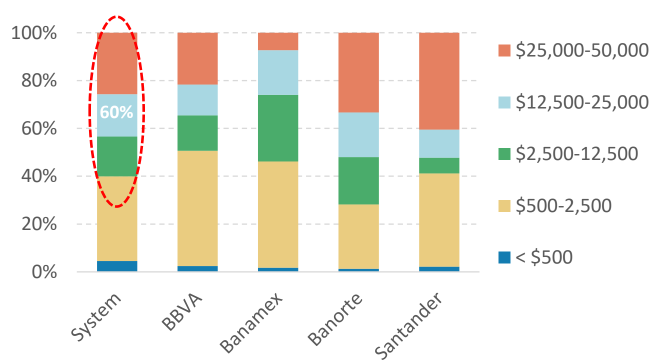
... but we think fintech companies could reach the masses. Technology and mobile communications are highly developed in Mexico, offering an effective way to reach middle- and low-income consumers. Fintech companies are leveraging technology to build innovative and low-cost products and services that have simple onboarding processes.

Exhibit 85: Mexico — Number of bank accounts by size (US$)



Source: CNBV, Morgan Stanley Research. Note: excludes accounts with an outstanding balance of more than US$50,000.

Exhibit 86: Mexico — Balance in bank accounts by size (US$)

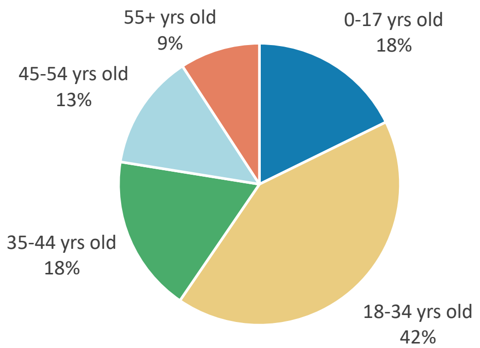


Source: CNBV, Morgan Stanley Research. Note: excludes accounts with an outstanding balance of more than US$50,000.

A large young digitalized population offers another path for fintech companies like Nubank. Indeed, Mexico has a total population of 126 million, of which 34% are between 0-19 years old and 31% are young adults between 20-39 years old. In addition, smartphone user penetration in Mexico is high at 55% of the population, compared to 40% penetration worldwide and 73% in the United States. But it is particularly high among young Mexicans, at 77% for individuals between 12-17 years old, 87% for those 18-24 years old, and 78% for those 25-34 years old. Unsurprisingly, young adults and teenagers represent 69% of smartphone users in Mexico, most of which are unbanked or underserved and more likely to use digital financial services.

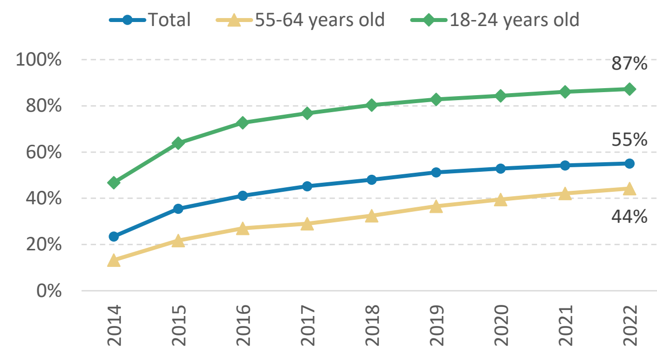
We estimate ~11.5 million individuals already applied to Nubank, suggesting that its 3.6 million client base could grow much faster. Indeed, 20% of the smartphone-connected adult population in Mexico already applied to receive a Nubank credit card or account, according to management here. And, we estimate ~57 million of adults have a smartphone, equal to 65% of adults 18 years or older.

Exhibit 87: Young adults and teenagers represent over two-thirds of smartphone users in Mexico.



Source: INEGI, Morgan Stanley Research.

Exhibit 88: Smartphone user penetration is high, particularly among younger consumers in Mexico.



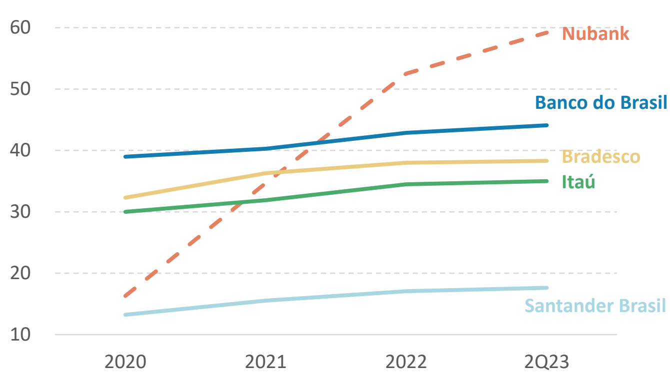
Source: INEGI, Morgan Stanley Research.

iv) Benchmarking Nubank's clients growth in Brazil.

In Brazil, Nubank captured a very significant amount of clients in a three-year span (~50 million), becoming one of the largest financial institutions in the country. However, we note that both number of accounts and total clients at the incumbent banks has not declined, despite Nubank's strong growth. In our view, this is explained by two things:

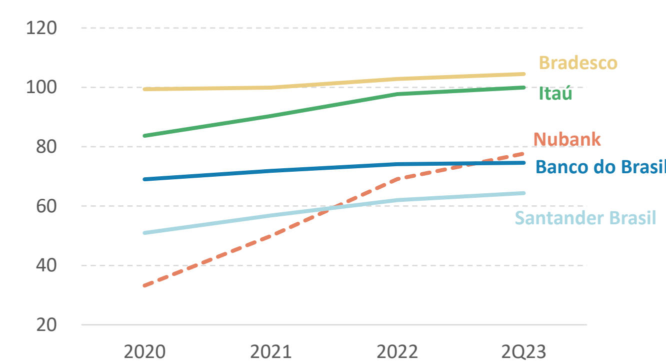
- **Financial Inclusion:** The total number of clients in the Brazilian financial system grew from 85 million in 2020 to 110 million in 2022, driven by both new entrants like Nubank and incumbent banks through digital only channels. As a result, most banks reported positive client growth in this period, despite being outpaced by Nubank. We think something similar could happen in Mexico.
- **Share of Wallet:** Brazilians typically have more than one bank, and Nubank has been able to capture significant share of wallet. However, this is not necessarily reflected in the number of accounts or clients at the incumbent banks. And this, we think, is likely happening already in Mexico. As we mentioned before, the majority of Nubank's 3.6 million clients in Mexico already had access to banking services.

Exhibit 89: Brazil — Active bank accounts by company (Million)



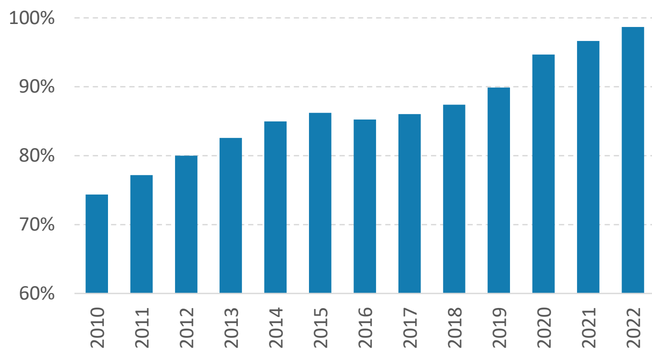
Source: Company Data, Morgan Stanley Research. Note: Bradesco and Banco do Brasil report unique active account holders, as per the central bank regulation. Itaú reports debit cards considering only account holders and excluding additional cards. Santander Brasil active current accounts, does not disclosed if it is based on unique clients.

Exhibit 90: Brazil — Total number of clients by company (Million)



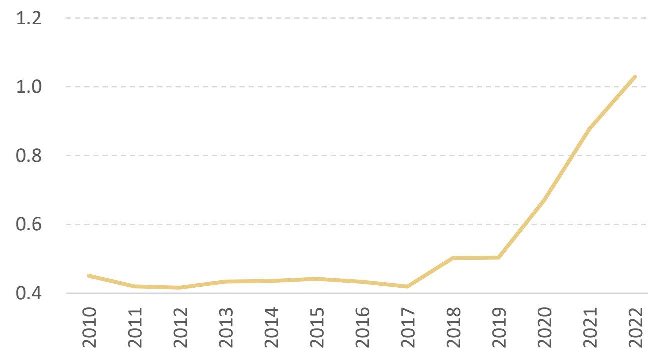
Source: Central Bank, Morgan Stanley Research. Note: total clients includes individuals and companies in different units, such as baking, payments, insurance, asset and wealth management, among others. It combines the data from the Client Registry of the National Financial System (CCS) and the Central Bank Credit Information System (SCR).

Exhibit 91: Brazil — Adults with a Banking Relationship.



Source: Central Bank, Morgan Stanley Research.

Exhibit 92: Brazil — Active Credit Cards per Inhabitant.

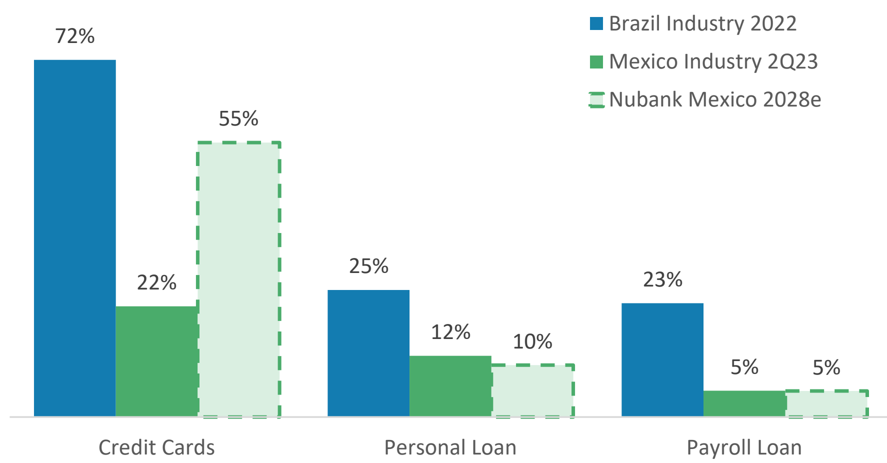


Source: Central Bank, Morgan Stanley Research.

2. What product penetration is achievable in Mexico?

We think Nubank can cross-sell 55% of credit cards and 5 to 10% of personal and payroll loans to its client base in Mexico over the next five years. Our forecast considers the following: i) the current cross-sell ratios at the incumbent banks in Mexico, ii) benchmarking product penetration in Brazil vs. Mexico, and iii) Nubank's development in Brazil.

Exhibit 93: We forecast penetration of personal and payroll loans in Mexico can reach 5-10% by 2028.



Source: BCB, CNBV, Morgan Stanley Research Estimates (e).

i) Incumbent bank clients are still underserved; penetration remains strikingly low.

As mentioned before, consumer loans are still highly underpenetrated in Mexico. And when looking at the large banks, most fail to provide many of their clients with basic credit products. For example, only 22% of active bank account holders have a credit card, and cross-sell drops off much further when looking at personal loans, payroll loans, and other key products.

In our view, the low level of credit products penetration is positive for Nubank as it offers an attractive growth avenue to extend credit to individuals already in the banking system. And of course, we expect Nubank to broaden penetration beyond individuals that are currently banked.

Exhibit 94: Mexico — The number of loans and credit cards as a percentage of active bank accounts suggests that product penetration is low.

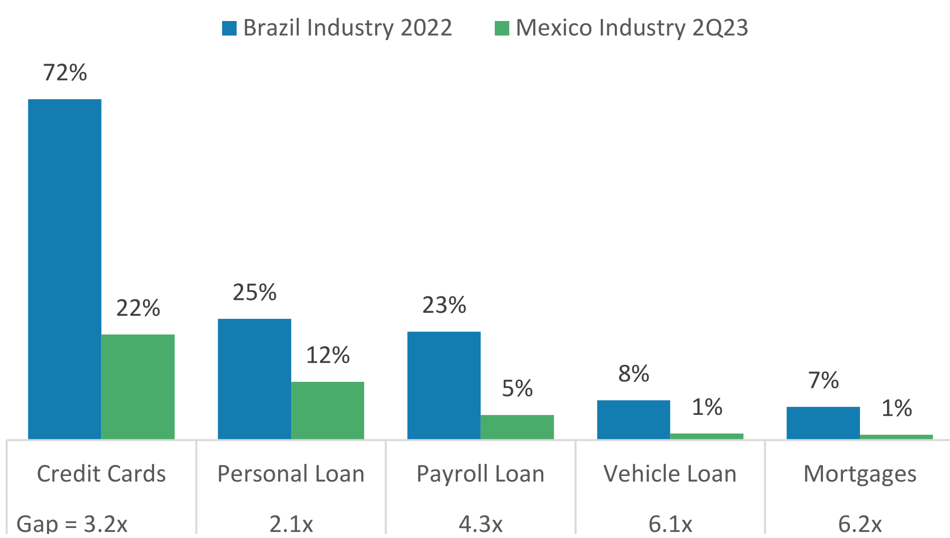
	# of Active Accounts	Active Credit Cards	Personal Loans	Payroll Loans	Vehicle Loans	Mortgages
System	**109,100,555**	**22%**	**12%**	**5%**	**1%**	**1%**
BBVA	28,841,650	23%	2%	8%	1%	1%
Banamex	13,580,399	37%	1%	6%	0%	1%
Banorte	11,159,589	17%	0%	8%	2%	2%
Santander	10,307,198	33%	1%	7%	1%	2%
Azteca	15,221,389	6%	64%	1%	0%	0%
Coppel	11,238,497	16%	7%	1%	0%	0%

Source: CNBV, Morgan Stanley Research.

ii) Penetration in Mexico could converge toward that in Brazil over time; there is plenty of upside.

Comparing consumer lending penetration in Mexico to that in Brazil highlights the lack of financial inclusion, and the significant opportunity available to new entrants. Across key consumer products, penetration in Brazil ranges from 2 to 6x higher than in Mexico. Credit card penetration, for example, runs at 72% of all banking clients in Brazil, vs. only 22% Mexico. As such, we think it is quite reasonable to assume that over time, penetration in Mexico can see a significant increase. Indeed, in our recent Insight report Dream the Nearshoring Dream, we introduce long-term, bull case estimates for the industry that have consumer lending penetration tripling by 2035.

Exhibit 95: Product penetration is significantly higher in Brazil — 2 to 6 times greater than Mexico.

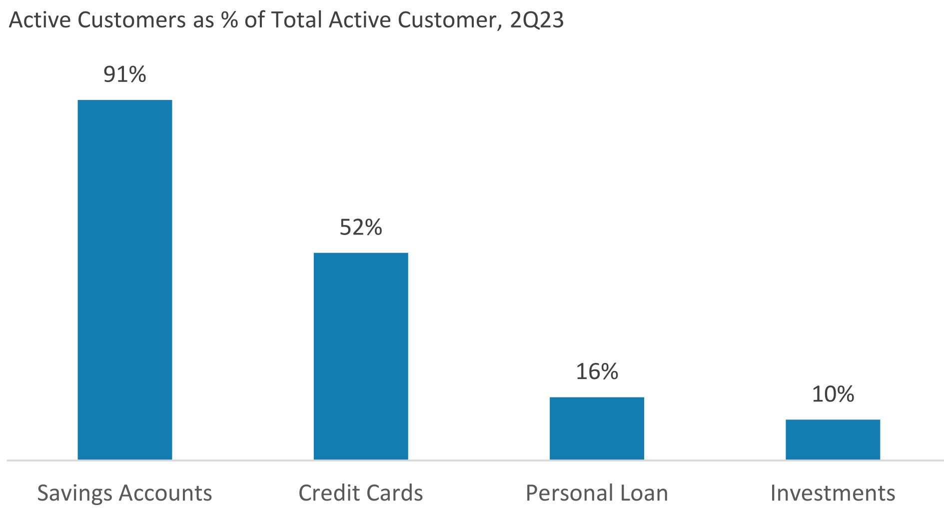


Source: BCB, CNBV, Morgan Stanley Research.

iii) Nubank has successfully demonstrated its ability to cross-sell credit products in Brazil.

In Brazil, more than half of Nubank's active customers hold an active credit card and over 15% already have an active personal loan, despite rolling out personal loans in 2020. And importantly, these penetration and activity rate figures have likely been diluted as a result of the rapid client growth in Brazil. All told, we think, Nubank has the tools and track record to deliver similar or greater levels of product penetration in Mexico, particularly given the highly underbancarized state of Mexico today.

Exhibit 96: Nubank — Product Penetration in Brazil.



Source: Company Data, Morgan Stanley Research Estimates. Activity Rate = Active product users as a percentage of total active customers.

Nubank Bottom-Up Build by Country

We built detailed bottom-up models for Brazil, Mexico, and Colombia. Our models couple unit economics with user growth and activity rate forecasts of key products, including savings accounts, credit cards, personal loans, payroll loans, BNPL, among others. Below, we provide the key takeaways and a detailed overview of the assumptions behind our new estimates for Nubank.

KEY TAKEAWAYS

1) We are increasing our 2028 EPS forecast by 46%, implying a 2028 ROE of 36% vs. 30% before. Our new estimates are mostly driven by user growth and product penetration in Mexico and Colombia, as well as higher operating leverage. More on this further below.

2) In our new model, Mexico and Colombia make up 20% of total revenue in 2028, up from 6% currently. This means Mexico and Colombia revenue should grow at five-year CAGRs of 67% and 82%, respectively, reaching US$5.1 and US$1.2 billion in 2028. So, our total revenue forecast is now 34% higher, reaching US$31 billion in 2028.

3) We think Nubank can reach 125 million active customers by 2028, with product penetration levels ~7% higher compared to today. Our revenue forecasts are based on reasonable customer growth and product penetration assumptions in each country. We see active customers reaching 40% of the combined adult population — 53% in Brazil, 25% in Mexico, and 20% in Colombia. In addition, we assume the average product penetration will increase over the next five years; for example, we see penetration of personal loans increasing from 10 to 14% and payroll loans from 1 to 8%.

4) Our new estimates incorporate attractive gross profit margin expansion. Our model yields a gross profit margin of 58% in 2028, up from 42% this year. The expansion is mostly due to revenue growth outpacing that of loan-loss provisions and financial expenses, as a result of the following assumptions:

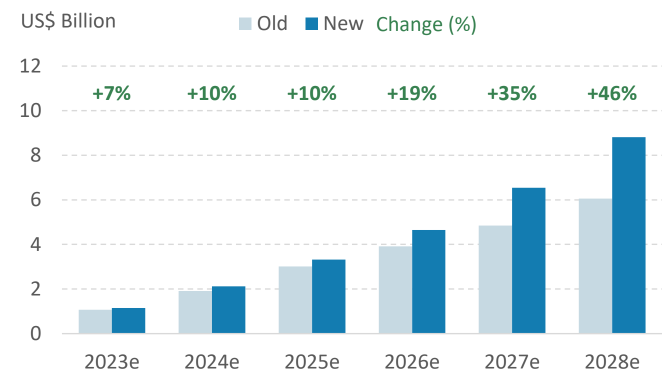
- **Lower cost of risk.** In Brazil, we expect lower cost of risk due to improving household financial conditions and a shift in loan mix to lower-risk longer-term payroll loans that should help build a back-book with stable and lower delinquency levels. In Mexico and Colombia, we assume lower cost of risk relative to Brazil, on the back of our unit economics analysis by product. However, cost of risk in these countries should improve at a later stage, as new vintages with higher provisioning should make most of the total portfolio over the next two years. Importantly, our long-term cost of risk forecasts in all countries trend lower toward, but slightly above, the historical industry average of each product.
- **Lower cost of funding.** We forecast lower funding costs on the back of falling interest rates, a growing low-cost deposit base in Brazil, and higher contribution from even lower-cost deposits in Mexico and Colombia.
- **Higher leverage.** We assume leverage will increase as loan growth continues to outpace deposits — we see the loan to deposit ratio going from ~80% in 2023 to ~120% in 2028 and interest-earning assets over interest-bearing liabilities increasing from 1.3x to 1.4x in the same period.

To make sure our gross profit margin assumptions are reasonable, we look at recent performance of risk-adjusted NIM relative to our expectations. Over the past year, risk-adjusted NIM increased ~6 percentage points, from 2% in 2Q22 to 8% in 2Q23, driven by three factors: loan growth outpaced deposits, cost of funding fell sharply due to the success of money boxes (i.e., new deposit accounts), and loan mix shifted toward higher margin personal loans. In contrast, we forecast risk-adjusted NIM will only expand less than 1 percentage point per year over the next five years, from ~8% in 2023 to ~12% in 2028, a fraction of what it increased over the past 12 months.

5) A highly scalable business model with a high percentage of fixed costs offers further operating leverage. For instance, we forecast that the monthly cost to serve (per client) should only increase from US$0.8 to 1.5 over the next five years, while the monthly ARPAC should jump from US$9.5 to 21.3 over the same period. In other words, we see the efficiency ratio improving from 36% in 2023 to 23% in 2028. As a reference, the efficiency ratio significantly improved over the past year, from 58% in 2Q22 to 35% in 2Q23.

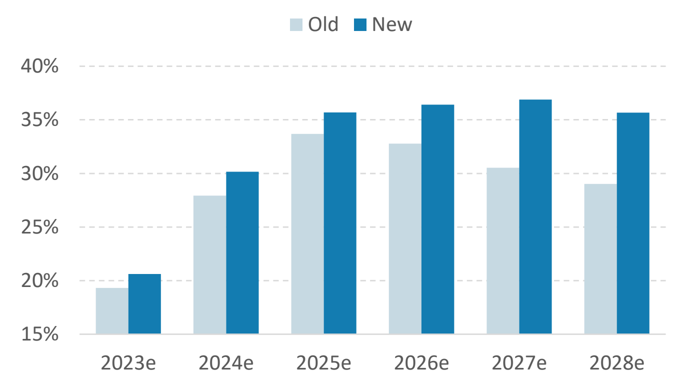
Key Takeaways in Charts

Exhibit 97: Our new 2028 adjusted NI estimate is 46% higher...



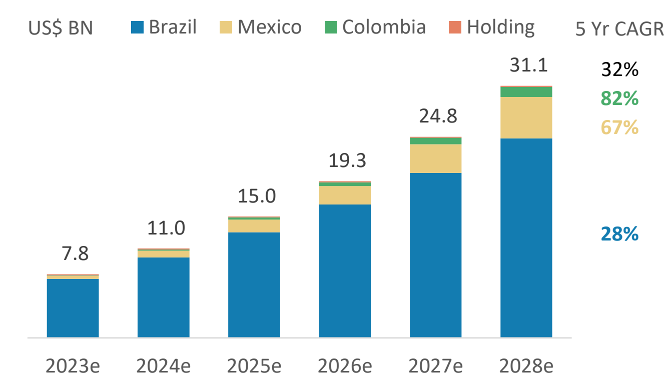
Source: Morgan Stanley Research Estimates (e).

Exhibit 98: ... implying 2028 adjusted ROE of 36% vs. 30% before.



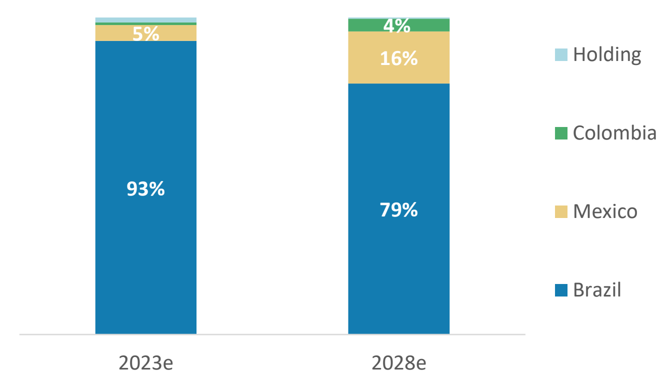
Source: Morgan Stanley Research Estimates (e).

Exhibit 99: We increased our 2028 revenue forecast by 34%, as we incorporate Mexico and Colombia.



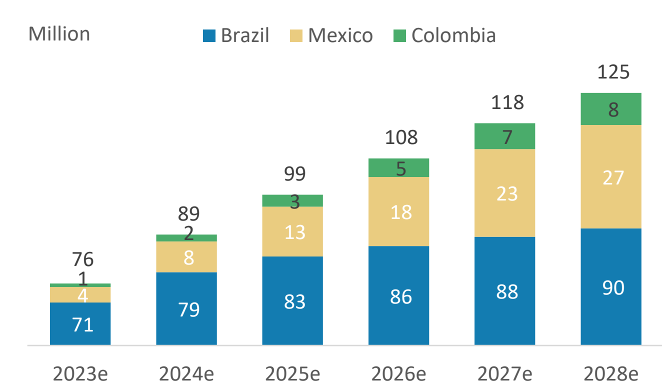
Source: Morgan Stanley Research Estimates (e).

Exhibit 100: We see Mexico and Colombia reaching 20% of revenue in 2028, up from 6% currently.



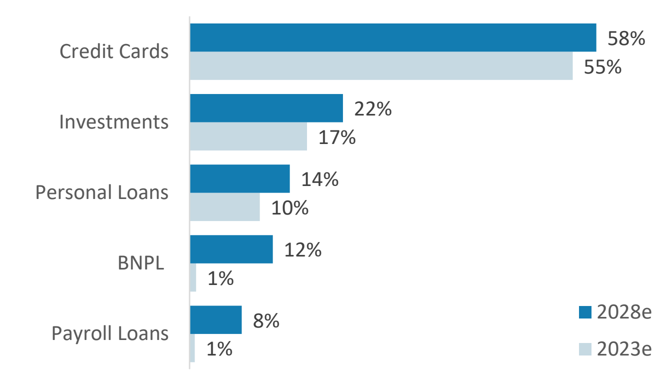
Source: Morgan Stanley Research Estimates (e).

Exhibit 101: In our view, Nubank can reach 125 million active customers by 2028; ~40% of the adult population.



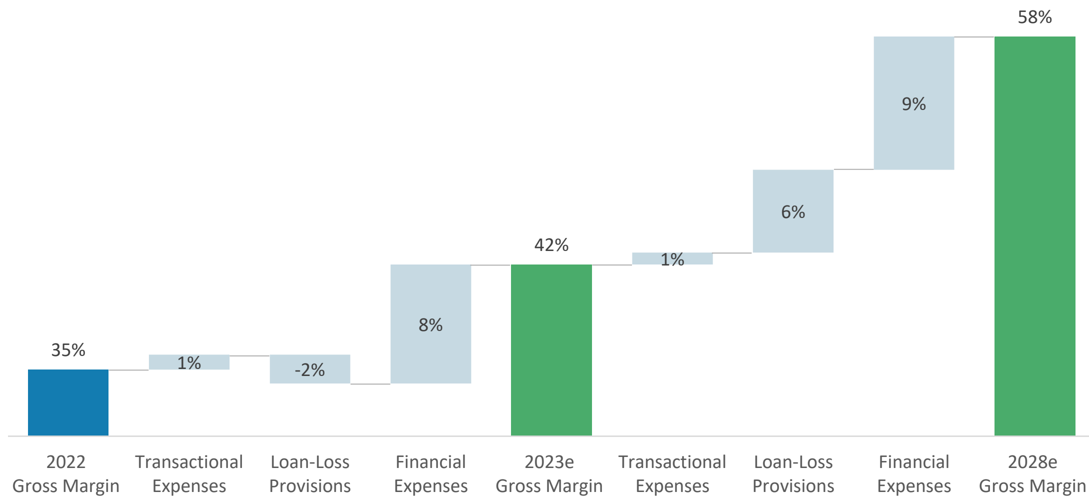
Source: Morgan Stanley Research Estimates (e).

Exhibit 102: Our revenue forecast also considers higher product penetration, on average, up ~7 percentage points by 2028.



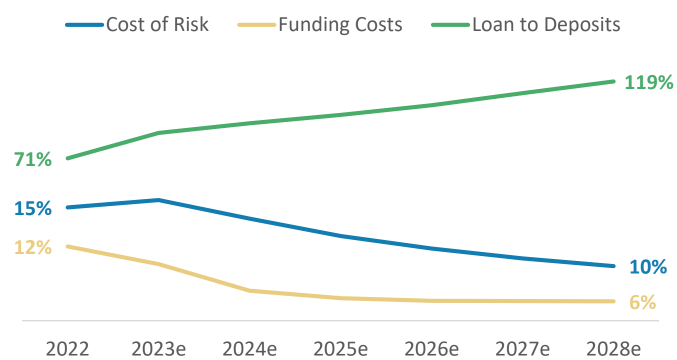
Source: Morgan Stanley Research Estimates (e).

Exhibit 103: Our new estimates incorporate attractive gross profit margin expansion, on the back of lower cost of risk, lower cost of funding, and higher leverage.



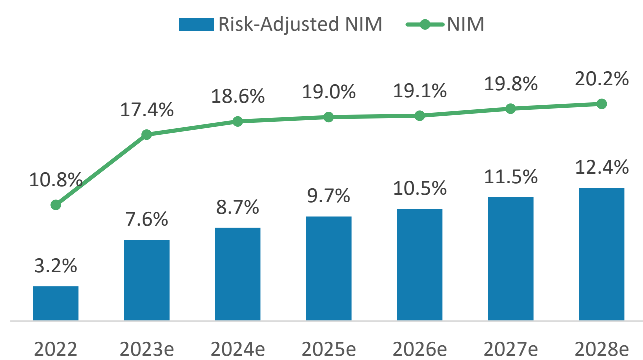
Source: Morgan Stanley Research Estimates (e).

Exhibit 104: Our long-term cost of risk, cost of funding, and leverage forecasts are in-line with peers and historical industry levels...



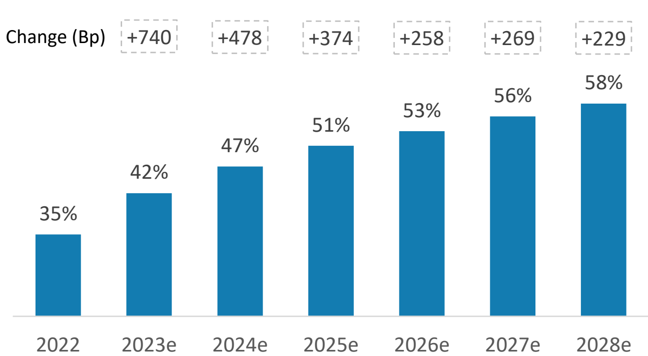
Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 105: ... reasonable assumptions, in our view, considering recent NIM performance and limited expansion going forward.



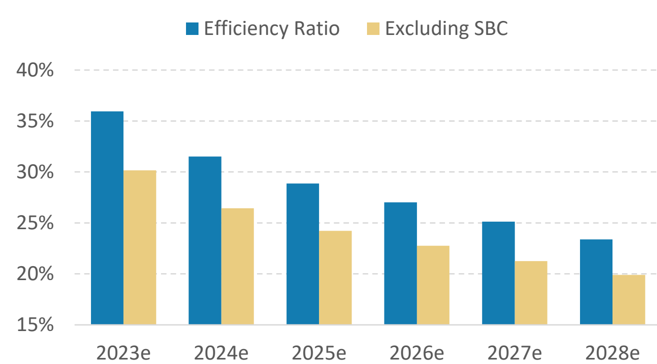
Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 106: In addition to attractive gross profit margin expansion...



Source: Morgan Stanley Research Estimates (e).

Exhibit 107: ... operating leverage should also boost earnings growth and ROE.



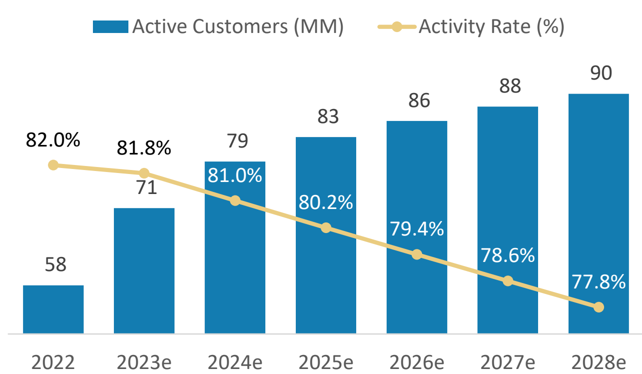
Source: Morgan Stanley Research Estimates (e). Note: Expenses / (NII + Fees)

BRAZIL: KEY DRIVERS

Our revenue forecast is mostly driven by growth in active customers and product penetration:

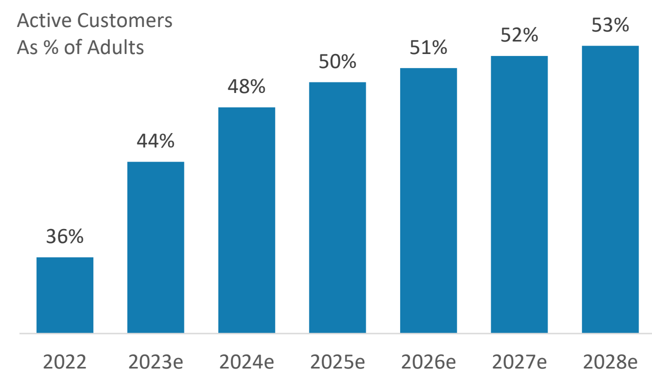
i) Active Customers. We forecast a deceleration in active customers growth, as we expect the company's client base will reach ~50% of the Brazilian adult population in 2025. Indeed, we see net adds staying positive but falling from 13 million in 2023 to 4 million in 2025 and 2 million in 2028 — reaching a total of 90 million active customers by 2028.

Exhibit 108: We expect a sharp deceleration in active customers growth...



Source: Morgan Stanley Research Estimates (e).

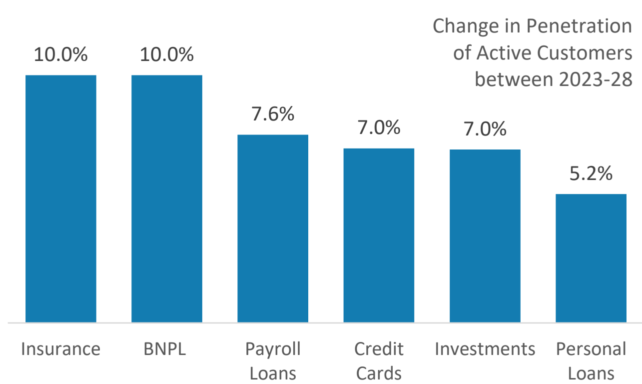
Exhibit 109: ... as the client base gets to ~50% of the adult population in 2025.



Source: Morgan Stanley Research Estimates (e).

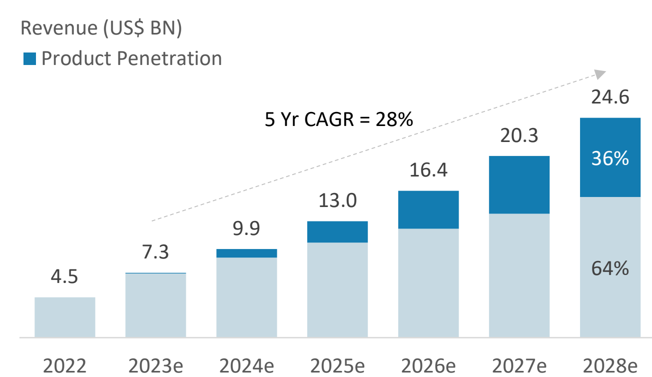
ii) Product Penetration. We forecast product penetration will increase, on average, 8 percentage points over the next five years. We think our assumptions are reasonable and even conservative, particularly on highly profitable personal and payroll loans. We see penetration of personal and payroll loans reaching 16% and 8% of active clients in 2028, up from 11 and 1% currently. All told, we think penetration could add 36% more revenue in Brazil by 2028.

Exhibit 110: We forecast product penetration will increase 5-10 percentage points...



Source: Morgan Stanley Research Estimates (e).

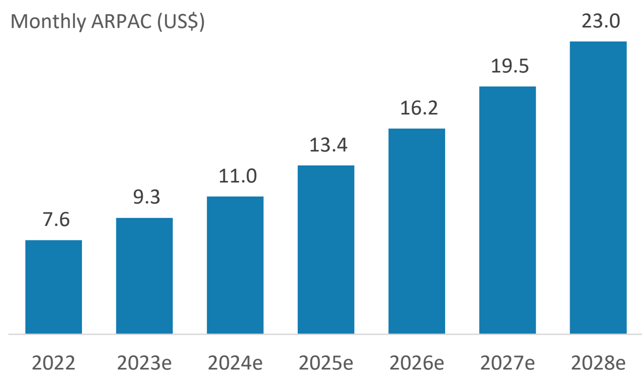
Exhibit 111: .... having a strong impact in revenue growth over the next 5 years.



Source: Morgan Stanley Research Estimates (e). Note: *Assumes no customer growth and all else equal.

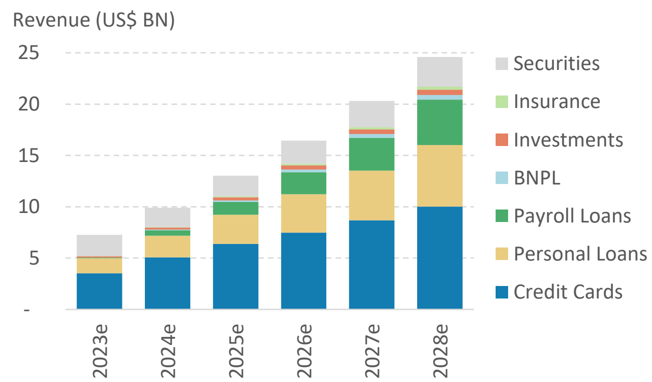
Putting all products together, our revenue estimate in Brazil grows at 5-year CAGR of 28%, reaching US$24 billion in 2028. Our forecast implies a monthly ARPAC of ~US$23 in 2028, up from ~US$9 currently, mostly due to cross-selling and higher credit limits of payroll loans, personal loans, and credit cards. We forecast revenues for each product by coupling our unit economics models with customer growth and product penetration. We provide key performance indicators of each product in the summary tables further below.

Exhibit 112: The ARPAC in Brazil could more than double by 2028...



Source: Morgan Stanley Research Estimates (e).

Exhibit 113:... driven by cross-selling new credit products and increasing credit limits.

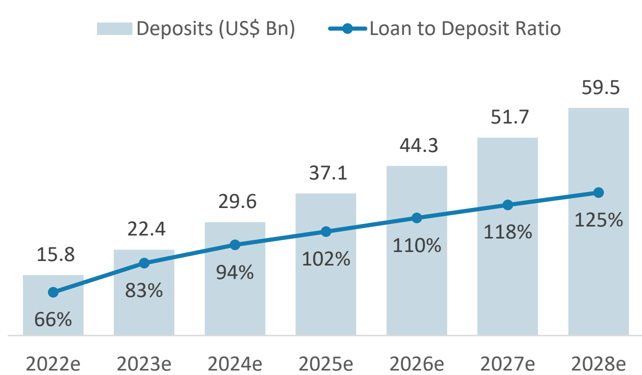


Source: Morgan Stanley Research Estimates (e).

On the cost side, we forecast the following for financial expenses and loan-loss provisions:

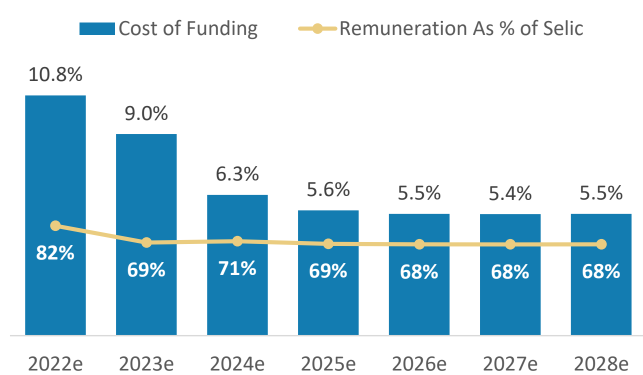
i) Financial expenses on deposits. First, we believe deposits can grow at a five-year CAGR of 22%, reaching US$60 billion in 2028. Our estimate is based on the assumption that 90% of active customers continue to have a savings account in this period and that the average deposit per account doubles from ~US$350 currently to ~US$700 in 2028. We also assume the remuneration on deposits as a percentage of the Selic rate remains stable around 68% going forward, after strong improvements delivered earlier this year. Then we assume the cost of funding improves from 9.0% in 2023 to 5.5% in 2028, as the average Selic rate comes down from ~13% to ~8%. Lastly, our deposit forecast implies higher leverage — loans to deposit ratio goes from 83% in 2023 to 125% 2028, close to industry historical levels of ~130%, on average, over the past 10 years.

Exhibit 114: Loan growth should outpace deposits, increasing leverage close to industry levels.



Source: Morgan Stanley Research Estimates (e).

Exhibit 115: Cost of funding should improve as remuneration remains stable and Selic falls.

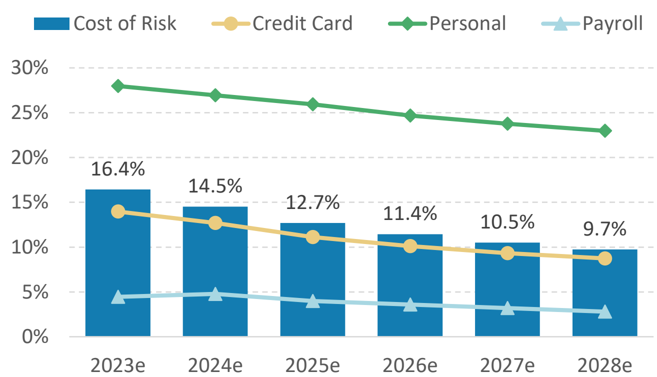


Source: Morgan Stanley Research Estimates (e). Note: includes deposits and excludes borrowing and financing instruments.

ii) Loan-loss provisions. We expect lower cost of risk due to improving household financial conditions and a mix sift toward low-risk, longer-term payroll loans that should help build a back-book with stable and lower delinquency levels. Our long-term forecasts trend toward, but are slightly above, the historical industry average for each product. For instance, we forecast the cost of risk of credit cards will reach 8.7% in 2028, compared to 8.4% from our unit economics model. And, even though it is not fully comparable to the cost of risk, we note that the average 90-day NPL ratio of credit cards was only 7.0% over the past 13 years.

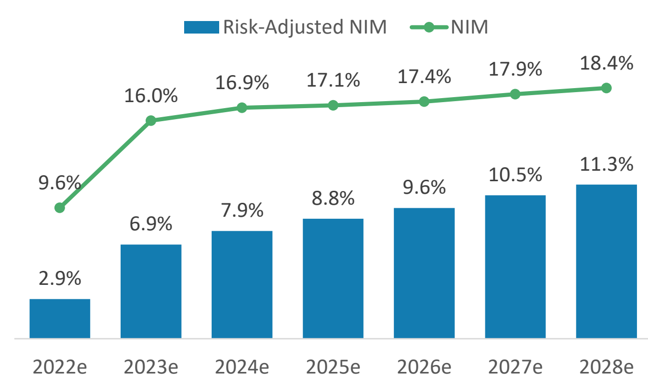
We expect gradual and reasonable NIM expansion. Over the next five years, we see NIM expanding 2 percentage points and risk-adjusted NIM 4 percentage points, on the back of loan growth outpacing deposits, further improvements in funding costs as interest rates fall, shift in loan mix to higher margin less rate sensitive products, and a normalization in the cost of risk.

Exhibit 116: Cost of risk should benefit from a shift in loan mix to payroll loans and a normalization in other products.



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 117: We expect a gradual improvement in net interest margins, after a sharp improvement in the past year.



Source: Company Data, Morgan Stanley Research Estimates (e). Note: Net interest income only considers interest expense from deposits.

Credit Card KPIs

Exhibit 118: Brazil — Key Drivers of Credit Cards.

Credit Cards	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)	30	38	43	46	49	52	54	8%
Penetration of Active Customers	52%	53%	54%	56%	57%	58%	60%	
Revenue (US$ BN)	1.9	3.5	5.1	6.4	7.5	8.7	10.0	23%
% of Brazil Total	43%	48%	51%	49%	45%	43%	41%	
Monthly ARPAC (US$)	5.9	8.6	10.5	11.9	13.1	14.4	15.8	13%
Cost of Risk	NA	14.0%	12.7%	11.1%	10.1%	9.3%	8.7%	
Loan Portfolio (US$ BN)	8.4	14.0	18.7	22.4	26.1	30.3	34.6	20%
Avg. Loan per Active Customer (US$)	284	381	442	487	537	591	645	11%
Implicit Market Share	9%	13%	15%	16%	17%	19%	20%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Personal Loan KPIs

Exhibit 119: Brazil — Key Drivers of Personal Loans.

Personal Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)	5	8	9	11	12	13	14	13%
Penetration of Active Customers	9%	11%	12%	13%	14%	15%	16%	
Revenue (US$ BN)	0.9	1.5	2.1	2.9	3.8	4.8	6.0	32%
% of Brazil Total	21%	20%	21%	22%	23%	24%	24%	
Monthly ARPAC (US$)	19.7	18.7	20.3	23.2	26.8	31.2	35.9	14%
Cost of Risk	NA	28.0%	26.9%	25.9%	24.7%	23.8%	23.0%	
Loan Portfolio (US$ BN)	2.0	3.4	5.2	7.4	9.8	12.2	15.1	34%
Avg. Loan per Active Customer (US$)	376	464	564	685	801	920	1,055	18%
Implicit Market Share	4%	6%	9%	11%	14%	16%	18%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Payroll Loan KPIs

Exhibit 120: Brazil — Key Drivers of Payroll Loans.

Payroll Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)	-	1	2	3	4	6	8	68%
Penetration of Active Customers	-	1%	2%	4%	5%	7%	8%	
Revenue (US$ BN)	-	0.1	0.5	1.2	2.2	3.2	4.4	136%
% of Brazil Total	-	1%	6%	10%	13%	16%	18%	
Monthly ARPAC (US$)	-	27.5	42.6	45.2	48.2	51.3	54.4	15%
Cost of Risk	-	4.4%	4.8%	4.0%	3.6%	3.2%	2.8%	
Loan Portfolio (US$ BN)	-	1.2	3.9	7.9	12.5	17.9	24.0	83%
Avg. Loan per Active Customer (US$)	-	2,531	2,660	2,796	2,938	3,088	3,245	5%
Implicit Market Share	-	1%	3%	6%	8%	11%	14%	

Source: Company Data, Morgan Stanley Research Estimates (e).

BNPL, Investments, & Insurance KPIs

Exhibit 121: Brazil — Key Drivers of BNPL, Investments, and Insurance.

Other Products	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)								
BNPL	-	1	2	4	6	8	10	69%
Investments	7	13	17	21	21	22	23	12%
Insurance	1	1	3	5	7	9	11	50%
Penetration of Active Customers								
BNPL	0%	1%	3%	5%	7%	9%	11%	
Investments	13%	18%	22%	25%	25%	25%	25%	
Insurance	2%	2%	4%	6%	8%	10%	12%	
Revenue (US$ BN)	0.1	0.2	0.3	0.5	0.8	1.0	1.3	53%
BNPL	-	0.0	0.1	0.2	0.3	0.4	0.5	126%
Investments	0.1	0.1	0.2	0.3	0.4	0.5	0.5	34%
Insurance	0.0	0.0	0.0	0.1	0.2	0.2	0.3	67%
Revenue Mix, % of Brazil Total	2%	2%	3%	4%	5%	5%	5%	
BNPL	0%	0%	1%	1%	2%	2%	2%	
Investments	2%	2%	2%	2%	2%	2%	2%	
Insurance	0%	0%	1%	1%	1%	1%	1%	
Monthly ARPAC (US$)								
BNPL	-	3.1	3.9	4.1	4.1	4.2	4.3	7%
Investments	1.5	1.0	1.0	1.2	1.6	1.8	1.9	15%
Insurance	1.5	1.7	1.8	2.0	2.1	2.3	2.5	9%
Implicit Market Share								
BNPL	-	1%	6%	10%	12%	14%	17%	
Investments	1%	2%	2%	3%	3%	3%	3%	
Insurance	1%	1%	2%	3%	4%	6%	7%	

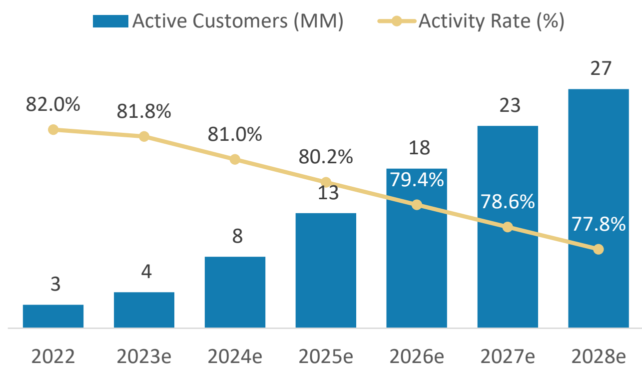
Source: Company Data, Morgan Stanley Research Estimates (e).

MEXICO: KEY DRIVERS

Our revenue forecast is mostly driven by growth in active customers and product penetration:

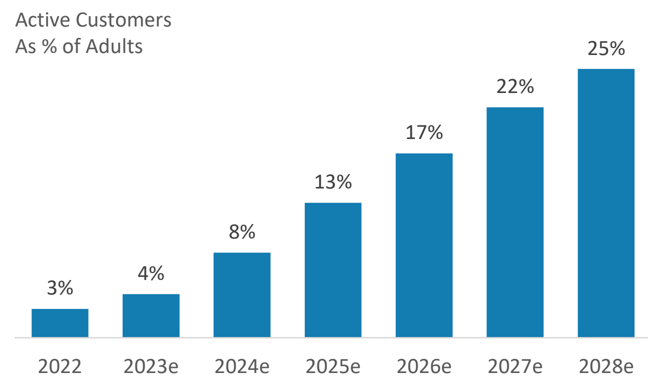
i) Active Customers. Leveraging our work from Rightsizing NU's Mexico Opportunity , we assume Nubank will capture 40 million clients in the next five years. But, to be conservative, we reduce the activity rate to similar levels in Brazil. As a result, we expect the active client base will reach 27 million in 2028 or ~25% of the adult population in Mexico. In terms of net adds, our numbers for 2024 and beyond also look reasonable — between 4 and 5 million per year.

Exhibit 122: To be conservative, we assume a reduction in the activity rate, but we still expect strong client growth...



Source: Company Data, Morgan Stanley Research Estimates (e).

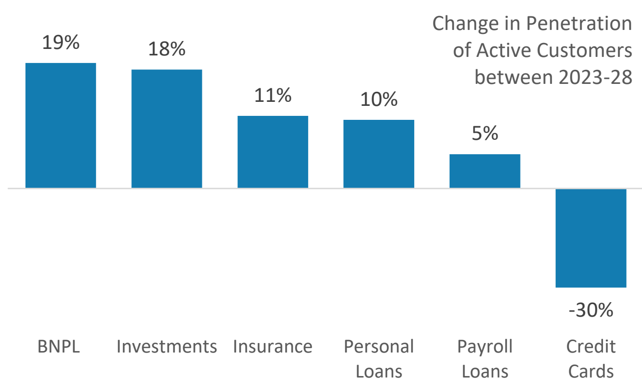
Exhibit 123: ... with active customers reaching 27 million in 2028 or 25% of the adult population in Mexico.



Source: Company Data, Morgan Stanley Research Estimates (e).

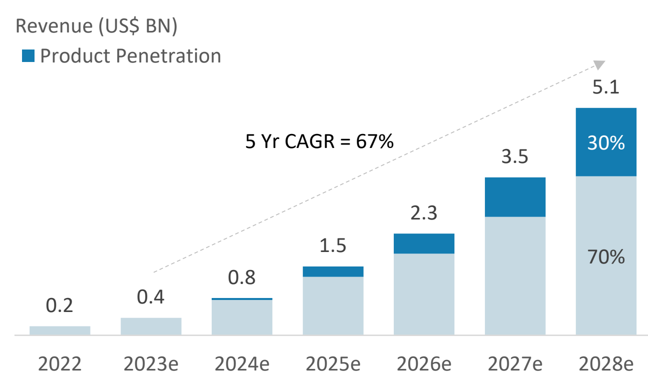
ii) Product Penetration. We forecast product penetration will increase, on average, 5 percentage points over the next five years. Importantly, penetration of credit cards among active clients in Mexico is very high — likely over 95% — because it was the only product offered until recently. So we assume credit card penetration declines to 55% in 2028, similar to current levels in Brazil. Excluding cards, penetration of other products would increase 13 percentage points, on average, driving attractive long-term revenue growth. For instance, we see penetration of personal and payroll loans reaching 10% and 5% of active clients in 2028, respectively, up from 0% currently.

Exhibit 124: Penetration of new products, such as personal and payroll loans, should ramp-up in the next couple of years...



Source: Company Data, Morgan Stanley Research Estimates (e).

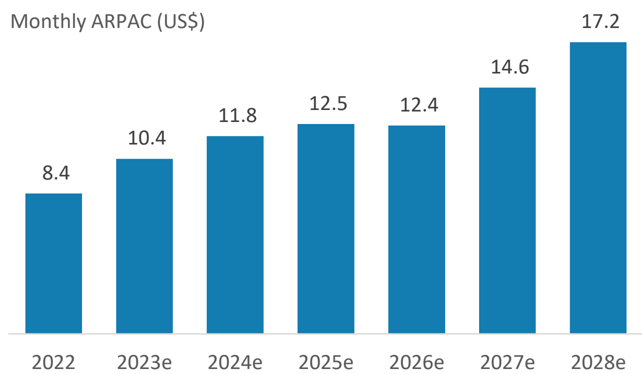
Exhibit 125: ... having a strong impact on revenue growth. We think penetration could add 30% more revenue by 2028.



Source: Company Data, Morgan Stanley Research Estimates (e).

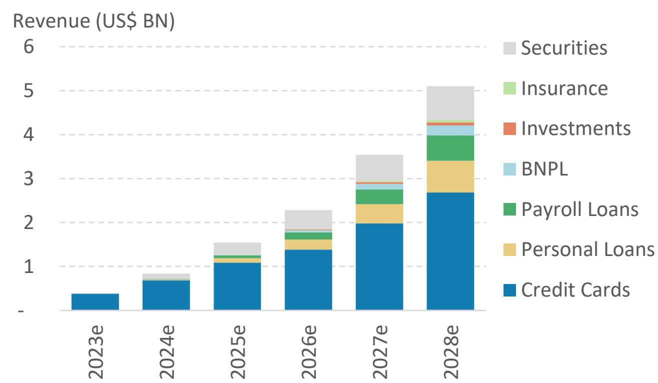
We see revenue in Mexico growing at five-year CAGR of 67%, reaching US$5.1 billion in 2028. This means Mexico alone could add over 20% more revenue to the company in the next five years. Our forecast implies a monthly ARPAC of ~US$17 in 2028, up from ~US$10 currently, mostly due to strong client growth, cross-selling new products, and higher credit limits of payroll loans, personal loans, and credit cards.

Exhibit 126: We cautiously assume the ARPAC in Mexico grows at a faster pace until 2027...



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 127: ... on the back of lower net adds and higher penetration of payroll & personal loans.



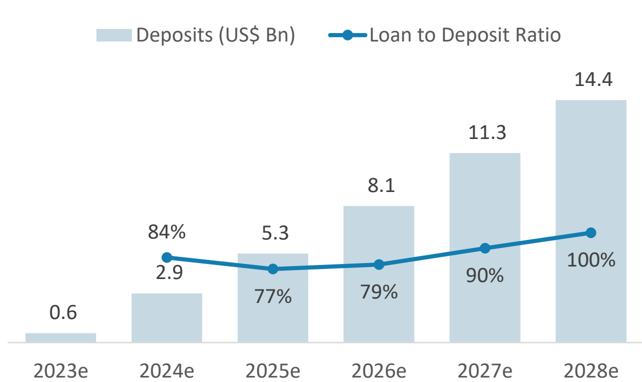
Source: Company Data, Morgan Stanley Research Estimates (e).

On the cost side, we forecast the following for financial expenses and loan-loss provisions:

i) Financial expenses on deposits. First, we believe deposits can grow at a five-year CAGR of 90%, reaching US$14 billion in 2028. Our estimate is based on the assumption that 90% of active customers continue to have a savings account in this period and that the average deposit per account doubles from ~US$300 currently to ~US$600 in 2028. We also assume the remuneration on deposits as a percentage of the benchmark interest rate remains stable around 49% going forward. As a reference, incumbent banks in Mexico pay 30-40% of the benchmark rate. Then we assume the cost of funding improves from ~5% in 2023 to ~4% in 2028, as the average interest rate comes down from ~11% to ~7.5%.

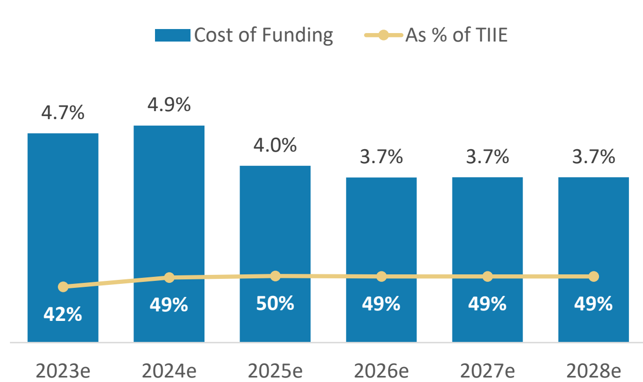
Importantly, our deposit forecast implies higher leverage — loans to deposit ratio goes from 84% in 2024 to 100% in 2028, below the industry historical levels of ~160%, on average, over the past 10 years. However, we estimate lower leverage in Mexico relative to Brazil because the credit card business requires more funding/deposits in Mexico, as we explain in detail in our credit card unit economics section here .

Exhibit 128: Loan growth should outpace deposits, increasing leverage but remaining below levels in Brazil on higher funding needs.



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 129: Cost of funding should improve as remuneration of deposits remains stable at 49% and the benchmark interest rate falls.

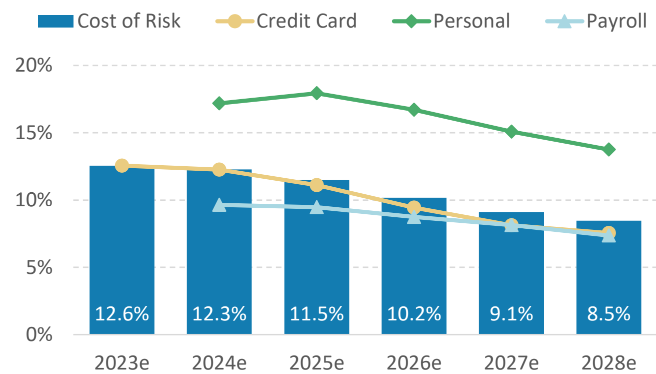


Source: Company Data, Morgan Stanley Research Estimates (e).

ii) Loan-loss provisions. We expect cost of risk will improve at a later stage relative to Brazil, as new vintages with higher provisioning should make up most of the total portfolio over the next two years. Our long-term forecasts trend toward, but are slightly above, the historical industry average for each product. For instance, we forecast the cost of risk of credit cards will reach 7.5% in 2028, compared to 6.9% from our unit economics model. And, even though it is not fully comparable to the cost of risk, we note that the average 90-day NPL ratio of credit cards was only 5.2% over the past 10 years.

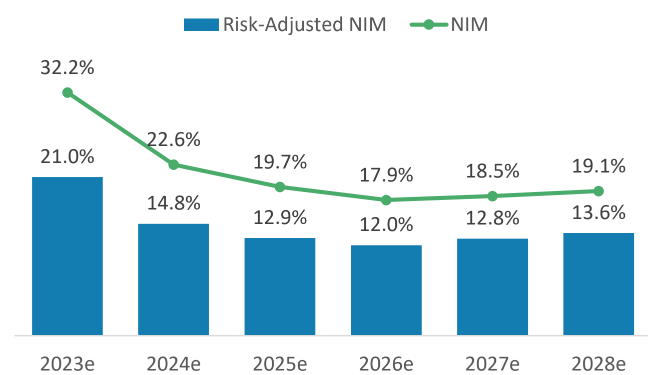
NIM should normalize as deposits ramp-up over the next three years, followed by reasonable expansion. We see risk-adjusted NIM normalizing around 12% in 2026, as deposits continue to build up. And gradually expanding going forward, on the back of loan growth outpacing deposits, shift in loan mix to higher margin less rate-sensitive products, and a normalization in the cost of risk.

Exhibit 130: Cost of risk should improve at a later stage, as the business continues to ramp-up in the next 2 years.



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 131: We expect a normalization in margins as deposits build-up, followed by a gradual expansion beyond 2026.



Source: Company Data, Morgan Stanley Research Estimates (e). Note: Net interest income only considers interest expense from deposits.

We forecast revenues for each product coupling our unit economics models with active customer growth and product penetration. We provide key performance indicators of each product in the summary tables further below.

Credit Card KPIs

Exhibit 132: Mexico — Key Drivers of Credit Cards.

Credit Cards	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)	2	4	6	8	10	12	15	33%
Penetration of Active Customers	95%	88%	80%	60%	55%	55%	55%	
Revenue (US$ BN)	0.2	0.4	0.7	1.1	1.4	2.0	2.7	48%
% of Mexico Total	100%	97%	82%	71%	61%	56%	53%	
Monthly ARPAC (US$)	8.9	11.1	11.6	12.8	13.5	14.8	16.5	8%
Cost of Risk	NA	12.6%	12.3%	11.1%	9.4%	8.1%	7.5%	
Loan Portfolio (US$ BN)	0.7	1.1	2.2	3.1	4.2	6.0	7.8	37%
Avg. Loan per Active Customer (US$)	295	328	365	406	447	493	543	10%
Implicit Market Share	3%	4%	6%	8%	10%	12%	14%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Personal Loan KPIs

Exhibit 133: Mexico — Key Drivers of Personal Loans.

Personal Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	4 Yr CAGR
Active Customers (MM)	-	-	0.2	0.7	1.2	1.9	2.8	85%
Penetration of Active Customers	-	-	3%	5%	7%	9%	10%	
Revenue (US$ BN)	-	-	0.0	0.1	0.2	0.4	0.7	167%
% of Mexico Total	-	-	2%	6%	10%	12%	14%	
Monthly ARPAC (US$)	-	-	12.9	18.4	20.7	23.2	25.6	19%
Cost of Risk	-	-	17.2%	17.9%	16.7%	15.1%	13.8%	
Loan Portfolio (US$ BN)	-	-	0.1	0.4	0.9	1.6	2.6	116%
Avg. Loan per Active Customer (US$)	-	-	619	710	799	882	955	11%
Implicit Market Share	-	-	1%	3%	5%	9%	12%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Payroll Loan KPIs

Exhibit 134: Mexico — Key Drivers of Payroll Loans.

Payroll Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	4 Yr CAGR
Active Customers (MM)	-	-	0.1	0.2	0.5	1.0	1.4	110%
Penetration of Active Customers	-	-	1%	2%	3%	4%	5%	
Revenue (US$ BN)	-	-	0.0	0.1	0.2	0.3	0.6	166%
% of Mexico Total	-	-	1%	4%	7%	10%	11%	
Monthly ARPAC (US$)	-	-	31	34	36	38	41	7%
Cost of Risk	-	-	9.6%	9.5%	8.7%	8.1%	7.4%	
Loan Portfolio (US$ BN)	-	-	0.1	0.5	1.1	2.2	3.4	128%
Avg. Loan per Active Customer (US$)	-	-	2,076	2,182	2,293	2,410	2,532	5%
Implicit Market Share	-	-	1%	2%	4%	7%	9%	

Source: Company Data, Morgan Stanley Research Estimates (e).

BNPL, Investments, & Insurance KPIs

Exhibit 135: Mexico — Key Drivers of BNPL, Investments, and Insurance.

Other Prodcuts	2022	2023e	2024e	2025e	2026e	2027e	2028e	4 Yr CAGR
Active Customers (MM)								
BNPL	-	-	0.2	0.9	2.0	3.4	5.1	115%
Investments	-	0.0	0.4	1.1	2.1	3.5	5.1	93%
Insurance	-	-	0.2	0.6	1.3	2.0	2.9	87%
Penetration of Active Customers								
BNPL	-	-	3%	7%	11%	15%	19%	
Investments	-	1%	5%	8%	12%	15%	19%	
Insurance	-	-	3%	5%	7%	9%	11%	
Revenue (US$ BN)	-	-	0.0	0.0	0.1	0.2	0.4	136%
BNPL	-	-	0.0	0.0	0.1	0.1	0.2	149%
Investments	-	-	0.0	0.0	0.0	0.0	0.1	125%
Insurance	-	-	0.0	0.0	0.0	0.0	0.1	113%
Revenue Mix, % of Mexico Total	-	-	1%	1%	4%	5%	7%	
BNPL	-	-	1%	1%	2%	3%	4%	
Investments	-	-	0%	0%	1%	1%	1%	
Insurance	-	-	0%	0%	1%	1%	1%	
Monthly ARPAC (US$)								
BNPL	-	-	5.4	5.9	6.1	6.2	6.3	4%
Investments	-	-	1.3	1.6	1.8	1.9	2.0	12%
Insurance	-	-	1.9	2.0	2.2	2.4	2.5	8%
Implicit Market Share								
BNPL	-	-	0%	2%	5%	8%	11%	
Investments	-	-	0%	0%	1%	1%	2%	
Insurance	-	-	0%	0%	1%	1%	2%	

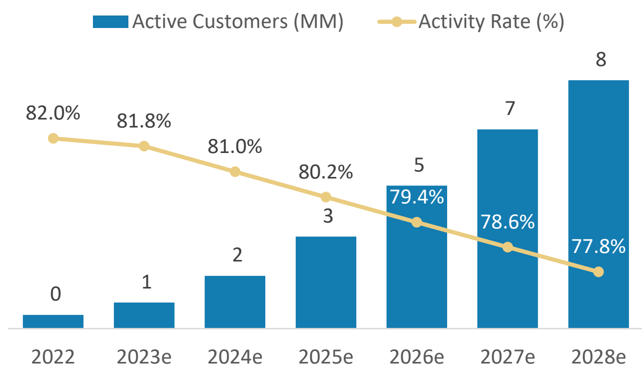
Source: Company Data, Morgan Stanley Research Estimates (e).

COLOMBIA: KEY DRIVERS

Our revenue forecast is backed by active customer growth and product penetration assumptions:

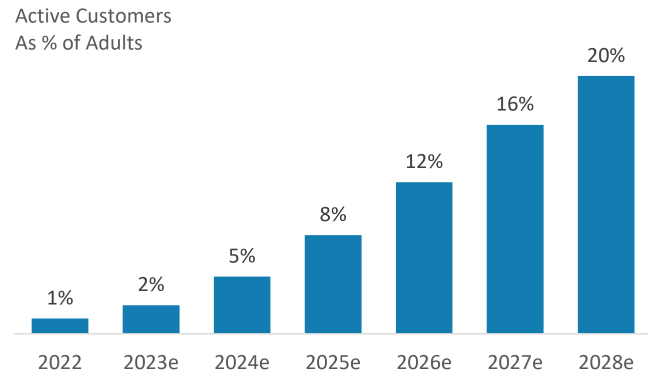
i) Active Customers. Similar to Mexico, we take a conservative approach and assume the activity rate falls to levels in-line with Brazil. As such, we think, Nubank will capture ~8 million active clients in Colombia, or ~20% of the adult population, over the next five years. In terms of net adds, our numbers for 2024 and beyond look reasonable — between 1 and 2 million per year.

Exhibit 136: We expect strong client growth in Colombia, despite lower activity rate.



Source: Company Data, Morgan Stanley Research Estimates (e).

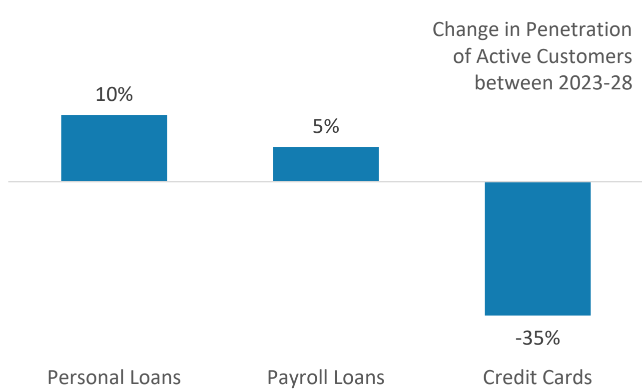
Exhibit 137: We estimate 8 million active clients, or 20% of the adult population, by 2028.



Source: Company Data, Morgan Stanley Research Estimates (e).

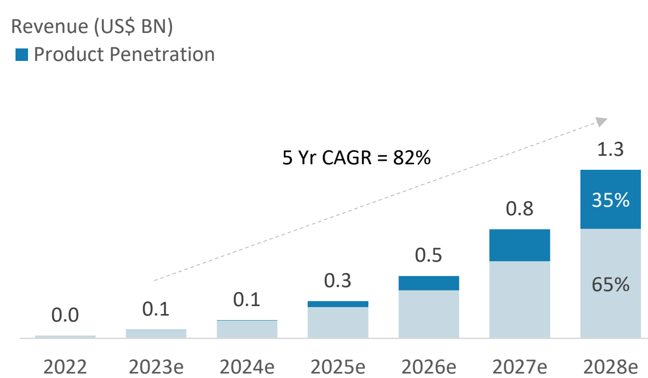
ii) Product Penetration. Penetration of credit cards among active clients in Colombia is very high — likely over 95% — because it's still the only product offered. So we assume credit card penetration declines to 55% in 2028, similar to current levels in Brazil. In addition, we think, the company will roll out personal and payroll loans in mid-to-late 2024. And we see penetration of active clients reaching 10% for personal loans and 5% payroll loans in 2028, driving attractive long-term revenue growth. Indeed, we think penetration could add 35% more revenue in Colombia by 2028.

Exhibit 138: Penetration of personal/payroll loans should ramp-up at a later stage...



Source: Company Data, Morgan Stanley Research Estimates (e).

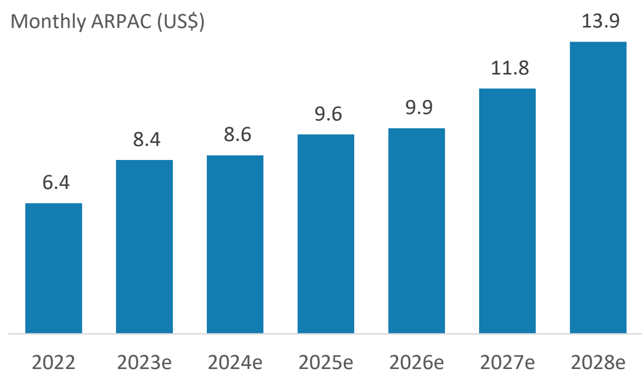
Exhibit 139: ... still adding a meaningful amount of revenue to our 2028 forecast.



Source: Company Data, Morgan Stanley Research Estimates (e).

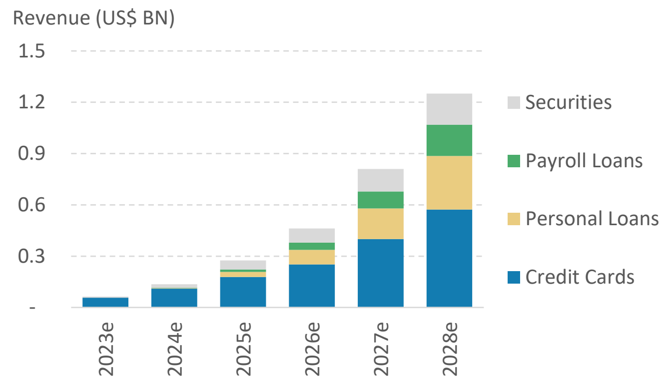
We see revenue in Colombia growing at five-year CAGR of 82%, reaching US$1.3 billion in 2028. This means Colombia alone could add ~5% more revenue to the company in the next five years. Our forecast implies a monthly ARPAC of ~US$14 in 2028, up from ~US$8 currently, mostly due to strong client growth, cross-selling new products, and higher credit limits of payroll loans, personal loans, and credit cards.

Exhibit 140: We assume the ARPAC in Colombia grows at a faster pace in 2027...



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 141: ... as higher penetration of payroll and personal loans kicks in.



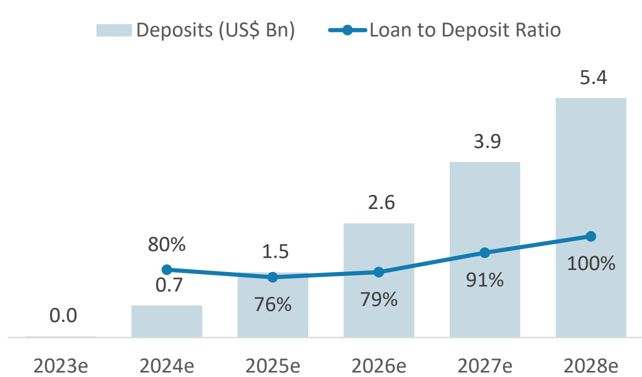
Source: Company Data, Morgan Stanley Research Estimates (e).

On the cost side, we forecast the following for financial expenses and loan-loss provisions:

i) Financial expenses on deposits. According to news reports, the company is expected to launch its savings account later this year or early in 2024. As such, we see deposits growing at a four-year CAGR of 193%, from US$0.7 billion in 2024 to US$5.4 billion in 2028. We assume 90% of active customers have a savings account in this period and that the average deposit per account almost doubles from ~US$400 in 2024 to ~US$700 in 2028. We also assume the remuneration on deposits as a percentage of the benchmark interest rate remains stable around 49% going forward. And that the cost of funding improves from ~4% in 2024 to ~3% in 2028, as the average interest rate comes down from ~9% to ~6%.

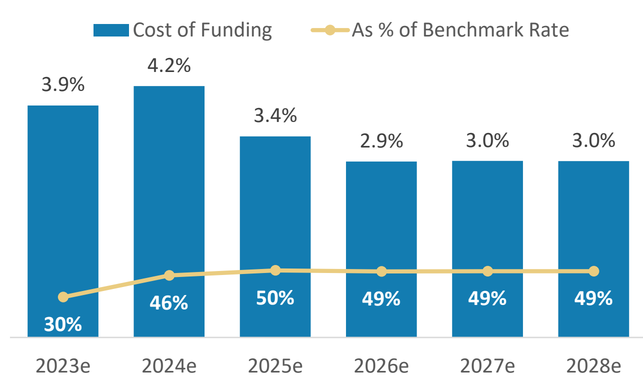
Importantly, our deposit forecast implies higher leverage — loans to deposit ratio goes from 80% in 2024 to 100% in 2028, below the industry historical levels of ~115%, on average, over the past 10 years. Similar to Mexico, we estimate leverage is lower relative to Brazil because the credit card business requires more funding/deposits.

Exhibit 142: Loan growth should outpace deposits, increasing leverage but remaining below levels in Brazil on higher funding needs.



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 143: Cost of funding should improve as remuneration of deposits remains stable at 49% and the benchmark interest rate falls.

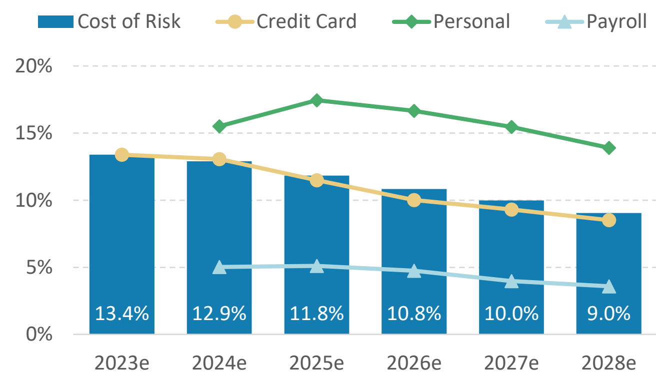


Source: Company Data, Morgan Stanley Research Estimates (e).

ii) Loan-loss provisions. We expect cost of risk will improve gradually and at a later stage, as new vintages with higher provisioning should make up most of the total portfolio over the next two years. Our long-term forecasts trend toward, but are slightly above, the historical industry average for each product. For instance, we forecast the cost of risk of credit cards will reach 8.5% in 2028, while the industry average 90-day NPL ratio of credit cards was ~6% over the past 10 years.

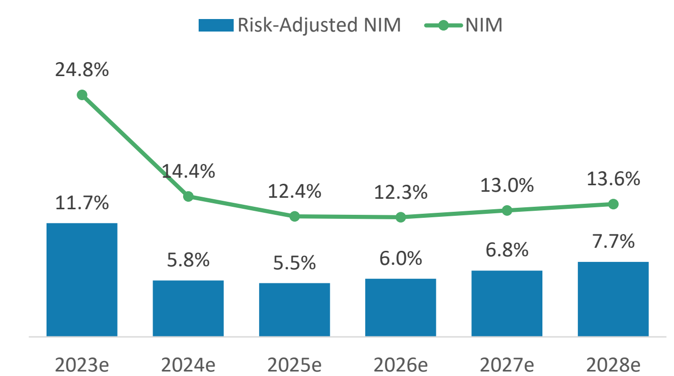
NIM should normalize as deposits ramp-up over the next three years, followed by reasonable expansion. We see risk-adjusted NIM normalizing to around 5.5% in 2025, as deposits continue to build up. And we expect a gradual expansion going forward, on the back of loan growth outpacing deposits, shift in loan mix to higher margin less rate sensitive products, and a normalization in the cost of risk.

Exhibit 144: Cost of risk should improve at a later stage, as the business continues to ramp-up in the next 2 years.



Source: Company Data, Morgan Stanley Research Estimates (e).

Exhibit 145: We expect a normalization in margins as deposits build-up, followed by a gradual expansion beyond 2026.



Source: Company Data, Morgan Stanley Research Estimates (e). Note: Net interest income only considers interest expense from deposits.

We forecast revenues for each product by coupling our unit economics models with active customer growth and product penetration. We provide key performance indicators of each product in the summary tables further below.

Credit Card KPIs

Exhibit 146: Colombia — Key Drivers of Credit Cards.

Credit Cards	2022	2023e	2024e	2025e	2026e	2027e	2028e	5 Yr CAGR
Active Customers (MM)	0	1	1	2	3	4	5	42%
Penetration of Active Customers	95%	90%	80%	60%	55%	55%	55%	
Revenue (US$ BN)	0.0	0.1	0.1	0.2	0.3	0.4	0.6	59%
% of Colombia Total	96%	90%	81%	65%	55%	49%	46%	
Monthly ARPAC (US$)	6.5	8.1	8.4	9.0	9.7	10.6	11.6	7%
Cost of Risk	NA	13.4%	13.1%	11.5%	10.0%	9.3%	8.5%	
Loan Portfolio (US$ BN)	0.6	0.7	0.7	0.8	0.9	1.0	1.1	9%
Avg. Loan per Active Customer (US$)	315	343	375	409	446	486	530	9%
Implicit Market Share	2%	3%	5%	6%	8%	11%	14%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Personal Loan KPIs

Exhibit 147: Colombia — Key Drivers of Personal Loans.

Personal Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	4 Yr CAGR
Active Customers (MM)	-	-	0.0	0.1	0.3	0.5	0.8	120%
Penetration of Active Customers	-	-	2%	5%	6%	8%	10%	
Revenue (US$ BN)	-	-	0.0	0.0	0.1	0.2	0.3	233%
% of Colombia Total	-	-	2%	11%	18%	22%	25%	
Monthly ARPAC (US$)	-	-	18.6	27.6	32.0	35.2	38.2	20%
Cost of Risk	-	-	15.5%	17.4%	16.7%	15.5%	13.9%	
Loan Portfolio (US$ BN)	-	-	0.0	0.2	0.5	1.0	1.7	155%
Avg. Loan per Active Customer (US$)	-	-	1,591	1,757	1,901	2,038	2,163	8%
Implicit Market Share	-	-	0%	1%	2%	4%	6%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Payroll Loan KPIs

Exhibit 148: Colombia — Key Drivers of Payroll Loans.

Payroll Loans	2022	2023e	2024e	2025e	2026e	2027e	2028e	4 Yr CAGR
Active Customers (MM)	-	-	0.0	0.1	0.1	0.3	0.4	128%
Penetration of Active Customers	-	-	1%	2%	3%	4%	5%	
Revenue (US$ BN)	-	-	0.0	0.0	0.0	0.1	0.2	208%
% of Colombia Total	-	-	1%	5%	9%	12%	15%	
Monthly ARPAC (US$)	-	-	26.5	33.6	37.0	40.1	43.2	13%
Cost of Risk	-	-	5.0%	5.1%	4.7%	4.0%	3.6%	
Loan Portfolio (US$ BN)	-	-	0.0	0.1	0.4	0.8	1.3	148%
Avg. Loan per Active Customer (US$)	-	-	2,563	2,693	2,831	2,975	3,126	5%
Implicit Market Share	-	-	0%	1%	2%	3%	5%	

Source: Company Data, Morgan Stanley Research Estimates (e).

Bull & Bear Scenarios

Our bull/bear scenarios provide more visibility on the potential growth paths given the early state of product penetration in Brazil and the business in Mexico and Colombia. Indeed, we think our bull/bear scenarios around our base case are value added, as they are built to consider multiple growth drivers and a comprehensive number of risks such as, a potential inability to appropriately price for credit risk and future pressure on lending spreads, among other things.

We gain greater conviction based on our work in this report; hence, we are moving our base case scenario closer to our bull case. Indeed, our 2028 bull case net income of US$11 billion is ~30% above our base case of US$8.8 billion, while our bear case net income of US$5.4 billion is ~40% below our base case.

Exhibit 149: Key Outcomes of Our Bull, Bear, and Base Scenarios.

	2022	2023e	2024e	2025e	2026e	2027e	2028e
Revenue (US$MM)							
Base	4,792	7,840	11,032	14,969	19,298	24,787	31,059
Bear	4,792	7,804	10,473	13,411	16,261	19,551	23,106
Bull	4,792	7,864	11,377	16,101	21,832	28,896	37,074
Gross Profit Margin							
Base	35%	42%	47%	51%	53%	56%	58%
Bear	35%	42%	46%	48%	50%	51%	52%
Bull	35%	43%	48%	53%	55%	58%	60%
Net Income (US$MM)							
Base	(9)	916	1,752	2,856	4,096	5,890	8,070
Bear	(9)	896	1,556	2,260	2,956	3,837	4,876
Bull	(9)	961	1,936	3,325	5,052	7,421	10,357
Adjusted Net Income (US$MM)							
Base	204	1,148	2,117	3,319	4,649	6,541	8,812
Bear	204	1,127	1,904	2,675	3,421	4,349	5,427
Bull	204	1,195	2,313	3,824	5,678	8,181	11,244
Adjusted ROE							
Base	4%	21%	30%	36%	36%	37%	36%
Bear	4%	20%	27%	30%	30%	29%	28%
Bull	4%	21%	32%	39%	41%	41%	39%

Source: Company Data, Morgan Stanley Research Estimates (e)

Exhibit 150: Key Drivers of Our Bull, Bear, and Base Scenarios.

	2023e	2028e	2023e	2028e	2023e	2028e	2023e	2028e
Active Customers								
Base	71	90	4	27	1	8	76	125
Bear	70	82	4	16	1	4	75	102
Bull	72	94	4	37	1	12	77	143
Active Customers as % of Adults								
Base	44%	53%	4%	25%	2%	20%	25%	40%
Bear	43%	48%	4%	15%	2%	10%	25%	32%
Bull	44%	55%	4%	35%	2%	30%	26%	45%
Penetration of Credit Cards								
Base	53%	60%	88%	55%	90%	55%	55%	58%
Bear	53%	55%	88%	55%	90%	55%	55%	55%
Bull	53%	60%	88%	60%	90%	60%	55%	60%
Penetration of Personal Loans								
Base	11%	16%	0%	10%	0%	10%	10%	14%
Bear	10%	12%	0%	6%	0%	6%	10%	11%
Bull	11%	18%	0%	13%	0%	13%	10%	16%
Penetration of Payroll Loans								
Base	1%	8%	0%	5%	0%	5%	1%	8%
Bear	1%	6%	0%	3%	0%	3%	1%	6%
Bull	1%	10%	0%	7%	0%	7%	1%	9%
Monthly ARPAC (US$)								
Base	9.3	23.0	10.4	17.2	8.4	13.9	9.5	21.3
Bear	9.3	20.0	10.5	15.5	8.7	11.3	9.5	19.1
Bull	9.3	24.4	10.4	19.5	8.3	16.2	9.5	22.6
Cost of Risk								
Base	16.4%	9.7%	12.6%	8.5%	13.4%	9.0%	16.2%	9.5%
Bear	16.7%	11.6%	12.8%	10.4%	13.5%	11.1%	16.3%	11.4%
Bull	15.7%	9.0%	12.3%	8.1%	13.1%	8.7%	15.8%	8.8%

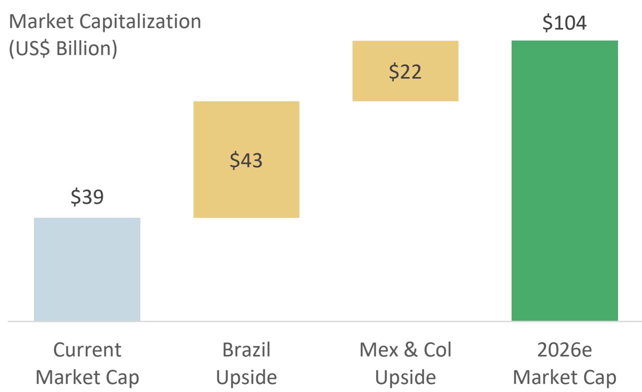
Source: Morgan Stanley Research Estimates (e)

The Math to US$100 Billion

We believe Nubank could reach over US$100 billion valuation by 2026. Increasing product penetration in Brazil and success in Mexico and Colombia could unlock US$65 billion in market cap. Our new product-by-product and country-by-country model suggests higher product penetration in Brazil — mainly personal, payroll, and BNPL loans — and successful customer growth and product penetration in Mexico and Colombia could add an additional US$65 billion in market cap. As mentioned before, we are gradually increasing our earnings estimates on the back of our new bottom-up model, starting with 7% in 2023 and as much as 46% by 2028.

We are rolling over our price target to year-end 2024 and increasing it to US$16 per share, up from US$11 before. Our valuation work and price target are based on residual income analysis. In our residual income model, the value of the firm is mainly a function of the magnitude of the spread between return on equity and cost of equity (performance spread) and the time period in which management can generate returns in excess of the cost of equity (economic growth horizon). Our valuation model uses a discount rate of 14.2%, long-term ROE of ~35%, and economic growth horizon of 20 years.

Exhibit 1: Increasing penetration in Brazil and the rise of Mexico and Colombia...



Source: Morgan Stanley Research Estimates (e).

So, how do we get to US$100 billion market cap by 2026? We use our discount rate to push forward our 2024 PT of US$16 to year-end 2026 — implying Nubank's fair value will reach US$104 billion, or US$21 per share. Relative to our 2026 EPS estimate of U$0.93, this implies P/E of 22.5x. A fair multiple, in our view, compared to where global profitable mature tech-platforms currently trade, e.g., Google 24x forward P/E, Visa 27x, and Amazon 55x.

For Brazil, we get to a 2026 value of US$82 billion or US$16.4 per share. We use our discount rate to push forward the NPV from our previous standalone Brazil model to year-end 2026 — implying a fair value of US$82 billion. In other words, we think the cross-selling opportunity in Brazil could unlock US$43 billion in market cap.

For Mexico and Colombia, we get to a 2026 value of US$22 billion, or US$4.5 per share. We calculate the fair value of Mexico and Colombia as the difference between the value of the new consolidated model (Brazil + Mexico + Colombia) and the value of our previous standalone Brazil model. Importantly, we are not explicitly modeling the expenses of the new countries' operations, given the lack of disclosure and because some will be shared with the headquarters. However, we think this is a good enough proxy.

Exhibit 2: ... can make Nubank a US$100 billion market cap stock by 2026.

Discount Rate	14.2%

US$	2024e	2025e	2026e	2027e	2028e
Fair Value / Share	16.0	18.3	20.9	23.9	27.3
EPS	0.42	0.66	0.93	1.31	1.76
P/E	37.8x	27.5x	22.5x	18.3x	15.5x
Market Cap ($Bn)	80	91	**104**	119	136

Source: Morgan Stanley Research Estimates (e).

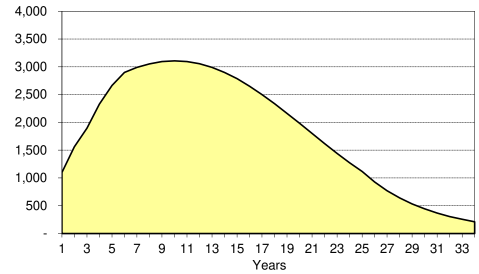
Exhibit 3: Residual Income Valuation Model.

Assumptions

Per Share Information	
Expected 12 Month Dividend	-
Current Stock Price	7.72
Issued Shares (million)	4,996
Reported Book Value Per Share	1.13
Cost of Capital	
10-yr. Govt. Bond Yield (%)	4.40
Beta	1.30
Market Risk Premium (%)	6.00
Country Risk Premium (%)	2.00
Cost of Common Equity (%)	14.20
Maturity Phase	
Number of Years	20
Target Capital Ratio (%)	24.0
End of Phase RWA Growth Rate (%)	6.0
Decline Phase	
Decay Factor (0-10%)	5.0
Growth in RWA (%)	5.0
Goodwill	
Current Goodwill	-
BV Adjustment	-

Valuation Analysis

Present Value of Residual Income ($)



Key Valuation Measures

Current Stock Price	7.72
Est. Current Fair Value ($)	14.08
12 Month Target Price ($)	16.08
12 Month Total Return (%)	108.3
Price/Book Value	6.8
Price/Economic Book Value	6.0
2024E Return on Economic Equity (%)	31.2
Cost of Common Equity (%)	14.2
Current Operations Value™/Est. Fair Value (%)	22

Valuation Drivers

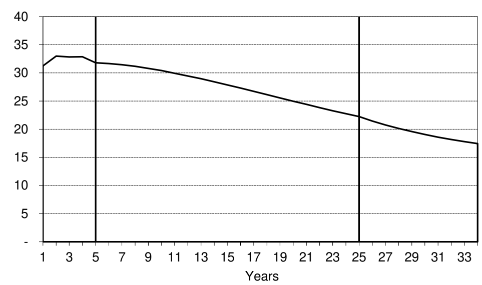
		CAGR (%)		End of Phase		% of Fair Value	
Phase	Years	Operating Earnings	RWA	ROEE (%)	Payout Ratio (%)	Phase	Cumulative
Economic BV	-	-	-	-	-	9.2	9.2
Development	5	42.1	27.9	31.8	0.0	13.6	22.7
Maturity	20	11.7	13.7	22.2	74.1	69.4	92.2
Decline	11	2.4	5.0	16.9	71.2	7.8	100.0

Valuation Parameters

	Est. Current Fair Value as a Multiple of			
	Bk Val	Econ Bk	Earnings	Op. Earn
2024	8.9	6.8	40.2	31.9
2025	6.5	4.8	24.6	20.5
2026	4.7	3.5	17.2	14.7
2027	3.4	2.5	11.9	10.5
2028	2.4	2.0	8.7	7.8

Return on Economic Equity

ROEE (%)



Distribution of Forecasted ROEE

ROEE Frequency (%)	ROEE Threshold (%)
75	25
65	26
55	28
50	28
45	29
35	30
25	31

Source: Morgan Stanley Research Estimates (e).

Latam Comps

Exhibit 154: Latam Financial Institutions Trading Comparables.

LatAm Banks - Summary Comps

		MS	Price	Price	Upside		EPS			P / E			ROE (%)	Price Performance in US$ (%)								52-week		Market Cap	Daily Vol. (US$ MM)		Div. Yield
	Coverage	Rating	17-Oct-23	Target	(%)	P/BV	22	23e	24e	22	23e	24e	23e	1d	1w	1m	3m	6m	12m	YTD	2022	High	Low	(US$ MM)	1-Year	1-Month	23e (%)
ARGENTINA																											
BBVA Argentina	$/ADR	U	4.3	UR	--	0.6x	1.81	1.84	1.09	2.4x	2.3x	4.0x	15.9	3	4	(14)	(21)	(1)	52	17	34	6.2	2.4	880	3	3	10.5
GF Galicia	$/ADR	U	12.5	UR	--	0.7x	1.34	3.13	1.02	9.4x	4.0x	12.2x	10.8	(1)	4	(21)	(25)	13	82	39	3	18.5	6.5	1,842	12	16	1.6
Macro	$/ADR	U	19.5	UR	--	0.6x	3.77	7.24	3.31	5.2x	2.7x	5.9x	14.0	3	7	(16)	(18)	15	48	30	18	27.6	11.2	1,248	6	5	5.8
Supervielle	$/ADR	U	2.1	UR	--	0.4x	(0.40)	0.27	0.27	NA	7.5x	7.7x	4.0	(0)	4	(21)	(34)	(13)	13	(5)	11	3.6	1.5	182	1	1	(1.9)
Argentine Financials														1	5	(18)	(22)	10	62	30	14			4,152	23	26	4.6
MSCI Argentina			3,652											4	8	(2)	(11)	17	63	28	32	4,292	2,194				
BRAZIL																											
Banco do Brasil	R$/ON	O	49.4	81.0	64	0.8x	11.13	12.46	13.13	4.4x	4.0x	3.8x	20.4	(1)	2	0	1	19	46	61	42	50.8	28.0	28,053	112	86	9.0
Bradesco	$/ADR	O	2.9	5.1	76	1.0x	0.38	0.36	0.56	7.6x	7.9x	5.2x	11.7	(2)	-	(6)	(17)	7	(23)	0	(7)	4.0	2.3	30,756	231	131	5.9
Itaú Unibanco	$/ADR	E	5.4	6.8	25	1.5x	0.61	0.72	0.76	8.9x	7.6x	7.2x	19.6	(1)	0	(4)	(6)	8	-	15	26	6.2	4.2	53,116	320	201	5.1
Santander Brasil	$/ADR	O	5.3	8.0	51	1.2x	0.67	0.55	0.79	7.9x	9.6x	6.7x	11.8	(2)	(3)	(2)	(14)	(1)	(8)	(2)	0	6.6	4.8	19,871	17	13	6.4
Banrisul	R$/PNB	NC	11.3	--	--	0.5x	1.88	2.37	2.97	6.0x	4.8x	3.8x	N/A	(1)	(2)	(12)	(26)	14	12	32	15	15.0	8.4	918	4	4	-
Banco BTG Pactual	R$/ON	O	30.2	30.0	(1)	2.5x	2.05	2.47	2.76	14.8x	12.2x	11.0x	18.7	(2)	(3)	(11)	(12)	35	22	35	24	34.1	17.7	22,860	66	53	1.7
Banco Pan	R$/ON	O	7.7	9.0	17	1.2x	0.54	0.56	0.70	14.3x	13.9x	11.1x	8.7	(4)	(2)	(17)	(20)	50	15	33	(37)	9.7	4.6	1,950	4	3	2.5
Nubank	$/O	O	7.7	16.0	107	6.8x	0.04	0.23	0.42	185.4x	33.6x	18.2x	14.8	(2)	2	5	1	58	76	90	(57)	6.6	3.0	38,570	154	150	-
Inter & Co	$/O	U	4.1	2.5	(39)	1.2x	(0.01)	0.13	0.20	NM	32.7x	20.1x	3.4	(2)	(3)	(5)	20	147	36	73	N/A	4.6	1.4	1,648	8	9	(0.0)
XP Inc	$/O	O	21.4	33.0	54	3.1x	1.38	1.58	2.04	15.5x	13.5x	10.5x	20.9	(1)	(4)	(19)	(10)	58	20	39	(47)	27.7	10.3	11,365	101	102	0.3
B3	R$/ON	E	11.3	14.5	29	3.2x	0.73	0.76	0.89	15.5x	14.8x	12.6x	23.0	(4)	(6)	(17)	(26)	(2)	(14)	(8)	30	15.5	10.0	12,902	110	85	8.1
Stone	$/O	E	10.0	13.0	29	1.3x	0.25	0.80	0.94	40.9x	12.6x	10.7x	8.6	3	(3)	(12)	(18)	(13)	(6)	6	(44)	14.8	8.1	3,423	58	47	-
PagSeguro	$/O	E	7.8	11.0	41	1.0x	0.94	1.05	1.14	8.3x	7.4x	6.9x	12.0	1	(10)	(13)	(19)	(14)	(45)	(11)	(67)	15.0	7.5	2,539	41	32	-
Cielo	R$/ON	E	3.5	6.0	69	0.8x	0.58	0.81	0.79	6.1x	4.3x	4.5x	16.1	(4)	(1)	(4)	(28)	(28)	(31)	(27)	155	5.7	3.2	1,893	28	23	7.4
Porto Seguro	R$/ON	O	25.8	33.0	28	1.5x	1.76	3.04	3.43	14.6x	8.5x	7.5x	17.1	(1)	(2)	(8)	(13)	9	26	21	23	29.4	19.8	3,308	10	6	3.4
BB Seguridade	R$/ON	U	31.6	30.0	(5)	7.8x	3.03	3.64	3.98	10.4x	8.7x	7.9x	60.7	(0)	0	(4)	5	(4)	31	9	84	34.5	25.1	12,519	35	27	8.5
Caixa Seguridade	R$/ON	O	11.2	14.0	25	2.8x	0.98	1.14	1.28	11.4x	9.8x	8.7x	55.2	(1)	(0)	(6)	6	22	47	49	14	11.6	7.0	6,676	5	6	7.7
IRB	R$/ON	NC	39.8	--	--	0.8x	(9.31)	2.38	4.12	NA	16.7x	9.7x	N/A	(1)	(9)	(5)	(21)	53	31	62	(77)	55.5	18.3	645	16	16	-
Brazil Financials														(1)	(1)	(5)	(8)	21	18	31	6			253,010	1,321	992	4.6
MSCI Brazil			1,547											0	1	(5)	(6)	8	(1)	6	2	2,429	1,036				
CHILE																											
Banco de Chile	$/ADR	E	20.1	25.0	24	2.0x	3.16	2.74	2.39	6.4x	7.4x	8.4x	22.1	1	1	(5)	(9)	(1)	10	(3)	33	23.5	17.3	10,162	12	8	11.0
Santander Chile	$/ADR	E	17.5	23.0	31	1.9x	1.95	1.58	2.16	9.0x	11.1x	8.1x	13.3	(1)	(3)	(8)	(16)	(2)	23	10	(3)	21.8	13.6	8,245	15	11	6.7
Chile Financials														1	(1)	(6)	(12)	(1)	16	3	17			18,407	26	19	9.1
MSCI Chile			998											1	1	(7)	(20)	(15)	(4)	(10)	15	1,548	698				
MEXICO																											
Banorte	M$/O	O	148.4	200.0	35	1.8x	15.75	18.47	20.43	9.4x	8.0x	7.3x	20.0	(2)	(2)	(5)	(8)	(3)	7	15	11	167.8	135.0	23,748	49	44	6.4
Banco del Bajio	M$/O	E	54.7	69.0	26	1.7x	6.78	9.35	8.28	8.1x	5.8x	6.6x	29.0	(2)	(3)	(7)	(10)	(10)	7	(4)	76	77.7	48.9	3,612	9	6	6.2
Banregio	M$/O	NC	130.3	--	--	1.6x	14.77	16.59	18.14	8.8x	7.9x	7.2x	N/A	(2)	(4)	0	(8)	(2)	12	1	39	170.6	118.7	2,372	5	6	-
Gentera	M$/O	NC	21.2	--	--	1.4x	2.85	3.14	3.59	7.4x	6.8x	5.9x	N/A	(1)	(5)	(2)	7	2	28	5	75	24.0	17.0	1,862	4	2	-
Bolsa Mexicana	M$/O	NC	32.2	--	--	2.7x	2.85	2.75	2.92	11.3x	11.7x	11.0x	N/A	(3)	(5)	(11)	(14)	(18)	4	(8)	2	41.5	32.0	1,029	3	3	-
Inbursa	M$/O	NC	36.5	--	--	1.1x	3.43	4.39	4.46	10.6x	8.3x	8.2x	N/A	(1)	0	(4)	(20)	(9)	13	20	41	45.2	32.1	12,390	4	3	-
Mexico Financials														(2)	(2)	(4)	(11)	(5)	10	13	28			45,014	74	64	3.9
MSCI Mexico			5,713											(0)	(0)	(7)	(14)	(9)	18	10	(5)	5,085	2,733				
OTHER																											
dLocal	$/O	O	18.5	25.0	35	14.6x	0.35	0.56	0.86	53.2x	32.9x	21.6x	36.0	(1)	(5)	(9)	28	36	(10)	19	(56)	24.7	9.0	5,683	38	17	-
Credicorp	$/ADR	E	126.2	158.0	25	1.3x	15.21	17.09	18.62	8.3x	7.4x	6.8x	15.4	(1)	(0)	(7)	(19)	(6)	(3)	(7)	11	160.2	120.8	10,038	48	29	4.2
Intercorp	$/ADR	NC	20.5	--	--	1.0x	3.76	3.03	4.02	5.4x	6.8x	5.1x	N/A	-	(2)	(15)	(18)	(14)	(8)	(13)	(11)	29.5	20.4	2,366	1	1	-
Bancolombia	$/ADR	E	25.3	37.0	46	0.7x	6.55	5.79	5.75	3.9x	4.4x	4.4x	16.5	1	(3)	(8)	(16)	(10)	(2)	(11)	(10)	30.9	21.6	6,079	11	9	12.6
Grupo Aval	$/ADR	E	2.4	3.0	23	0.7x	0.53	0.29	0.36	4.6x	8.3x	6.8x	10.5	0	(0)	(2)	(8)	(6)	0	(4)	(50)	2.7	2.0	2,897	0	0	12.2
Davivienda	Co$/PFD	NC	16,500	--	--	0.5x	3,527	3,607	4,631	4.7x	4.6x	3.6x	N/A	1	3	(2)	(23)	(17)	(40)	(31)	(27)	31,300	15,680	1,779	0	0	-
Grupo Sura	Co$/PFD	NC	12,440	--	--	0.2x	2,857	4,174	4,787	4.4x	3.0x	2.6x	N/A	1	4	10	(13)	17	12	9	(50)	15,490	9,600	1,720	0	0	-
Bladex	$/ADR	NC	22.5	--	--	0.7x	2.33	4.31	4.37	9.6x	5.2x	5.1x	N/A	1	3	(4)	0	20	59	39	(2)	24.7	14.1	820	2	2	-
Other Latam Financials														0	(1)	(7)	(8)	2	(4)	(3)	(18)			31,380	101	58	4.9
MSCI LatAm			2,252											0	0	(6)	(9)	0	4	6	(0)	2,984	1,383				
Latin America Financials														(1)	(1)	(5)	(8)	14	16	24	8			351,963	1,545	1,158	4.8

Source: Morgan Stanley Research. Nubank price as of October 18, 2023. Note (1): O = Overweight E = Equal-weight U = Underweight NR = Not rated NA = Not available UR = Under Review neg = Negative NM = Not Meaningful NC = Not Covered. Morgan Stanley Estimates for Banco do Brasil, Bradesco, Itaú, Santander Brasil, BTG Pactual, Banco Pan, Nubank, Inter & Co, Boa Vista, XP Inc, B3, Stone, PagSeguro, Cielo, Porto Seguro, BB Seguridade, Caixa Seguridade, Banco de Chile, Santander Chile, Banorte, Banco del Bajio, dLocal, Credicorp, Banco Macro, GF Galicia, Banco Frances, Supervielle, Bancolombia, and Grupo Aval. All other from Thomson Reuters. Country samples are market cap. weighted. Average Trading Volumes are combined total for ADRs plus Local Shares. Note(2): ++ Data for this Company have been removed from consideration in this report because, under applicable law and/or Morgan Stanley policy, Morgan Stanley may be precluded from issuing such information with respect to this company at this time. Note (3): EPS and P/E for Nubank, XP Inc, PagSeguro, and Stone are based on adjusted numbers. Note (4): P/BV and P/E of Argentine banks is based on local shares. Please note that all important disclosures including personal holdings disclosures and Morgan Stanley disclosures appear on the Morgan Stanley public website at www.morganstanley.com/researchdisclosures.

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(as of September 30, 2023)

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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1350	37%	281	43%	21%	608	39%
Equal-weight/Hold	1673	46%	303	46%	18%	709	46%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	608	17%	68	10%	11%	225	15%
Total	3,634		652			1543	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index, on a risk-adjusted basis over the next 12-18 months.
Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index, on a risk-adjusted basis over the next 12-18 months.
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Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Stock Rating History: 10/1/18 : NA/NR; 1/3/22 : O/NR

Price Target History: 1/3/22 : 16; 5/24/22 : 14; 8/17/22 : 12.5; 11/16/22 : 10.75; 3/3/23 : 10; 5/1/23 : 11

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Brazil Financial Institutions

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/17/2023)
Jorge Echevarria		
BB Seguridade Participacoes S.A. (BBSE3.SA)	U (06/25/2023)	R$31.61
Caixa Seguridade Participacoes S.A. (CXSE3.SA)	O (06/10/2021)	R$11.22
Porto Seguro (PSSA3.SA)	O (11/02/2022)	R$25.77
Jorge Kuri		
B3 SA BRASIL BOLSA BALCAO (B3SA3.SA)	E (07/21/2017)	R$11.27
Banco Bradesco (BBD.N)	O (08/07/2019)	US$2.85
Banco BTG Pactual SA (BPAC11.SA)	O (03/28/2017)	R$30.24
Banco do Brasil (BBAS3.SA)	O (03/07/2018)	R$49.36
Banco Pan SA (BPAN4.SA)	O (11/29/2022)	R$7.72
Banco Santander Brasil (BSBR.N)	O (08/07/2019)	US$5.21
Cielo S.A. (CIEL3.SA)	E (12/18/2020)	R$3.54
Inter & Co Inc. (INTR.O)	U (06/13/2018)	US$4.10
Itau Unibanco Holding S.A. (ITUB.N)	E (05/19/2023)	US$5.27
Nu Holdings Ltd. (NU.N)	O (01/03/2022)	US$7.71
PagSeguro Digital (PAGS.N)	E (12/06/2022)	US$7.56
StoneCo Ltd. (STNE.O)	E (12/06/2022)	US$10.04
XP Inc (XP.O)	O (01/06/2020)	US$21.38

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

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